

Unit Functional Design Specification – CPKO Fractionation

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1 EQUIPMENT MODULE CLASS DESIGN – EM_CF_OF_CRYSTAL (F1002A)

1.1 Functional Description

EM_CF_OF_CRYSTAL is a class based equipment module that is used for Oil Feed and Crystallization with Mobilizers as a part of Discharge Line Blow. EM_CF_OF_CRYSTAL is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. All devices used in EM shall be acquired in Aux On Command and shall be released on completion of Aux Off Command. Hold Request shall be enabled for all acquired devices at the end of HOLD command.

1.2 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name
EM	EM_CF_OF_CRYSTAL	CF1EQ_1002A
DEV1	_MRSS11_130_CUR	CF1_PF1001A_B
DEV2	_MRSS11_130_SE	CF1_PU105002A
DEV3	_MVS11_IL_130_SO	CF1_PU1002A
DEV4	_V12_IP_130_SE	CF1_V1002AFO
DEV5	_V12_IP_130_SE	CF1_V1002ADO
DEV6	_V12_IP_130_SE	CF1_V1002TIW
DEV7	_V12_IP_130_SE	CF1_V1002THW
DEV8	_V12_IP_130_SE	CF1_V1002AIW
DEV9	_V12_IP_130_SE	CF1_V1002AHW
DEV10	_CSD_130_SE	CF1_TT1050A1
DEV11	_CSD_130_SE	CF1_TT1021A1
DEV12	_V12_IP_130_SE	CF1_V1001AD
DEV13	_V12_IP_130_SE	CF1_V1001AR
DEV14	_V12_IP_130_SE	CF1_V1002DO
DEV15	EM_CF_STATOLIZR1	CF1EQ_1002S1
DEV16	EM_CF_STATOLIZR1	CF1EQ_1002S2
DEV17	EM_CF_DSCHRG_BLW	CF1EQ_1002_BLW
DEV18	_O_RAMP_130_SE	CF1_TT1050A1_RMP

Table 1: Instances

1.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

1.3.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Command Hold Start	A0010	Store Fail Message
		A0020	Reset Hold Request
		A0030	Hold Interval Timer and Pause Ramp module
		A0040	Save Previous Step Time
T1000	Actions Complete or Time Out & Hold=True		If Hold = True then go to step H1000
T2000	Actions Complete or Time Out & Hold=False		If Hold = False then go to step H2000
H1000	Hold	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Stopping DEV2
		A0030	Stopping and Disabling Conditional Logic for DEV3
T9900A	Actions Complete or Time Out		Go to step H9900A
H2000	General Reset	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Stopping DEV2
		A0030	Stopping and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7

Step	Description	Action	Action Description
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Set DEV10 Mode to Remote Out
		A0110	Set DEV10 output to Closed and Disable Conditional Logic
		A0120	Set DEV11 Mode to Remote Out
		A0130	Set DEV11 output to Closed and Disable Conditional Logic
		A0140	Closing and Disabling Conditional Logic for DEV12
		A0150	Closing and Disabling Conditional Logic for DEV13
		A0160	Closing and Disabling Conditional Logic for DEV14
T9900B	Actions Complete or Time Out		Go to step H9900B
H9900A	Command Hold End	A0010	Completed
END	Command Complete		True
H9900B	Release Devices	A0010	Set Message: Disable HM and SM
		A0020	Release Private Devices
		A0030	Release Private Devices
		A0040	Release Private Devices
		A0050	Release Private Devices
		A0060	Release Private Devices
		A0070	Release Private Devices
		A0080	Release Private Devices
		A0090	Release Private Devices
		A0100	Release Private Devices
		A0110	Release Private Devices
		A0120	Release Private Devices
		A0130	Release Private Devices
		A0140	Release Private Devices
		A0150	Release Private Devices

Step	Description	Action	Action Description
		A0160	Reset Parameters
END1	Command Complete		True

Table 2: Hold Sequence

1.3.2 Command 01 – Aux On

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Aux On Command
T0010	First Pass = False		If First Pass = False then go to step S0010
T0020A	First Pass = True		If First Pass = True then go to step S0020
S0010	Initializations	A0010	First Pass Logic
		A0020	First Pass = True
		A0030	Reset Interval timer
T0020B	Acquire Devices		If no pending actions in S0010 then go to step S0020
S0020	Acquire Devices	A0010	Acquire Devices Message
		A0020	Acquire Private Devices
		A0030	Acquire Private Devices
		A0040	Acquire Private Devices
		A0050	Acquire Private Devices
		A0060	Acquire Private Devices
		A0070	Acquire Private Devices
		A0080	Acquire Private Devices
		A0090	Acquire Private Devices
		A0100	Acquire Private Devices
		A0110	Acquire Private Devices
		A0120	Acquire Private Devices
		A0130	Acquire Private Devices
		A0140	Acquire Private Devices
		A0150	Acquire Private Devices
		A0160	Acquire Private Devices
		A0170	Reset Hold Request
T9900	Complete		If no pending actions in S0020 then go to step S9900
S9900	Complete	A0010	Set Message: Aux On Command Completed

END	Command Complete		True
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Table 3: Aux On Sequence

1.3.3 Command 02 – Aux Off

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Aux off Command
T0100	Aux Off		If no pending actions in S0000 then go to step S0100
S0100	Aux Off	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Stopping DEV2
		A0030	Stopping and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Set DEV10 Mode to Remote Out
		A0110	Set DEV10 output to Closed and Disable Conditional Logic
		A0120	Set DEV11 Mode to Remote Out
		A0130	Set DEV11 output to Closed and Disable Conditional Logic
		A0140	Closing and Disabling Conditional Logic for DEV12
		A0150	Closing and Disabling Conditional Logic for DEV13
		A0160	Closing and Disabling Conditional Logic for DEV14
		A0170	Reset Aux Parameter
T0200	Release Devices		If no pending actions in S0100 then go to step S0200
S0200	Release Devices	A0020	Release Private Devices
		A0030	Release Private Devices
		A0040	Release Private Devices
		A0050	Release Private Devices
		A0060	Release Private Devices

Step	Description	Action	Action Description
		A0070	Release Private Devices
		A0080	Release Private Devices
		A0090	Release Private Devices
		A0100	Release Private Devices
		A0110	Release Private Devices
		A0120	Release Private Devices
		A0130	Release Private Devices
		A0140	Release Private Devices
		A0150	Release Private Devices
T9900	Complete		If no pending actions in S0200 then go to step S9900
S9900	Complete	A0010	Set Message: Aux Off Command Completed
END	Command Complete		True

Table 4: Aux Off Sequence

1.3.4 Command 03 – Crystallization

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Crystallization Command
		A0020	Reset Interval Timer
T0100	Standby		
T0200B	Step 1		If Task Pointer = 2 then go to step S0200
T0300B	Step 2		If Task Pointer = 3 then go to step S0300
T0400B	Step 3		If Task Pointer = 4 then go to step S0400
T0500B	Step 4		If Task Pointer = 5 then go to step S0500
T0600B	Step 5		If Task Pointer = 6 then go to step S0600
T0700B	Step 6		If Task Pointer = 7 then go to step S0700
T0800B	Step 7		If Task Pointer = 8 then go to step S0800
T0900B	Step 8		If Task Pointer = 9 then go to step S0900
T1000B	Step 9		If Task Pointer = 10 then go to step S1000
T1100B	Step 10		If Task Pointer = 11 then go to step S1100
T1200B	Step 11		If Task Pointer = 12 then go to step S1200
T1300B	Step 12		If Task Pointer = 13 then go to step S1300
S0100	Standby	A0010	Set Interval timer
		A0020	Stopping and Disabling Conditional Logic for DEV1
		A0030	Stopping DEV2

Step	Description	Action	Action Description
		A0040	Stopping and Disabling Conditional Logic for DEV3
		A0050	Closing and Disabling Conditional Logic for DEV4
		A0060	Closing and Disabling Conditional Logic for DEV5
		A0070	Closing DEV6
		A0080	Closing and Disabling Conditional Logic for DEV7
		A0090	Closing DEV8
		A0100	Closing and Disabling Conditional Logic for DEV9
		A0110	Set DEV10 Mode to Remote Out
		A0120	Set DEV10 output to Closed and Disable Conditional Logic
		A0130	Set DEV11 Mode to Remote Out
		A0140	Set DEV11 output to Closed and Disable Conditional Logic
		A0150	Closing and Disabling Conditional Logic for DEV12
		A0160	Closing and Disabling Conditional Logic for DEV13
		A0170	Closing and Disabling Conditional Logic for DEV14
		A0180	Starting Interval timer
		A0190	Load Next Recipe
T0200A	Step 1		If Actual interval time >= Interval time SP and Push button Tank Farm Ready (OP002) is ON and V1001A closed feedback and Push Button F1002A (OP001) Auto then go to step S0200
S0200	Step 1	A0005	Reset Actual Interval Time and Set Task Pointer
		A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Starting DEV2
		A0030	Stopping and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Closing DEV8
		A0090	Enabling Conditional Logic for DEV9

Step	Description	Action	Action Description
		A0100	Set DEV10 Mode to Remote Cascade
		A0110	Set DEV10 output to Temp.H2O RSP and Enabling Conditional Logic
		A0120	Set DEV11 Mode to Remote Out
		A0130	Set DEV11 output to Closed and Disable Conditional Logic
		A0140	Closing and Disabling Conditional Logic for DEV12
		A0150	Closing and Disabling Conditional Logic for DEV13
		A0160	Closing and Disabling Conditional Logic for DEV14
		A0170	Acquire Private Devices
		A0180	Starting Blow Command DEV17
		A0190	Set Hot water flushing Time
		A0200	Start Hot water flushing Time
		A0210	Hot water flushing Time Message
T0300A	Step 2		If Step Time is over then go to step S0300
S0300	Step 2	A0010	Enabling Conditional Logic for DEV1 and Set Task Pointer
		A0020	Starting DEV2
		A0030	Enabling Conditional Logic for DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Closing DEV8
		A0090	Enabling Conditional Logic for DEV9
		A0100	Set DEV10 Mode to Remote Cascade
		A0110	Set DEV10 output to Temp.H2O RSP
		A0120	Set DEV11 Mode to Remote Cascade
		A0130	Setting DEV11 setpoint to RSP and Enabling Conditional Logic
		A0140	Enabling Conditional Logic for DEV12
		A0150	Closing and Disabling Conditional Logic for DEV13
		A0160	Closing and Disabling Conditional Logic for DEV14
		A0170	Set Oil Filling Time
		A0180	Start Oil Filling Time
		A0190	Oil Filling Time Message

Step	Description	Action	Action Description
T0400A	Step 3		If Step Time is over and (Crystalliser Fill level reached or LSH1002A1 triggered) then go to step S0400
S0400	Step 3	A0010	Stopping and Disabling Conditional Logic for DEV1 and Set Task Pointer
		A0020	Starting DEV2
		A0030	Starting and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Closing DEV8
		A0090	Enabling Conditional Logic for DEV9
		A0100	Set DEV10 Mode to Remote Cascade
		A0110	Set DEV10 output to Temp.H2O RSP
		A0120	Set DEV11 Mode to Remote Output
		A0130	Set DEV11 output to Closed and Disable Conditional Logic
		A0140	Opening and Disabling Conditional Logic for DEV12
		A0150	Opening and Disabling Conditional Logic for DEV13
		A0160	Set Message: Wait for 10 Sec
		A0170	Closing and Disabling Conditional Logic for DEV12
		A0180	Closing and Disabling Conditional Logic for DEV13
		A0190	Closing and Disabling Conditional Logic for DEV14
		A0200	Set Homogenizing Time
		A0210	Start Homogenizing Time
		A0220	Homogenizing Time Message
T0500A	Step 4		If Step Time is over then go to step S0500
S0500	Step 4	A0005	Take PID to OOS
		A0006	Set PID to Direct Action
		A0010	Stopping and Disabling Conditional Logic for DEV1 and Set Task Pointer
		A0020	Starting DEV2
		A0030	Starting and Disabling Conditional Logic for DEV3

Step	Description	Action	Action Description
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Enabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Enabling Conditional Logic for DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Set DEV10 Mode to Remote Cascade
		A0110	Set DEV10 output to Temp.H2O RSP
		A0120	Set DEV11 Mode to Remote Output
		A0130	Set DEV11 output to Closed and Disable Conditional Logic
		A0140	Closing and Disabling Conditional Logic for DEV12
		A0150	Closing and Disabling Conditional Logic for DEV13
		A0160	Closing and Disabling Conditional Logic for DEV14
		A0170	Set Cooling 1 Time
		A0180	Start Cooling 1 Time
		A0190	Cooling 1 Time Message
		A0200	Action on LO Mode CF1_TT1050A1
T0600A	Step 5		If Step Time is over then go to step S0600
S0600	Step 5	A0005	Take PID to OOS Mode
		A0006	Set PID to Direct Action
		A0010	Stopping and Disabling Conditional Logic for DEV1 and Set Task Pointer
		A0020	Starting DEV2
		A0030	Starting and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Enabling DEV6 Opening Condition
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Enabling DEV8 Opening Condition
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Set DEV10 Mode to Remote Cascade

Step	Description	Action	Action Description
		A0110	Set DEV10 output to Temp.H2O RSP
		A0120	Set DEV11 Mode to Remote Output
		A0130	Set DEV11 output to Closed and Disable Conditional Logic
		A0140	Closing and Disabling Conditional Logic for DEV12
		A0150	Closing and Disabling Conditional Logic for DEV13
		A0160	Closing and Disabling Conditional Logic for DEV14
		A0170	Set Cooling 2 Time
		A0180	Start Cooling 2 Time
		A0190	Cooling 2 Time Message
		A0200	Action on LO Mode CF1_TT1050A1
T0700A	Step 6		If Step Time is over then go to step S0700
S0700	Step 6	A0005	Take PID to OOS Mode
		A0006	Set PID to Direct Direction
		A0010	Stopping and Disabling Conditional Logic for DEV1 and Set Task Pointer
		A0020	Starting DEV2
		A0030	Starting and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Enabling DEV6 Opening Condition
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Enabling DEV8 Opening Condition
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Set DEV10 Mode to Remote Cascade
		A0110	Set DEV10 output to Temp.H2O RSP
		A0120	Set DEV11 Mode to Remote Output
		A0130	Set DEV11 output to Closed and Disable Conditional Logic
		A0140	Closing and Disabling Conditional Logic for DEV12
		A0150	Closing and Disabling Conditional Logic for DEV13
		A0160	Closing and Disabling Conditional Logic for DEV14
		A0170	Set Cooling 3 Time

Step	Description	Action	Action Description
		A0180	Start Cooling 3 Time
		A0190	Cooling 3 Time Message
		A0200	Action on LO Mode CF1_TT1050A1
T0800A	Step 7		If Step Time is over then go to step S0800
T1100D	Step 10		If TT1002A1 < Fill Stato Temp then go to step S1100
S0800	Step 7	A0005	Take PID to OOS
		A0006	Set PID to Direct Direction
		A0010	Stopping and Disabling Conditional Logic for DEV1 and Set Task Pointer
		A0020	Starting DEV2
		A0030	Starting and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Enabling DEV6 Opening Condition
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Enabling DEV8 Opening Condition
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Set DEV10 Mode to Remote Cascade
		A0110	Set DEV10 output to Temp.H2O RSP
		A0120	Set DEV11 Mode to Remote Output
		A0130	Set DEV11 output to Closed and Disable Conditional Logic
		A0140	Closing and Disabling Conditional Logic for DEV12
		A0150	Closing and Disabling Conditional Logic for DEV13
		A0160	Closing and Disabling Conditional Logic for DEV14
		A0170	Set Cooling 4 Time
		A0180	Start Cooling 4 Time
		A0190	Cooling 4 Time Message
		A0200	Action on LO Mode CF1_TT1050A1
T0900A	Step 8		If Step Time is over then go to step S0900
T1100E	Step 10		If TT1002A1 < Fill Stato Temp then go to step S1100
S0900	Step 8	A0005	Take PID in OOS Mode
		A0006	Set PID to Direct Direction

Step	Description	Action	Action Description
		A0010	Stopping and Disabling Conditional Logic for DEV1 and Set Task Pointer
		A0020	Starting DEV2
		A0030	Starting and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Enabling DEV6 Opening Condition
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Enabling DEV8 Opening Condition
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Set DEV10 Mode to Remote Cascade
		A0110	Set DEV10 output to Temp.H2O RSP
		A0120	Set DEV11 Mode to Remote Output
		A0130	Set DEV11 output to Closed and Disable Conditional Logic
		A0140	Closing and Disabling Conditional Logic for DEV12
		A0150	Closing and Disabling Conditional Logic for DEV13
		A0160	Closing and Disabling Conditional Logic for DEV14
		A0170	Set Cooling 5 Time
		A0180	Start Cooling 5 Time
		A0190	Cooling 5 Time Message
		A0200	Action on LO Mode CF1_TT1050A1
T1000A	Step 9		If Step Time is over then go to step S1000
T1100F	Step 10		If TT1002A1 < Fill Stato Temp then go to step S1100
S1000	Step 9	A0005	Take PID in OOS Mode
		A0006	Set PID to Direct Direction
		A0010	Stopping and Disabling Conditional Logic for DEV1 and Set Task Pointer
		A0020	Starting DEV2
		A0030	Starting and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Enabling DEV6 Opening Condition

Step	Description	Action	Action Description
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Enabling DEV8 Opening Condition
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Set DEV10 Mode to Remote Cascade
		A0110	Set DEV10 output to Temp.H2O RSP
		A0120	Set DEV11 Mode to Remote Output
		A0130	Set DEV11 output to Closed and Disable Conditional Logic
		A0140	Closing and Disabling Conditional Logic for DEV12
		A0150	Closing and Disabling Conditional Logic for DEV13
		A0160	Closing and Disabling Conditional Logic for DEV14
		A0170	Set Cooling 6 Time
		A0180	Start Cooling 6 Time
		A0190	Cooling 6 Time Message
		A0200	Action on LO Mode CF1_TT1050A1
T1100A	Step 10		If Step Time is over then go to step S1100
S1100	Step 10	A0005	Take PID in OOS Mode and set Task Pointer
		A0006	Set PID to Direct Direction
		A0010	Acquire DEV15
		A0015	Acquire DEV16
		A0020	Set Message: Starting Blow Command of DEV15 or DEV16.
		A0030	Stopping and Disabling Conditional Logic for DEV1
		A0040	Starting DEV2
		A0050	Starting and Disabling Conditional Logic for DEV3
		A0060	Closing and Disabling Conditional Logic for DEV4
		A0070	Closing and Disabling Conditional Logic for DEV5
		A0080	Enabling DEV6 Opening Condition
		A0090	Closing and Disabling Conditional Logic for DEV7
		A0100	Enabling DEV8 Opening Condition
		A0110	Closing and Disabling Conditional Logic for DEV9
		A0120	Set DEV10 Mode to Remote Cascade

Step	Description	Action	Action Description
		A0130	Set DEV10 output to Temp.H2O RSP
		A0140	Set DEV11 Mode to Remote Output
		A0150	Set DEV11 output to Closed and Disable Conditional Logic
		A0160	Closing and Disabling Conditional Logic for DEV12
		A0170	Closing and Disabling Conditional Logic for DEV13
		A0180	Closing and Disabling Conditional Logic for DEV14
		A0190	Set Cooling 7 & Stato check Time
		A0200	Start Cooling 7 & Stato check Time
		A0210	Cooling 7 & Stato check Time Message
		A0220	Action on LO Mode CF1_TT1050A1
T1200A	Step 11		If Step Time is over and selected F1002Sx <On Pressure> then go to step S0800
S1200	Step 11	A0005	Take PID in OOS Mode
		A0006	Set PID to Direct Direction
		A0010	Stopping and Disabling Conditional Logic for DEV1 and Set Task Pointer
		A0020	Starting DEV2
		A0030	Starting and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Enabling DEV6 Opening Condition
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Enabling DEV8 Opening Condition
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Set DEV10 Mode to Remote Cascade
		A0110	Set DEV10 output to Temp.H2O RSP
		A0120	Set DEV11 Mode to Remote Output
		A0130	Set DEV11 output to Closed and Disabe Conditional Logic
		A0140	Closing and Disabling Conditional Logic for DEV12
		A0150	Closing and Disabling Conditional Logic for DEV13
		A0160	Closing and Disabling Conditional Logic for DEV14
		A0170	Set Cooling 8 & Safe check Time

Step	Description	Action	Action Description
		A0180	Start Cooling 8 & Safe check Time
		A0190	Start Cooling 8 & Safe check Time Message
		A0200	Action on LO Mode CF1_TT1050A1
T1300A	Step 12		If Step Time is over and selected F1002Sx <Temp Regulation> elapsed and XVF02A-BA close feedback received then go to step S0800
S1300	Step 12	A0005	Take PID in OOS Mode
		A0006	Set PID to Direct Direction
		A0010	Stopping and Disabling Conditional Logic for DEV1 and Set Task Pointer
		A0020	Starting DEV2
		A0030	Enabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Enabling DEV6 Opening Condition
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Enabling DEV8 Opening Condition
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Set DEV10 Mode to Remote Cascade
		A0110	Set DEV10 output to Temp.H2O RSP
		A0120	Set DEV11 Mode to Remote Output
		A0130	Set DEV11 output to Closed and Disable Conditional Logic
		A0140	Enabling Conditional Logic for DEV12
		A0150	Closing and Disabling Conditional Logic for DEV13
		A0160	Enabling Conditional Logic for DEV14
		A0170	Set Message: Waiting for LT1002A1 >= Crystallizer Low Level SP (TP026).
		A0180	Delay Timer (5 sec).
		A0190	Action on LO Mode CF1_TT1050A1
T9900	Complete		If LT1002A1 < Crystallizer Low Level SP (TP001) + 5sec delay then go to step S9900
S9900	Complete	A0010	Set Message: Crystallization Command Complete
		A0020	Take PID to OOS
		A0030	Set PID to Reverse action

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Step	Description	Action	Action Description
		A0040	Set all time parameters
END	Command Complete		True

Table 5: Crystallization Sequence

1.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value/Reference
Operating Parameters				
OP001	F1002A Mode	Floating Point	NA	0
OP002	Tank Farm Ready	Floating Point	NA	0
OP003	HOLD	Named Set	NA	OFF
OP004	AUX	Named Set	NA	OFF
OP005	Operator Confirmation	Boolean	NA	False
Tuning Parameters				
TP001	Crystallizer Low Level SP	Floating Point	sec	5
TP002	Not Used	Floating Point	NA	0
TP003	Crystalliser Fill Level	Floating Point	%	10
TP004	Agitator Level	Floating Point	%	0
TP005	Fill Stato Temp	Floating Point	°C	0
TP006	Interval time SP	Floating Point	min	1
TP007	Actual Recipe Number	Floating Point	NA	0
TP008	Hot Water Flushing	External Reference	min	CRYS_RECIPE_AB/STEP_TIME_AB_1
TP009	Oil Filling	External Reference	min	CRYS_RECIPE_AB/STEP_TIME_AB_2
TP010	Homogenizing	External Reference	min	CRYS_RECIPE_AB/STEP_TIME_AB_3
TP011	Cooling 1	External Reference	min	CRYS_RECIPE_AB/STEP_TIME_AB_4
TP012	Cooling 2	External Reference	min	CRYS_RECIPE_AB/STEP_TIME_AB_5
TP013	Cooling 3	External Reference	min	CRYS_RECIPE_AB/STEP_TIME_AB_6
TP014	Cooling 4	External Reference	min	CRYS_RECIPE_AB/STEP_TIME_AB_7
TP015	Cooling 5	External Reference	min	CRYS_RECIPE_AB/STEP_TIME_AB_8
TP016	Cooling 6	External Reference	min	CRYS_RECIPE_AB/STEP_TIME_AB_9



Parameter	Description	Type	Units	Default Value/Reference
TP017	Cooling 7+ Stato Check	External Reference	min	CRYS_RECIPE_AB/STEP_TIME_AB_10
TP018	Cooling 8 + Safe Check	External Reference	min	CRYS_RECIPE_AB/STEP_TIME_AB_11
TP019	Oil Draining	External Reference	min	CRYS_RECIPE_AB/STEP_TIME_AB_12
TP020	Next Batch Number	Floating Point	NA	1
TP021	Min PF1002A Speed	Floating Point	NA	0

Table 6: Operating and Tuning Parameters

1.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value/Reference
MY_TAG		CF1EQ_1002A
OPR_CMD_MASK		15
DY_REF_1	Dynamic Reference	
V1001A	External Reference	CF1_V1001A/EDC1/PV_D
LT1002A1	External Reference	CF1_LT1002A1/PV
LSH1002A1	External Reference	CF1_LSH1002A1/DI1/DISC_ACT
TT1002A1	External Reference	CF1_TT1002A1/PV
LSH1360V1	External Reference	CF1-LSH1360V1/DI1/DISC_ACT
DY_REF_2	Dynamic Reference	
STATOLIZER_S1	String	CF1EQ_1002S1
STATOLIZER_S2	String	CF1EQ_1002S2
STEP_TIME	Floating Point	0
COMPLETED_TIME	Floating Point	0
RESET_COMP_TM	Boolean	False
TT1050A1	External Reference	CF1_TT1050A1/PID1/CONTROL_OPTS
HOLD	Floating Point	0
TIM_SEC	Floating Point	0

Table 7: Configurable Parameters

1.6 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_CF_OF_CRYSTAL	CPKO Fractionation Area: Oil Feed & Crystallization with Mobilizers

Table 8: Named Set Details

Details of states for NS_CF_OF_CRYSTAL named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Aux On
02	Aux Off
03	Crystallization

Table 9: Name Set State Details

2 EQUIPMENT MODULE CLASS DESIGN – EM_CF_DSCHARGE_BLW (F1002A DISCHARGE LINE BLOW)

2.1 Functional Description

EM_CF_DSCHARGE_BLW is a class based equipment module that is used for Oil Feed and Crystallization with Mobilizers as a part of Discharge Line Blow. EM_CF_DSCHARGE_BLW is derived from the Command Driven equipment module class EM_SFC16_130. Refer FDS for the functional design requirements including faceplate and detail display. Hold Request shall be enabled for all acquired devices at the end of HOLD command.

2.2 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name
EM	EM_CF_DSCHARGE_BLW	CFXEQ_1002A
DEV1	V12_IP_130_SE	Ignored
DEV2	V12_IP_130_SE	CF1_V1102ABY

Table 10: Instances

2.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

2.3.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Command Hold Start	A0010	Store Fail Message
		A0020	Reset Hold Request
		A0030	Save Previous Step Time
		A0040	Hold Blow On Timer
T1000	Actions Complete or Time Out & Hold=True		Go to step H1000
T2000	Actions Complete or Time Out & Hold=False		Go to step H2000

Step	Description	Action	Action Description
H1000	Hold	A0010	Msg: Executing Hold Command
T9900A	Actions Complete or Time Out		Go to step H9900A
H2000	General Reset	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Closing DEV2
T9900B	Actions Complete or Time Out		Go to step H9900B
H9900A	Command Hold End	A0010	Completed
END	Command Complete		True
H9900B	Release Devices	A0010	Release Devices Message
		A0020	Release Private Devices
		A0030	Release Private Devices
		A0040	Reset Parameters
END1	Command Complete		True

Table 11: Hold Sequence

2.3.2

Command 01 – Blow

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Blow Command
T0010	First Pass = False		If First Pass = False then go to step S0010
T0020A	First Pass = True		If First Pass = True then go to step S0020
S0010	Initializations	A0010	First Pass Logic
		A0020	First Pass = True
		A0030	Reset Blow On timer
T0020B	Actions Complete		If no pending actions in S0010 then go to step S0020
S0020	Acquire Devices	A0010	Acquire Devices Message
		A0020	Acquire Private Devices
		A0030	Acquire Private Devices
		A0040	Reset Hold Request
T0100	Start		If Task Pointer = 1 then go to step S0100



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Step	Description	Action	Action Description
T0200B	Fill		If Task Pointer = 2 then go to step S0200
T0300B	Production		If Task Pointer = 3 then go to step S0300
T0400B	Drain		If Task Pointer = 4 then go to step S0400
S0100	Standby	A0010	Set Message: F1002A Discharge Line Blow Enable
		A0020	Reset Blow On Timer and Actual Preset Count
		A0030	Closing DEV1
		A0040	Closing DEV2
T0200A	Blow 1_ON		If LT_F02A-1 < Crystalliser Low Level SP (TP001) & F1002A is not in <Step 12> for 5seconds then go to step S0200
S0200	Blow 1_ON	A0010	Enabling Conditional Logic for DEV1 and Set task Pointer
		A0020	Opening DEV2
		A0030	Starting Blow On timer
		A0040	Blow On Time
		A0060	Condition for Blow On timer
T0300	Blow 1_OFF		If Blow On Time is over (TP002) then go to step S0300
S0300	Blow 1_OFF	A0005	Increment Actual Blow Count & Set Task Pointer
		A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Opening DEV2
		A0030	Set Blow off Time
		A0040	Start Blow Off Time
		A0050	Blow Off Time Message
T0200B	Blow 1_ON		Blow Off Time (TP003) Over & Actual Blow count (TP004) < Preset Blow Count (TP005)
T0400	Blow Over		Blow Off Time (TP003) Over & Actual Blow count (TP004) >= Preset Blow Count (TP005)
S0400	Blow Over	A0010	Closing and Disabling Conditional Logic for DEV1 and Set Task Pointer
		A0020	Closing DEV2
T0500	Close Devices		If LT_F02A-1 > Crystalliser Low Level SP (TP001) and 10% & F1002Sx is in <Blow Over> (EM_1002S1/EM_1002S2) and 5seconds completed then go to step S0500



Step	Description	Action	Action Description
S0500	Close Devices	A0010	Closing and Disabling Conditional Logic for DEV1
		A0020	Closing DEV2
T0600	Release Devices		If no pending actions in S0500 then go to step S0600
S0600	Release Devices	A0010	Release Devices Message
		A0020	Release Private Devices
		A0030	Release Private Devices
T9900	Complete		Go to step S9900
S9900	Complete	A0010	Set Message: Blow Command Complete
END	Command Complete		True

Table 12: Blow Sequence

2.4

Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value/Reference
Operating Parameters				
N/A				
Tuning Parameters				
TP001	Crystallizer Low Level SP	Floating Point	sec	5
TP002	Blow On	Floating Point	sec	20
TP003	Blow Off	Floating Point	sec	0
TP004	Actual Blow Count	Floating Point	%	0
TP005	Preset Blow Count	Floating Point	%	0

Table 13: Operating and Tuning Parameters

2.5

Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value/Reference
MY_TAG		CF1EQ_1002_BLW

Parameter Name	Parameter Type	Value/Reference
OPR_CMD_MASK		3
LT_002A1	External Reference	CF1_LT1002A1/PV
EM_F1002A	External Reference	CF1EQ_1002A/COMMAND_00003/S1300/ACTIVE
EM_1002S1	External Reference	CF_STX_BLW_S1/STEP_STATUS
EM_1002S2	External Reference	CF_STX_BLW_S2/STEP_STATUS
STEP_STATUS	String	
DELAY_TMR_1	String	5
STEP_TIME	Floating Point	0
COMPLETED_TIME	Floating Point	0
RESET_COMP_TM	Boolean	False
HOLD	External Reference	CF1EQ_1002A/OP003_VALUE

Table 14: Configurable Parameters

2.6

Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_CF_DSCHARGE_BLW	CPKO Fractionation Area: Oil Feed & Crystallization with Mobilizers

Table 15: Named Set Details

Details of states for NS_CF_DSCHARGE_BLW named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Blow

Table 16: Name Set State Details

3 EQUIPMENT MODULE CLASS DESIGN – EM_CF_STATOLIZR1 (F1002S1/S2)

3.1 Functional Description

EM_CF_STATOLIZR1 is a class based equipment module that is used for Statolizer as a part of Discharge Line Blow. EM_CF_STATOLIZR1 is derived from the CommandDriven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. All devices used in EM shall be acquired in Blow Command and shall be released on completion of the same. Hold Request shall be enabled for all acquired devices at the end of Hold command.

3.2 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name	Instance Name
EM	EM_CF_STATOLIZR1	CFEQ_1002S1	CFEQ_1002S2
DEV1	_MVS11_IL_130_SO	CF1_PU1102A	
DEV2	_MRSS11_130_SE	CF1_PU11502S1	CF1_PU1150S2
DEV3	_V12_IP_130_SE	CF1_V1102S1FO	CF1_V1102S2FO
DEV4	_CSL_130_SE	CF1_PT1102S	
DEV5	_V12_IP_130_SE	Ignored	
DEV6	_V12_IP_130_SE	Ignored	
DEV7	_V12_IP_130_SE	Ignored	
DEV8	_V12_IP_130_SE	CF1_V11021DW	CF1_V1102DW
DEV9	_V12_IP_130_SE	CF1_V1102S1FW	Ignored
DEV10	_V12_IP_130_SE	CF1_V1102S1RW	Ignored
DEV11	_V12_IP_130_SE	CF1_V1102S1	CF1_V1102S2
DEV12	_V12_IP_130_SE	CF1_V1102S1V2	CF1_V1102S2V2
DEV13	_CSD_130_SE	CF1_TT1150S11	CF1_TT1150S21
DEV14	_V12_IP_130_SE	CF1_V1102S1HO	
DEV15	_V12_IP_130_SE	CF1_V1102SH1	
DEV16	_V12_IP_130_SE	CF1_V1102SH2	
DEV17	_CSL_130_SE	Ignored	
DEV18	_O_RAMP_130_SE	CF1_TT1150S1_RMP	CF1_TT1150S2_RM P
DEV19	EM_CF_STX_BLW	CF_STX_BLW_S1	CF_STX_BLW_S2

Table 17: Instances

3.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

3.3.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Command Hold Start	A0010	Store Fail Message
		A0020	Reset Hold Request
		A0030	Hold Retentive Timers
		A0040	Save Previous Step Time
T1000	Actions Complete or Time Out		Go to step H1000
H1000	Hold Command 1	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Stopping DEV2
		A0030	Closing DEV3
		A0040	Set DEV4 Mode to Remote Out
		A0050	Set DEV4 output to Closed and Disable Conditional Logic
		A0060	Closing DEV5
		A0070	Closing DEV6
		A0080	Closing DEV7
		A0090	Closing and Disabling Conditional Logic for DEV8
		A0100	Closing and Disabling Conditional Logic for DEV9
		A0110	Closing and Disabling Conditional Logic for DEV10
		A0120	Closing DEV11
		A0130	Opening DEV12
		A0140	Set DEV13 Mode to Remote Out
		A0150	Set DEV13 output to Closed and Disable Conditional Logic
		A0160	Opening DEV14
		A0170	Opening DEV15
		A0180	Opening DEV16

Step	Description	Action	Action Description
T9900	Actions Complete or Time Out		Go to step H9900
H9900	Command Hold End	A0010	Completed
END	Command Complete		True

Table 18: Hold Sequence

3.3.2 Command 01 – Blow

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Blow Command
T0010	First Pass = False		If First Pass = False then go to step S0010
T0020A	First Pass = True		If First Pass = True then go to step S0020
S0010	Initializations	A0010	First Pass Logic
		A0020	First Pass = True
		A0030	Reset Timers
		A0040	Set Task Pointer
T0020B	Acquire Devices		If no pending actions in S0010 then go to step S0020
S0020	Acquire Devices	A0010	Acquire Devices Message
		A0020	Acquire Shared Device
		A0030	Acquire Private Device
		A0040	Acquire Private Device
		A0050	Acquire Shared Device
		A0060	Acquire Private Device
		A0070	Acquire Private Device
		A0080	Acquire Private Device
		A0090	Acquire Private Device
		A0100	Acquire Private Device
		A0110	Acquire Private Device
		A0120	Acquire Shared Device
		A0130	Acquire Shared Device
		A0140	Acquire Private Device



Step	Description	Action	Action Description
		A0150	Acquire Private Device
		A0160	Acquire Private Device
		A0170	Acquire Private Device
		A0180	Acquire Private Device
		A0190	Acquire Private Device
		A0200	Set Delay timer
		A0210	Reset Hold Request
T0100	Standby		Go to step S0100
T0200B	Stato On Pressure		If Task Pointer = 2 then go to step S0200
T0300B	Step 1		If Task Pointer = 3 then go to step S0300
T0400B	Step 2		If Task Pointer = 4 then go to step S0400
T0500B	Step 3		If Task Pointer = 5 then go to step S0500
T0600B	Step 4		If Task Pointer = 6 then go to step S0600
T0700B	Step 5		If Task Pointer = 7 then go to step S0700
T0800B	Step 6		If Task Pointer = 8 then go to step S0800
T0900B	Step 7		If Task Pointer = 9 then go to step S0900
T1000C	Step 8		If Task Pointer = 10 then go to step S1000
T1100B	Step 9		If Task Pointer = 11 then go to step S1100
T1200B	Step 10		If Task Pointer = 12 then go to step S1200
T1300B	Step 11		If Task Pointer = 13 then go to step S1300
T1400B	Step 12		If Task Pointer = 14 then go to step S1400
T1500B	Closing Statoliser		If Task Pointer = 15 then go to step S1500
T1600B	Aux Off		If Task Pointer = 16 then go to step S1600
S0100	Standby	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Stopping DEV2
		A0030	Closing DEV3
		A0040	Set DEV4 Mode to Remote Out
		A0050	Set DEV4 output to Closed and Disable Conditional Logic
		A0060	Closing DEV5
		A0070	Closing DEV6
		A0080	Closing DEV7
		A0090	Closing and Disabling Conditional Logic for DEV8

Step	Description	Action	Action Description
		A0100	Closing and Disabling Conditional Logic for DEV9
		A0110	Closing and Disabling Conditional Logic for DEV10
		A0120	Closing DEV11
		A0130	Opening DEV12
		A0140	Set DEV13 Mode to Remote Out
		A0150	Set DEV13 output to Closed and Disable Conditional Logic
		A0160	Opening DEV14
		A0170	Opening DEV15
		A0180	Opening DEV16
		A0190	Load Recipe Number
T0200A	Stato On Pressure		If F1002A is in <Step10> and (not other F1002S Step1 or Step2) and F1002Sx selection based on lowest machine number if both F1002S1/S2 same step and selected F1002Sx [signal (no fault, Auto and Close Position) & Pb F1002Sx AUTO (OP001)] then go to step S0200
S0200	Stato On Pressure	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Stopping DEV2
		A0030	Closing DEV3
		A0040	Set DEV4 Mode to Remote Out
		A0050	Set DEV4 output to Closed and Disable Conditional Logic
		A0060	Closing DEV5
		A0070	Closing DEV6
		A0080	Closing DEV7
		A0090	Closing and Disabling Conditional Logic for DEV8
		A0100	Closing and Disabling Conditional Logic for DEV9
		A0110	Closing and Disabling Conditional Logic for DEV10
		A0120	Closing DEV11
		A0130	Opening DEV12
		A0140	Set DEV13 Mode to Remote Out

Step	Description	Action	Action Description
		A0150	Set DEV13 output to Closed and Disable Conditional Logic
		A0160	Opening DEV14
		A0170	Opening DEV15
		A0180	Opening DEV16
T0300A	Step 1		If F1002A is in <step 11> and selected F1002Sx <On Pressure> (OP006) and not any other F1002S in step1 or step2 and Pb F1002Sx AUTO (OP001) then go to step S0300
S0300	Step 1	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Starting DEV2
		A0030	Closing DEV3
		A0040	Set DEV4 Mode to Remote Out
		A0050	Set DEV4 output to Closed and Disable Conditional Logic
		A0060	Closing DEV5
		A0070	Closing DEV6
		A0080	Closing DEV7
		A0090	Closing and Disabling Conditional Logic for DEV8
		A0100	Enabling Conditional Logic for DEV9
		A0110	Enabling Conditional Logic for DEV10 (TP01)
		A0120	Closing DEV11
		A0130	Opening DEV12
		A0140	Set DEV13 Mode to Remote Cascade
		A0150	Set DEV13 output to Temp.H2O RSP and Enabling Conditional Logic
		A0160	Opening DEV14
		A0170	Opening DEV15
		A0180	Opening DEV16
		A0190	Reset ON Pressure
T0400A	Step 2		If F1002A is in <step12> and both V1102S_BY (V1102S_BY) and XV F02Sx-BA (CF1_V1102SBA) closed feedback and Pb F1002Sx AUTO (OP001) then go to step S0400

Step	Description	Action	Action Description
S0400	Step 2	A0010	Enabling Conditional Logic for DEV1
		A0020	Starting DEV2
		A0030	Opening DEV3
		A0040	Set DEV4 Mode to Remote Cascade
		A0050	Set DEV4 setpoint to RSP for DEV4
		A0060	Closing DEV5
		A0070	Closing DEV6
		A0080	Closing DEV7
		A0090	Closing and Disabling Conditional Logic for DEV8
		A0100	Closing and Disabling Conditional Logic for DEV9
		A0110	Enabling Conditional Logic for DEV10 (TP01)
		A0120	Closing DEV11
		A0130	Opening DEV12
		A0140	Set DEV13 Mode to Remote Cascade
		A0150	Set DEV13 output to Temp.H2O RSP and Enabling Conditional Logic
		A0160	Opening DEV14
		A0170	Opening DEV15
		A0180	Opening DEV16
T0500A	Step 3		If F1002A is not in <Step 12> and Pb F1002Sx AUTO (OP001) then go to step S0500
S0500	Step 3	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Starting DEV2
		A0030	Closing DEV3
		A0040	Set DEV4 Mode to Remote Out
		A0050	Set DEV4 output to Closed and Disable Conditional Logic
		A0060	Closing DEV5
		A0070	Closing DEV6
		A0080	Closing DEV7
		A0090	Closing and Disabling Conditional Logic for DEV8
		A0100	Closing and Disabling Conditional Logic for DEV9

Step	Description	Action	Action Description
		A0110	Enabling Conditional Logic for DEV10 (TP01)
		A0120	Closing DEV11
		A0130	Closing DEV12
		A0140	Set DEV13 Mode to Remote Cascade
		A0150	Set DEV13 output to Temp.H2O RSP and Enabling Conditional Logic
		A0160	Opening DEV14
		A0170	Opening DEV15
		A0180	Opening DEV16
		A0190	Set Homogenizing Time
		A0200	Start Homogenizing Time
		A0210	Homogenizing Time Message
		A0220	Acquiring DEV19
		A0230	Command to DEV19
T0600A	Step 4		If Step time (TP007) is over and Pb F1002Sx AUTO (OP001) then go to step S0600
S0600	Step 4	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Starting DEV2
		A0030	Closing DEV3
		A0040	Set DEV4 Mode to Remote Out
		A0050	Set DEV4 output to Closed and Disable Conditional Logic
		A0060	Closing DEV5
		A0070	Closing DEV6
		A0080	Closing DEV7
		A0090	Closing and Disabling Conditional Logic for DEV8
		A0100	Closing and Disabling Conditional Logic for DEV9
		A0110	Enabling Conditional Logic for DEV10 (TP01)
		A0120	Closing DEV11
		A0130	Opening DEV12
		A0140	Set DEV13 Mode to Remote Cascade
		A0150	Set DEV13 output to Temp.H2O RSP
		A0160	Opening DEV14
		A0170	Opening DEV15

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Step	Description	Action	Action Description
		A0180	Opening DEV16
		A0190	Set Cooling 1 Time
		A0200	Start Cooling 1 Time
		A0210	Cooling 1 Time Message
		A0220	Action in LO Mode of TT1150S11
T0700A	Step 5		If Step time (TP018) is over and Pb F1002Sx AUTO (OP001) then go to step S0700
S0700	Step 5	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Starting DEV2
		A0030	Closing DEV3
		A0040	Set DEV4 Mode to Remote Out
		A0050	Set DEV4 output to Closed and Disable Conditional Logic
		A0060	Closing DEV5
		A0070	Closing DEV6
		A0080	Closing DEV7
		A0090	Closing and Disabling Conditional Logic for DEV8
		A0100	Closing and Disabling Conditional Logic for DEV9
		A0110	Enabling Conditional Logic for DEV10 (TP01)
		A0120	Closing DEV11
		A0130	Opening DEV12
		A0140	Set DEV13 Mode to Remote Cascade
		A0150	Set DEV13 output to Temp.H2O RSP
		A0160	Opening DEV14
		A0170	Opening DEV15
		A0180	Opening DEV16
		A0190	Set Cooling 2 Time
		A0200	Start Cooling 2 Time
		A0210	Cooling 2 Time Message
		A0220	Action in LO Mode of TT1150S11
T0800A	Step 6		If Step time (TP019) is over and Pb F1002Sx AUTO (OP001) then go to step S0800
S0800	Step 6	A0010	Stopping and Disabling Conditional Logic for DEV1



Step	Description	Action	Action Description
		A0020	Starting DEV2
		A0030	Closing DEV3
		A0040	Set DEV4 Mode to Remote Out
		A0050	Set DEV4 output to Closed and Disable Conditional Logic
		A0060	Closing DEV5
		A0070	Closing DEV6
		A0080	Closing DEV7
		A0090	Closing and Disabling Conditional Logic for DEV8
		A0100	Closing and Disabling Conditional Logic for DEV9
		A0110	Enabling Conditional Logic for DEV10 (TP01)
		A0120	Closing DEV11
		A0130	Opening DEV12
		A0140	Set DEV13 Mode to Remote Cascade
		A0150	Set DEV13 output to Temp.H2O RSP
		A0160	Opening DEV14
		A0170	Opening DEV15
		A0180	Opening DEV16
		A0190	Set Cooling 3 Time
		A0200	Start Cooling 3 Time
		A0210	Cooling 3 Time Message
		A0220	Action in LO Mode of TT1150S11
T0900A	Step 7		If Step time (TP020) is over and Pb F1002Sx AUTO (OP001) then go to step S0900
S0900	Step 7	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Starting DEV2
		A0030	Closing DEV3
		A0040	Set DEV4 Mode to Remote Out
		A0050	Set DEV4 output to Closed and Disable Conditional Logic
		A0060	Closing DEV5
		A0070	Closing DEV6
		A0080	Closing DEV7

Step	Description	Action	Action Description
		A0090	Closing and Disabling Conditional Logic for DEV8
		A0100	Closing and Disabling Conditional Logic for DEV9
		A0110	Enabling Conditional Logic for DEV10 (TP01)
		A0120	Closing DEV11
		A0130	Opening DEV12
		A0140	Set DEV13 Mode to Remote Cascade
		A0150	Set DEV13 output to Temp.H2O RSP
		A0160	Opening DEV14
		A0170	Opening DEV15
		A0180	Opening DEV16
		A0190	Set Cooling 4 Time
		A0200	Start Cooling 4 Time
		A0210	Cooling 4 Time Message
		A0220	Action in LO Mode of TT1150S11
T1000A	Step 8		If Step time (TP021) is over and Pb F1002Sx AUTO (OP001) then go to step S1000
S1000	Step 8	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Starting DEV2
		A0030	Closing DEV3
		A0040	Set DEV4 Mode to Remote Out
		A0050	Set DEV4 output to Closed and Disable Conditional Logic
		A0060	Closing DEV5
		A0070	Closing DEV6
		A0080	Closing DEV7
		A0090	Closing and Disabling Conditional Logic for DEV8
		A0100	Closing and Disabling Conditional Logic for DEV9
		A0110	Enabling Conditional Logic for DEV10 (TP01)
		A0120	Closing DEV11
		A0130	Opening DEV12
		A0140	Set DEV13 Mode to Remote Cascade
		A0150	Set DEV13 output to Temp.H2O RSP
		A0160	Opening DEV14

Step	Description	Action	Action Description
		A0170	Opening DEV15
		A0180	Opening DEV16
		A0190	Set Cooling 5 Time
		A0200	Start Cooling 5 Time
		A0210	Cooling 5 Time Message
		A0220	Action in LO Mode of TT1150S11
T1100A	Step 9		If Step time (TP024) is over and Pb F1002Sx AUTO (OP001) then go to step S1100
S1100	Step 9	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Starting DEV2
		A0030	Closing DEV3
		A0040	Set DEV4 Mode to Remote Out
		A0050	Set DEV4 output to Closed and Disable Conditional Logic
		A0060	Closing DEV5
		A0070	Closing DEV6
		A0080	Closing DEV7
		A0090	Closing and Disabling Conditional Logic for DEV8
		A0100	Closing and Disabling Conditional Logic for DEV9
		A0110	Enabling Conditional Logic for DEV10 (TP01)
		A0120	Closing DEV11
		A0130	Opening DEV12
		A0140	Set DEV13 Mode to Remote Cascade
		A0150	Set DEV13 output to Temp.H2O RSP
		A0160	Opening DEV14
		A0170	Opening DEV15
		A0180	Opening DEV16
		A0190	Set Cooling 6 Time
		A0200	Start Cooling 6 Time
		A0210	Cooling 6 Time Message
		A0220	Action in LO Mode of TT1150S11
T1200A	Step 10		If Step time (TP025) is over and Pb F1002Sx AUTO (OP001) then go to step S1200

Step	Description	Action	Action Description
S1200	Step 10	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Starting DEV2
		A0030	Closing DEV3
		A0040	Set DEV4 Mode to Remote Out
		A0050	Set DEV4 output to Closed and Disable Conditional Logic
		A0060	Closing DEV5
		A0070	Closing DEV6
		A0080	Closing DEV7
		A0090	Closing and Disabling Conditional Logic for DEV8
		A0100	Closing and Disabling Conditional Logic for DEV9
		A0110	Enabling Conditional Logic for DEV10 (TP01)
		A0120	Closing DEV11
		A0130	Opening DEV12
		A0140	Set DEV13 Mode to Remote Cascade
		A0150	Set DEV13 output to Temp.H2O RSP
		A0160	Opening DEV14
		A0170	Opening DEV15
		A0180	Opening DEV16
		A0190	Set ON DW Timer
		A0200	Set ON FHW Timer
		A0210	Set H0 Open Timer
		A0220	Set H1 Open Timer
		A0230	Set H2 Open Timer
		A0240	Set Force Open H Valve Timer
		A0250	Set Delay PF1050/02S Timer
		A0260	Action in LO Mode of TT1150S11
T1300A	Step 11		If Hopper number < 5Lot and all TT1102S_x < Max Oil Temp to Discharge (TP005) (If Pb Bypass TTF02Sx (OP002/3/4) is OFF) and not any other F1002S in step 11 or step 12 and lowest machine number is selected if both F1002S in same step and Pb F1002Sx AUTO (OP001) then go to step S1300
S1300	Step 11	A0010	Stopping and Disabling Conditional Logic for DEV1

Step	Description	Action	Action Description
		A0020	Start ON DW Timer
		A0030	Start Force Open H valve, ON FHW and Delay PF1050/02S timer
		A0040	Starting DEV2
		A0050	Closing DEV3
		A0060	Set DEV4 Mode to Remote Out
		A0070	Set DEV4 output to Closed and Disable Conditional Logic
		A0080	Closing DEV5
		A0090	Closing DEV6
		A0100	Closing DEV7
		A0110	Enabling Conditional Logic for DEV8
		A0120	Enabling Conditional Logic for DEV9
		A0130	Enabling Conditional Logic for DEV10
		A0140	Closing DEV11
		A0150	Opening DEV12
		A0160	Set DEV13 Mode to Remote Cascade
		A0170	Set DEV13 output to Temp.H2O RSP
		A0180	Start H0 Open Timer
		A0190	Opening DEV14 and Closing DEV15 & DEV16
		A0200	Start H1 Open Timer
		A0210	Opening DEV15 and Closing DEV14 & DEV16
		A0220	Start H2 Open Timer
		A0230	Opening DEV16 and Closing DEV14 & DEV15
		A0240	Set Drain Water Time
		A0250	Start Drain Water Time
		A0260	Drain Water Time Message
		A0270	Action in LO Mode of TT1150S11
		A0280	Stopping DEV2
		A0290	Closing and Disabling Conditional Logic for DEV10
		A0300	Set DEV13 Mode to Remote Out
		A0310	Set DEV13 output to Closed and Disable Conditional Logic
T1400A	Step 12		If Step time is over and Pb F1002Sx AUTO (OP001) then go to step S1400

Step	Description	Action	Action Description
S1400	Step 12	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Start ON DW Timer
		A0030	Start Force Open H valve, ON FHW and Delay PF1050/02S timer
		A0040	Starting DEV2
		A0050	Closing DEV3
		A0060	Set DEV4 Mode to Remote Out
		A0070	Set DEV4 output to Closed and Disable Conditional Logic
		A0080	Closing DEV5
		A0090	Closing DEV6
		A0100	Closing DEV7
		A0110	Closing and Disabling Conditional Logic for DEV8
		A0120	Enabling Conditional Logic for DEV9
		A0130	Enabling Conditional Logic for DEV10
		A0140	Closing DEV11
		A0150	Closing DEV12
		A0160	Set DEV13 Mode to Remote Cascade
		A0170	Set DEV13 output to Temp.H2O RSP
		A0180	Start H0 Open Timer
		A0190	Opening DEV14
		A0200	Start H1 Open Timer
		A0210	Opening DEV15
		A0220	Start H2 Open Timer
		A0230	Opening DEV16
		A0240	Set Cake Discharge Time
		A0250	Start Cake Discharge Time
		A0260	Cake Discharge Time Message
		A0270	Action in LO Mode of TT1150S11
		A0280	Stopping DEV2
		A0290	Closing and Disabling Conditional Logic for DEV10
		A0300	Set DEV13 Mode to Remote Out
		A0310	Set DEV13 output to Closed and Disable Conditional Logic

Step	Description	Action	Action Description
T1500A	Closing Statoliser		If Step time is over and Pb F1002Sx AUTO (OP001) is selected and Stato last pack opened signal received then go to step S1500
S1500	Closing Statoliser	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Stopping DEV2
		A0030	Closing DEV3
		A0040	Set DEV4 Mode to Remote Out
		A0050	Set DEV4 output to Closed and Disable Conditional Logic
		A0060	Closing DEV5
		A0070	Closing DEV6
		A0080	Closing DEV7
		A0090	Closing and Disabling Conditional Logic for DEV8
		A0100	Closing and Disabling Conditional Logic for DEV9
		A0110	Closing and Disabling Conditional Logic for DEV10
		A0120	Closing DEV11
		A0130	Opening DEV12
		A0140	Set DEV13 Mode to Remote Out
		A0150	Set DEV13 output to Closed and Disable Conditional Logic
		A0160	Closing DEV14
		A0170	Closing DEV15
		A0180	Closing DEV16
		A002	Set Message: F1057CK/S Hopper number automatically update = Batch per stato discharge + exsiting hopper number
T1600	Aux Off		If Stato Closed signal (TP007) is received and Pb F1002Sx AUTO (OP001) then go to step S1600
S1600	Aux Off	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Stopping DEV2
		A0030	Closing DEV3
		A0040	Set DEV4 Mode to Remote Out

Step	Description	Action	Action Description
		A0050	Set DEV4 output to Closed and Disable Conditional Logic
		A0060	Closing DEV5
		A0070	Closing DEV6
		A0080	Closing DEV7
		A0090	Closing and Disabling Conditional Logic for DEV8
		A0100	Closing and Disabling Conditional Logic for DEV9
		A0110	Closing and Disabling Conditional Logic for DEV10
		A0120	Closing DEV11
		A0130	Opening DEV12
		A0140	Set DEV13 Mode to Remote Out
		A0150	Set DEV13 output to Closed and Disable Conditional Logic
		A0160	Opening DEV14
		A0170	Opening DEV15
		A0180	Opening DEV16
T1700	Release Devices		If no pending actions in S1600 then go to step S1700
S1700	Release Devices	A0190	Release Devices Message
		A0200	Release Shared Device
		A0210	Release Private Device
		A0220	Release Private Device
		A0230	Release Shared Device
		A0240	Release Private Device
		A0250	Release Private Device
		A0260	Release Private Device
		A0270	Release Private Device
		A0280	Release Private Device
		A0290	Release Private Device
		A0300	Release Shared Device
		A0310	Release Shared Device
		A0320	Release Private Device
		A0330	Release Private Device
		A0340	Release Private Device

Step	Description	Action	Action Description
		A0350	Release Private Device
		A0360	Release Private Device
		A0370	Release Private Device
T9900	Complete		If no pending actions in S1700 then go to step S9900
S9900	Complete	A0010	Set Message: Blow Command Complete
		A0020	Reset First Pass
		A0030	Self Release EM
END	Command Complete		True

Table 19: Blow Sequence

3.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value/Reference	
Operating Parameters					
OP001	Mode	Boolean	NA	False	
OP002	Bypass for TT1102Sx_C	Boolean	NA	False	
OP003	Bypass for TT1102Sx_E	Boolean	NA	False	
OP004	Bypass for TT1102Sx_F	Boolean	NA	False	
OP005	Push button F1002Sx	Boolean	NA	False	
OP006	ON Pressure Button	Boolean	NA	False	
OP007	Drain Water+Heat (RSP)	External Reference	NA	STAT_RE CIPE_AB/ TEMP_H2 O_AB_11	STAT2_RE CIPE_AB/T EMP_H2O _AB_11
OP008	Cake Discharge (RSP)	External Reference	NA	STAT_RE CIPE_AB/ TEMP_H2 O_AB_12	STAT2_RE CIPE_AB/ TEMP_H2 O_AB_12
OP009	Cooling 1 (RAMP)	External Reference	NA	STAT_RE CIPE_AB/ RAMP_AB _4	STAT2_RE CIPE_AB/ RAMP_AB _4

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Parameter	Description	Type	Units	Default Value/Reference	
OP010	Cooling 2 (RAMP)	External Reference	NA	STAT_RE CIPE_AB/ RAMP_AB _5	STAT2_RE CIPE_AB/ RAMP_AB _5
OP011	Cooling 3 (RAMP)	External Reference	NA	STAT_RE CIPE_AB/ RAMP_AB _6	STAT2_RE CIPE_AB/ RAMP_AB _6
OP012	Cooling 4 (RAMP)	External Reference	NA	STAT_RE CIPE_AB/ RAMP_AB _7	STAT2_RE CIPE_AB/ RAMP_AB _7
OP013	Cooling 5 (RAMP)	External Reference	NA	STAT_RE CIPE_AB/ RAMP_AB _8	STAT2_RE CIPE_AB/ RAMP_AB _8
OP014	Cooling 6 (RAMP)	External Reference	NA	STAT_RE CIPE_AB/ RAMP_AB _9	STAT2_RE CIPE_AB/ RAMP_AB _9
OP015	Cooling 7 + Safe Check (RAMP)	External Reference	NA	STAT_RE CIPE_AB/ RAMP_AB _10	STAT2_RE CIPE_AB/ RAMP_AB _10
OP016	Drain Water + Heat (RAMP)	External Reference	NA	STAT_RE CIPE_AB/ RAMP_AB _11	STAT2_RE CIPE_AB/ RAMP_AB _11
OP017	Cake Discharge (RAMP)	External Reference	NA	STAT_RE CIPE_AB/ RAMP_AB _12	STAT2_RE CIPE_AB/ RAMP_AB _12
Tuning Parameters					
TP001	Delay Timer	Floating Point	sec	5	
TP002	Cryst. Low Level SP	Floating Point	NA	0	
TP003	Actual Recipe Number	Floating Point	NA	0	
TP004	Hopper number	External Reference	NA	CF1EQ_1057CK/TP002 _VALUE	
TP005	Max Oil Temp to Discharge Step 10	External Reference	°C	STAT2_RECIPE_AB/MA X_OIL_TEMP	

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Parameter	Description	Type	Units	Default Value/Reference	
TP006	On DW Time	Floating Point	NA	1	
TP007	Stato Closed	Boolean	NA	False	
TP008	Stato last pack opened	Boolean	NA	False	
TP009	Force Open H Valve Time	Floating Point	sec	1	
TP010	H0 Open Time	Floating Point	sec	1	
TP011	H1 Open Time	Floating Point	sec	1	
TP012	H2 Open Time	Floating Point	sec	1	
TP013	Delay PF1050_02S	Floating Point	sec	1	
TP014	On FHW Time	Floating Point	sec	1	
TP015	Temp Regulation	External Reference	min	STAT_RE CIPE_AB/ STEP_TIM E_AB_1	STAT2_RE CIPE_AB/ STEP_TIM E_AB_1
TP016	Oil Filling	External Reference	min	STAT_RE CIPE_AB/ STEP_TIM E_AB_2	STAT2_RE CIPE_AB/ STEP_TIM E_AB_2
TP017	Homogenizing & Line Blow	External Reference	min	STAT_RE CIPE_AB/ STEP_TIM E_AB_3	STAT2_RE CIPE_AB/ STEP_TIM E_AB_3
TP018	Cooling 1	External Reference	min	STAT_RE CIPE_AB/ STEP_TIM E_AB_4	STAT2_RE CIPE_AB/ STEP_TIM E_AB_4
TP019	Cooling 2	External Reference	min	STAT_RE CIPE_AB/ STEP_TIM E_AB_5	STAT2_RE CIPE_AB/ STEP_TIM E_AB_5
TP020	Cooling 3	External Reference	min	STAT_RE CIPE_AB/ STEP_TIM E_AB_6	STAT2_RE CIPE_AB/ STEP_TIM E_AB_6

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Parameter	Description	Type	Units	Default Value/Reference	
TP021	Cooling 4	External Reference	min	STAT_RE CIPE_AB/ STEP_TIM E_AB_7	STAT2_RE CIPE_AB/ STEP_TIM E_AB_7
TP022	F1078D Water Balance Level	Floating Point		0	
TP023	RSP for PT-F02Sx-1	Floating Point		0.5	
TP024	Cooling 5	External Reference	min	STAT_RE CIPE_AB/ STEP_TIM E_AB_8	STAT2_RE CIPE_AB/ STEP_TIM E_AB_8
TP025	Cooling 6	External Reference	min	STAT_RE CIPE_AB/ STEP_TIM E_AB_9	STAT2_RE CIPE_AB/ STEP_TIM E_AB_9
TP026	Cooling 7 + Safe Check	External Reference	min	STAT_RE CIPE_AB/ STEP_TIM E_AB_10	STAT2_RE CIPE_AB/ STEP_TIM E_AB_10
TP027	Drain Water + Heat	External Reference	min	STAT_RE CIPE_AB/ STEP_TIM E_AB_11	STAT2_RE CIPE_AB/ STEP_TIM E_AB_11
TP028	Cake Discharge	External Reference	min	STAT_RE CIPE_AB/ STEP_TIM E_AB_12	STAT2_RE CIPE_AB/ STEP_TIM E_AB_12
TP029	Next Batch Number	Floating Point		2	
TP030	Statoliser Cycle Count	Floating Point		0	
TP031	Temp Regulation (RSP)	External Reference		STAT_RE CIPE_AB/ TEMP_H2 O_AB_1	STAT2_RE CIPE_AB/ TEMP_H2 O_AB_1
TP032	Oil Filling (RSP)	External Reference		STAT_RE CIPE_AB/ TEMP_H2 O_AB_2	STAT2_RE CIPE_AB/ TEMP_H2 O_AB_2
TP033	Homogenizing + Line Blow (RSP)	External Reference		STAT_RE CIPE_AB/ TEMP_H2 O_AB_3	STAT2_RE CIPE_AB/ TEMP_H2 O_AB_3



Parameter	Description	Type	Units	Default Value/Reference	
TP034	Cooling 1 (RSP)	External Reference		STAT_RE CIPE_AB/ TEMP_H2 O_AB_4	STAT2_RE CIPE_AB/ TEMP_H2 O_AB_4
TP035	Cooling 2 (RSP)	External Reference		STAT_RE CIPE_AB/ TEMP_H2 O_AB_5	STAT2_RE CIPE_AB/ TEMP_H2 O_AB_5
TP036	Cooling 3 (RSP)	External Reference		STAT_RE CIPE_AB/ TEMP_H2 O_AB_6	STAT2_RE CIPE_AB/ TEMP_H2 O_AB_6
TP037	Cooling 4 (RSP)	External Reference		STAT_RE CIPE_AB/ TEMP_H2 O_AB_7	STAT2_RE CIPE_AB/ TEMP_H2 O_AB_7
TP038	Cooling 5 (RSP)	External Reference		STAT_RE CIPE_AB/ TEMP_H2 O_AB_8	STAT2_RE CIPE_AB/ TEMP_H2 O_AB_8
TP039	Cooling 6 (RSP)	External Reference		STAT_RE CIPE_AB/ TEMP_H2 O_AB_9	STAT2_RE CIPE_AB/ TEMP_H2 O_AB_9
TP040	Cooling 7+Safe Check (RSP)	External Reference		STAT_RE CIPE_AB/ TEMP_H2 O_AB_10	STAT2_RE CIPE_AB/ TEMP_H2 O_AB_10

Table 20: Operating and Tuning Parameters

3.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value/Reference	
MY_TAG		CFEQ_1002S1	CFEQ_1002S2
OPR_CMD_MASK		3	3
EM_CRYSTAL	String	CF1EQ_1002A	
DY_REF_1	Dynamic Reference		
DY_REF_2	Dynamic Reference		

Parameter Name	Parameter Type	Value/Reference	
LT_478D	External Reference	CF1_LT1478D1/PV	CF1_LT1478D1/PV
LSH_1360V1	External Reference	CF1_LSH1360V1/DI1/DISC_ACT	CF1_LSH1360V1/DI1/DISC_ACT
F1002S	String	CF1EQ_1002S2	CF1EQ_1002S1
STEP_STATUS	String		
TT1102S_C	External Reference	CF1_TT1102S1C/PV	CF1_TT1102S2C/PV
TT1102S_E	External Reference	CF1_TT1102S1E/PV	CF1_TT1102S2E/PV
TT1102S_F	External Reference	CF1_TT1102S1F/PV	CF1_TT1102S2F/PV
HOLD	External Reference	CF1EQ_1002A/HOLD	
V1102S_BY	External Reference	CF_STX_BW_S1/D EV1/PV_D	CF_STX_BW_S2/DEV1/PV_D
BATCH_PER_STSTO	External Reference	CF1EQ_1057CK/TP003_VALUE	
TIME_SEC	Floating Point	0	
FAULT_SGNL	Boolean	FALSE	
CLS_POSITION	Boolean	FALSE	
STEP_TIME	Floating Point	0	
COMPLETED_TIME	Floating Point	0	
RESET_COMP_TM	Boolean	FALSE	

Table 21: Configurable Parameters

3.6

Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_CF_STATOLIZR1	CPKO Fractionation Area: Statolizer

Table 22: Named Set Details

Details of states for NS_CF_STATOLIZR1 named set are as follows,

Named Set Value	Named Set Name
00	Hold

Named Set Value	Named Set Name
01	Blow

Table 23: Name Set State Details

4 EQUIPMENT MODULE CLASS DESIGN – EM_CF_STX_BLW (STX BLOW)

4.1 Functional Description

EM_CF_STX_BLW is a class based equipment module that is used for Statolizer as a part of Discharge Line Blow. EM_CF_STX_BLW is derived from the CommandDriven equipment module class _EM_CMD16_130. Refer FDS for the functional design requirements including faceplate and detail display. All devices used in EM shall be acquired in Blow Command and shall be released on completion of the same. Hold Request shall be enabled for all acquired devices at the end of Hold command.

4.2 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name	Instance Name
EM	EM_CF_STX_BLW	CF_STX_BLW_S1	CF_STX_BLW_S2
DEV1	_V12_IP_130_SE	CF1_V1102SBY	CF1_V1102S2BY
DEV2	_V12_IP_130_SE	CF1_V1102SBA	CF1_V1102SBA
DEV3	_V12_IP_130_SE	CF1_V1102S1BA	CF1_V1102S2BA
DEV4	_CSD_130_SE	CF1_V1102S1	CF1_V1102S2
DEV5	_V12_IP_130_SE	CF1_V1102S1V2	CF1_V1102S2V2
DEV6	_V12_IP_130_SE	CF1_V1102ADO1	

Table 24: Instances

4.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

4.3.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Command Hold Start	A0010	Store Fail Message
		A0020	Reset Hold Request
		A0030	Hold Timers
		A0040	Save Previous Step Time
T1000	Actions Complete or Time Out		Go to step H1000
H1000		A0010	Closing DEV1

Step	Description	Action	Action Description
	Hold Command 1	A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing DEV6
T9900	Actions Complete or Time Out		Go to step H9900
H9900	Command Hold End	A0010	Completed
END	Command Complete		True

Table 25: Hold Sequence

4.3.2

Command 01 – Blow

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Blow Command
T0010	First Pass = False		If First Pass = False then go to step S0010
T0020A	First Pass = True		If First Pass = True then go to step S0020
S0010	Initializations	A0010	First Pass Logic
		A0020	First Pass = True
		A0030	Reset Actual Counts
		A0040	Reset Timers
		A0050	Set Task Pointer
T0020B	Acquire Devices		If no pending actions in S0010 then go to step S0020
S0020	Acquire Devices	A0010	Acquire Devices Message
		A0020	Acquire Private Device
		A0030	Acquire Private Device
		A0040	Acquire Private Device
		A0050	Acquire Shared Device
		A0060	Acquire Shared Device
		A0070	Acquire Shared Device
		A0080	Delay Timer
		A0100	Reset HOLD_REQ



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Step	Description	Action	Action Description
T0100	Standby		If Task Pointer = 1 then go to step S0100
T0200C	Line Blow ON		If Task Pointer = 2 then go to step S0200
T0300B	Line Blow OFF		If Task Pointer = 3 then go to step S0300
T0400C	Cross Blow ON		If Task Pointer = 4 then go to step S0400
T0500B	Cross Blow OFF		If Task Pointer = 5 then go to step S0500
T0600C	Feed Core Blow ON		If Task Pointer = 6 then go to step S0600
T0700B	Feed Core Blow OFF		If Task Pointer = 7 then go to step S0700
T0800C	Vent Core Blow ON		If Task Pointer = 8 then go to step S0800
T0900B	Vent Core Blow OFF		If Task Pointer = 9 then go to step S0900
T1000B	Blow Over		If Task Pointer = 10 then go to step S1000
T1100B	Aux Off		If Task Pointer = 11 then go to step S1100
S0100	Standby	A0010	Closing DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing DEV6
T0200A	Line Blow ON		If F1002A is in step <Blow Over> (F1002_BLW) then go to step S0200
S0200	Line Blow ON	A0010	Closing DEV1
		A0020	Enabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing DEV4
		A0050	Opening and Disabling Conditional Logic for DEV5
		A0060	Opening DEV6
		A0070	Set Blow ON time
		A0080	Start Blow ON Timer
		A0090	Blow ON time Message

Step	Description	Action	Action Description
T0300A	Line Blow OFF		If Blow On Time (TP002) Over then go to step S0300
S0300	Line Blow OFF	A0010	Closing DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing DEV4
		A0050	Opening and Disabling Conditional Logic for DEV5
		A0060	Opening DEV6
		A0070	Set Blow OFF time
		A0080	Start Blow OFF Timer
		A0090	Blow OFF time Message
		A0100	Increment Cross Blow Count count by 1
T0200B	Line Blow ON		If Blow OFF Time (TP003) Over & Actual Line Blow count (TP014) < Preset Line Blow Count (TP010) then go to step S0200
T0400A	Cross Blow ON		If Blow OFF Time (TP003) Over & Actual Line Blow count (TP014) >= Preset Line Blow Count (TP010) then go to step S0400
S0400	Cross Blow ON	A0010	Closing DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Enabling Conditional Logic for DEV3
		A0040	Closing DEV4
		A0050	Opening and Disabling Conditional Logic for DEV5
		A0060	Closing DEV6
		A0070	Set Blow ON time
		A0080	Start Blow ON Timer
		A0090	Blow ON time Message
T0500A	Cross Blow OFF		If Blow On Time (TP002) Over then go to step S0500
S0500	Cross Blow OFF	A0010	Closing DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing DEV4

Step	Description	Action	Action Description
		A0050	Opening and Disabling Conditional Logic for DEV5
		A0060	Closing DEV6
		A0070	Set Blow OFF time
		A0080	Start Blow OFF Timer
		A0090	Blow OFF time Message
		A0100	Increment Cross Blow Count count by 1
T0400B	Cross Blow ON		If Blow OFF Time (TP003) Over & Actual Line Blow count (TP015) < Preset Line Blow Count (TP011) then go to step S0400
T0600A	Feed Core Blow ON		If Blow OFF Time (TP003) Over & Actual Line Blow count (TP015) >= Preset Line Blow Count (TP011) then go to step S0600
S0600	Feed Core Blow ON	A0010	Closing DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Enabling Conditional Logic for DEV3
		A0040	Opening DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing DEV6
		A0070	Set Blow ON time
		A0080	Start Blow ON Timer
		A0090	Blow ON time Message
T0700A	Feed Core Blow OFF		If Blow On Time (TP002) Over then go to step S0700
S0700	Feed Core Blow OFF	A0010	Closing DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Opening DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing DEV6
		A0070	Set Blow OFF time
		A0080	Start Blow OFF Timer
		A0090	Blow OFF time Message
		A0100	Increment Actual feed core blow count by 1

Step	Description	Action	Action Description
T0600B	Feed Core Blow ON		If Blow OFF Time (TP003) Over & Actual Line Blow count (TP016) < Preset Line Blow Count (TP012) then go to step S0400
T0800A	Feed Core Blow ON		If Blow OFF Time (TP003) Over & Actual Line Blow count (TP016) >= Preset Line Blow Count (TP012) then go to step S0600
S0800	Vent Core Blow ON	A0010	Opening DEV1
		A0020	Enabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing DEV4
		A0050	Opening and Disabling Conditional Logic for DEV5
		A0060	Closing DEV6
		A0070	Set Blow ON time
		A0080	Start Blow ON Timer
		A0090	Blow ON time Message
T0900A	Vent Core Blow OFF		If Blow On Time (TP002) Over then go to step S0900
S0900	Vent Core Blow OFF	A0010	Opening DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing DEV4
		A0050	Opening and Disabling Conditional Logic for DEV5
		A0060	Closing DEV6
		A0070	Set Blow OFF time
		A0080	Start Blow OFF Timer
		A0090	Blow OFF time Message
		A0100	Increment Actual vent core blow count by 1
T0800B	Vent Core Blow ON		If Blow OFF Time (TP003) Over & Actual Line Blow count (TP017) < Preset Line Blow Count (TP013) then go to step S0400
T1000A	Blow Over		If Blow OFF Time (TP003) Over & Actual Line Blow count (TP017) >= Preset Line Blow Count (TP013) then go to step S0600

Step	Description	Action	Action Description
S1000	Blow Over	A0010	Closing DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing DEV4
		A0050	Opening and Disabling Conditional Logic for DEV5
		A0060	Closing DEV6
T1100A	Aux off		If "Stato Closed" (STATO_SIGNAL) signal received in step <Closing Statoliser> then go to step S1100
S1100	Aux off	A0010	Closing DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing DEV6
T1200	Release Devices		If no pending actions in S1100 then go to step S1200
S1200	Release Devices	A0010	Release Devices Message
		A0020	Release Private Device
		A0030	Release Private Device
		A0040	Release Private Device
		A0050	Release Shared Device
		A0060	Release Shared Device
		A0070	Release Private Device
T9900	Complete		Go to step S9900
S9900	Complete	A0010	Set Message: Blow Command Complete
END	Command Complete		True

Table 26: Blow Sequence

4.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Default Value	Units	Available on Detail Display
Operating Parameters				
NA				
Tuning Parameters				
TP001	Delay Timer	5	sec	Yes
TP002	Line Blow On	0	NA	Yes
TP003	Line Blow Off	0	sec	Yes
TP010	Line Blow Preset Count	20	sec	Yes
TP011	Cross Blow Preset Count	5	sec	Yes
TP012	Feed Core Blow Preset Count	5	sec	Yes
TP013	Vent Core Blow Preset Count	10	sec	Yes
TP014	Actual Line Blow Count	10	sec	Yes
TP015	Actual Cross Blow Count	120	sec	Yes
TP016	Actual Feed Core Blow Count	20	sec	Yes
TP017	Actual Vent Core Blow Count	5	NA	Yes

Table 27: Operating and Tuning Parameters

4.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value/Reference	
MY_TAG		CF_STX_BLOW_S1	CF_STX_BLOW_S2
OPR_CMD_MASK		3	3
F1002_BLOW	String	CF1EQ_1002_BLOW/COMMAND_00001/S0400/ACTIVE	
STEP_STATUS	String		
HOLD	External Reference	CF1EQ_1002A/HOLD	
STEP_TIME	Floating Point	0	
COMPLETED_TIME	Floating Point	0	
RESET_COMP_TM	Boolean	False	
STATO_SIGNAL	Boolean	False	

Table 28: Configurable Parameters

4.6 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_CF_STX_BLOW	CPKO Fractionation Area: Statolizer

Table 29: Named Set Details

Details of states for NS_CF_STX_BLOW named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Blow

Table 30: Name Set State Details

5 EQUIPMENT MODULE CLASS DESIGN – EM_CF_STATOLIZR2 (F1057CK)

5.1 Functional Description

EM_CF_STATOLIZR2 is a class based equipment module that is used for Statolizer as a part of Discharge Line Blow. EM_CF_STATOLIZR2 is derived from the CommandDriven equipment module class _EM_CMD16_130. Refer FDS for the functional design requirements including faceplate and detail display. All devices used in EM shall be acquired in Pumping Command and shall be released on completion of Pumping Command. Hold Request shall be enabled for all acquired devices at the end of Hold command.

5.2 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name
EM	EM_CF_STATOLIZR2	CFXEQ_1057CK
DEV1	_MVS11_IL_130_SO	CF1_PU1119S1
DEV2	_MRSS11_130_SE	CF1_PU1178TR
DEV3	_MVS11_IL_130_SO	CF1_PU1109S2
DEV4	_V12_IP_130_SE	CF1_V11TRTIW
DEV5	_V12_IP_130_SE	CF1_V11TRFW
DEV6	_V12_IP_130_SE	CF1_V11TRFIW
DEV7	_V12_IP_130_SE	CF1_V1116SAD1
DEV8	_V12_IP_130_SE	CF1_V1116ASRO
DEV9	_V12_IP_130_SE	CF1_V1116ASR
DEV10	_V12_IP_130_SE	CF1_FN1109S1
DEV11	_MVS11_IL_130_SO	CF1_FN1109S2

Table 31: Instances

5.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

5.3.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Start	A0010	Store Fail Message
		A0020	Set Hold Request
T1000	Actions Complete or Time Out & Hold=True		Go to step H1000
T2000	Actions Complete or Time Out & Hold=False		Go to step H2000
H1000	Hold Command 1	A0010	Stopping DEV1
		A0020	Stopping DEV2
		A0030	Stopping DEV3
T9900A	Actions Complete or Time Out		Go to step H9900
H2000	Hold Command 2	A0010	Stopping DEV1
		A0020	Stopping DEV2
		A0030	Stopping DEV3
		A0040	Closing DEV4
		A0050	Closing DEV5
		A0060	Closing DEV6
		A0070	Closing DEV7
		A0080	Closing DEV8
		A0090	Closing DEV9
		A0100	Stopping DEV10
		A0110	Stopping DEV11
T3000	Release Devices		Go to step H3000
H3000	Release Devices	A0010	Release Devices Message
		A0020	Release Private Devices



Step	Description	Action	Action Description
		A0030	Release Private Devices
		A0040	Release Private Devices
		A0050	Release Private Devices
		A0060	Release Private Devices
		A0070	Release Private Devices
		A0080	Release Private Devices
		A0090	Release Private Devices
		A0100	Release Private Devices
		A0110	Release Private Devices
		A0120	Release Private Devices
T9900B	Action Complete or Time Out		Go to step H9900
H9900	Command Hold End	A0010	Completed
End	Command Complete		True

Table 32: Hold Sequence

5.3.2 Command 01 – Pumping

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Pumping Command
T0010	First Pass = False		If First Pass = False then go to step S0010
T0020A	First Pass = True		If First Pass = True then go to step S0020
S0010	Initializations	A0010	First Pass Logic
		A0020	First Pass = True
T0020B	Acquire Devices		If no pending actions in S0010 then go to step S0020
S0020	Acquire Devices	A0010	Acquire Devices Message
		A0020	Acquire Private Devices
		A0030	Acquire Private Devices
		A0040	Acquire Private Devices
		A0050	Acquire Private Devices
		A0060	Acquire Private Devices

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Step	Description	Action	Action Description
		A0070	Acquire Private Devices
		A0080	Acquire Private Devices
		A0090	Acquire Private Devices
		A0100	Acquire Private Devices
		A0110	Acquire Private Devices
		A0120	Acquire Private Devices
T0100	Standby		If Task Pointer = 1 then go to step S0100
T0200B	Conveying & Pumping		If Task Pointer = 2 then go to step S0200
S0100	Standby	A0010	Stopping DEV1
		A0020	Starting DEV2
		A0030	Stopping DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Enabling Conditional Logic for DEV6
		A0070	Closing DEV7
		A0080	Opening DEV8
		A0090	Closing DEV9
		A0100	Stopping DEV10
		A0110	Stopping DEV11
T0200A	Conveying & Pumping		If F1016A1 (EM_FILTRATN_A1) or F1016A2 (EM_FILTRATN_A2)(selected filter in operation) is in step <Fill Ramp Up> or <Fill Constant Pressure> or <Fill Ramping> then go to step S0200
S0200	Conveying & Pumping	A0010	Set Message: Set F1057CK/S Hopper number and Set task Pointer
		A0020	Enabling Conditional Logic for DEV1 (Reverse)
		A0025	Enabling Conditional Logic for DEV1 (Forward)
		A0030	Starting DEV2
		A0040	Enabling Conditional Logic for DEV3 (Reverse)
		A0045	Enabling Conditional Logic for DEV3 (Forward)
		A0050	Enabling Conditional Logic for DEV4
		A0060	Enabling Conditional Logic for DEV5

Step	Description	Action	Action Description
		A0070	Enabling Conditional Logic for DEV6
		A0080	Opening DEV7
		A0090	Opening & Enabling Conditional Logic for DEV8
		A0100	Closing DEV9
		A0110	Enabling Conditional Logic for DEV10
		A0120	Enabling Conditional Logic for DEV11
T0300	Close Devices		If F1016A1 (EM_FILTRATN_A1) or F1016A2 (selected filter is in operation) is not in step <Fill Ramp Up> or <Fill Constant Pressure> or <Fill Ramping> then go to step S0300
S0300	Close Devices	A0010	Stopping DEV1
		A0020	Starting DEV2
		A0030	Stopping DEV3
		A0050	Enabling Conditional Logic for DEV4
		A0060	Enabling Conditional Logic for DEV5
		A0070	Enabling Conditional Logic for DEV6
		A0080	Closing DEV7
		A0090	Opening DEV8
		A0100	Closing DEV9
		A0110	Stopping DEV10
		A0120	Stopping DEV11
T0400	Realease Devices		If no pending actions in S0300 then go to step S0400
S0400	Release Devices	A0010	Release Devices Message
		A0020	Release Private Devices
		A0030	Release Private Devices
		A0040	Release Private Devices
		A0050	Release Private Devices
		A0060	Release Private Devices
		A0070	Release Private Devices
		A0080	Release Private Devices
		A0090	Release Private Devices
		A0100	Release Private Devices
		A0110	Release Private Devices
		A0120	Release Private Devices

Step	Description	Action	Action Description
T9900	Complete		If no pending actions in S0400 then go to step S9900
S9900	Complete	A0010	Set Message: Pumping Command Complete
		A0020	Set First Pass to False
END	Command Complete		True

Table 33: Pumping Sequence

5.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value/Reference
Operating Parameters				
OP001	Screw Mode	Named Set	NA	Ratio
OP002	Reverse Enable	Named Set	NA	No
Tuning Parameters				
TP001	Delay Timer	Floating Point	sec	5
TP002	Hopper Number	Floating Point	Lot	5
TP003	Batch per Stato Discharge	Floating Point	Lot	2
TP004	Hopper Temp.	Floating Point	°C	30
TP005	Screw Mode Ratio	Floating Point	%	30
TP006	Screw Mode Manual	Floating Point	%	20
TP007	Forced Reverse Timer	Floating Point	sec	5

Table 34: Operating and Tuning Parameters

5.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value/Reference
MY_TAG		CF1EQ_1057CK
OPR_CMD_MASK		3
TT_178TR1	External Reference	CF1_TT1178TR1/PV
DY_REF_1	Dynamic Reference	
DY_REF_2	Dynamic Reference	

Parameter Name	Parameter Type	Value/Reference
EM_FILTRATN_A1	String	CF1EQ_1016A1
EM_FILTRATN_A2	String	CF1EQ_1016A2
LSH_382B1	External Reference	CF1_LSH1382B1/DI1/DISC_ACT
LSL_109S1	External Reference	CF1_LSL1109S1/DI1/DISC_ACT
PT_116AS1	External Reference	CF1_PT1116AS1/AI1/HI_HI_ACT
HOLD	External Reference	CF1EQ_1002A/OP003_VALUE

Table 35: Configurable Parameters

5.6 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_CF_STATOLIZR2	CPKO Fractionation Area: Statolizer

Table 36: Named Set Details

Details of states for NS_CF_STATOLIZR2 named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Pumping

Table 37: Name Set State Details

6 EQUIPMENT MODULE CLASS DESIGN – EM_CF_FILTRATION (F1016A1/A2)

6.1 Functional Description

EM_CF_FILTRATION is a class based equipment module that is used for Filtration as a part of Discharge Line Blow. EM_CF_FILTRATION is derived from the CommandDriven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. All devices used in EM shall be acquired in Filter Command and shall be released on completion of Filter Command. Hold Request shall be enabled for all acquired devices at the end of Hold command.

6.2 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name	Instance Name
EM	EM_CF_FILTRATION	CFEQ_1016A1	CFEQ_1016A2
DEV1	_MVS11_IL_130_SO	CF1_PU1116AS	
DEV2	_CMT_130_SE	CF1_PT1216A11	CF1_PT1216A21
DEV3	_MVS11_IL_130_SO	CF1_PU1278161	CF1_PU1278162
DEV4	EM_CF_STATOLIZR2	CF1EQ_1057CK	
DEV5	_MRSS11_130_SE	CF1_PU125016	
DEV6	_V12_IP_130_SE	CF1_V1216A1FO	CF1_V1216A2FO
DEV7	_V12_IP_130_SE	CF1_V1216A101	CF1_V1216A2F1
DEV8	_V12_IP_130_SE	CF1_V1216A102	CF1_V1261A2F2
DEV9	_V12_IP_130_SE	CF1_V1216A1DO	CF1_V1216A2DO
DEV10	_V12_IP_130_SE	CF1_V1216A1FC	CF1_V1216A2FC
DEV11	_V12_IP_130_SE	CF1_V12161BA	CF1_V1216A1BA
DEV12	_V12_IP_130_SE	CF1_V1216ABC	CF1_V1216A2BC
DEV13	_V12_IP_130_SE	CF1_V1216A1BO	CF1_V1216A2BO
DEV14	_V12_IP_130_SE	CF1_V1216A1OL	CF1_V1216A2DL
DEV15	_V12_IP_130_SE	CF1_V1216A1DH	CF1_V1216A2DH
DEV16	_V12_IP_130_SE	CF1_V1216A1CL	CF1_V1216A2CL
DEV17	_V12_IP_130_SE	CF1_V1216A1CH	CF1_V1216A2CO
DEV18	_V12_IP_130_SE	CF1_V1216A1RC	CF1_V1216A2RC
DEV19	_V12_IP_130_SE	CF1_V1116SAD2	
DEV20	_CSD_130_SE	CF1_PT1216AS2	CF1_PT1216AS4
DEV21	_V12_IP_130_SE	CF1_V12161R1	CF1_V12162R1
DEV22	_O_RAMP_130_SE	CF1_PT1216AS2_RM	CF1_PT1216AS4_RM
DEV23	_O_RAMP_130	Ignored	

Table 38: Instances

6.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

6.3.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Command Hold Start	A0010	Store Fail Message
		A0020	Set Hold Request
T1000	Actions Complete or Time Out		Go to step H1000
H1000	Hold Command 1	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Set DEV2 Mode to Remote Out
		A0030	Set DEV2 output to Closed
		A0040	Stopping and Disabling Conditional Logic for DEV3
		A0050	Stopping DEV5
		A0060	Closing DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Closing and Disabling Conditional Logic for DEV14
		A0150	Opening DEV15
		A0160	Closing DEV16
		A0170	Closing DEV17
		A0180	Closing DEV18

Step	Description	Action	Action Description
		A0190	Closing DEV19
		A0200	Set DEV20 Mode to Remote Out
		A0210	Set DEV20 output to Closed
		A0220	Opening DEV21
T9900	Actions Complete or Time Out		Go to step H9900
H9900	Command Hold End	A0010	Completed
END	Command Complete		True

Table 39: Hold Sequence

6.3.2 Command 01 – Filter

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Filter Command
T0010	First Pass = False		If First Pass = False then go to step S0010
T0020A	First Pass = True		If First Pass = True then go to step S0020
S0010	Initializations	A0010	First Pass Logic
		A0020	First Pass = True
		A0030	Re-setting Wash Demand and Timers
T0020B	Acquire Devices		If no pending actions in S0010 then go to step S0020
S0020	Acquire Devices	A0010	Acquire Devices Message
		A0020	Acquire Shared Devices
		A0030	Acquire Shared Devices
		A0040	Acquire Private Devices
		A0050	Acquire Shared Devices
		A0060	Acquire Shared Devices
		A0070	Acquire Private Devices
		A0080	Acquire Private Devices
		A0090	Acquire Private Devices
		A0100	Acquire Private Devices
		A0110	Acquire Private Devices

Step	Description	Action	Action Description
		A0120	Acquire Private Devices
		A0130	Acquire Private Devices
		A0140	Acquire Private Devices
		A0150	Acquire Private Devices
		A0160	Acquire Private Devices
		A0170	Acquire Private Devices
		A0180	Acquire Private Devices
		A0190	Acquire Private Devices
		A0200	Acquire Shared Devices
		A0210	Acquire Private Devices
		A0220	Acquire Private Devices
		A0230	Acquire Private Devices
		A0240	Reset Hold Request
T0100	Standby		If no pending actions in S0020 then go to step S0100
S0100	Standby	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Set DEV2 Mode to Remote Out
		A0030	Set DEV2 output to Closed
		A0040	Stopping and Disabling Conditional Logic for DEV3
		A0050	Stopping DEV5
		A0060	Closing DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Closing and Disabling Conditional Logic for DEV14

Step	Description	Action	Action Description
		A0150	Opening DEV15
		A0160	Closing DEV16
		A0170	Closing DEV17
		A0180	Closing DEV18
		A0190	Closing DEV19
		A0200	Set DEV20 Mode to Remote Out
		A0210	Set DEV20 output to Closed
		A0220	Opening DEV21
		A0230	Re-setting Wash Demand and Timers
		A0240	Checking Conditions
T0200	Filter On Pressure		If no pending actions in S0100 then go to step S0200 after 5 sec delay
S0200	Filter On Pressure	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Set DEV2 Mode to Remote Out
		A0030	Set DEV2 output to Closed
		A0040	Stopping and Disabling Conditional Logic for DEV3
		A0050	Stopping DEV5
		A0060	Closing DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Closing and Disabling Conditional Logic for DEV14
		A0150	Opening DEV15
		A0160	Closing DEV16
		A0170	Closing DEV17

Step	Description	Action	Action Description
		A0180	Closing DEV18
		A0190	Closing DEV19
		A0200	Set DEV20 Mode to Remote Out
		A0210	Set DEV20 output to Closed
		A0220	Opening DEV21
		A0230	Set Message: Starting Filter On Pressure Process
		A0240	Delay 10 sec
T0300	Fill Ramp Up		If F1016Ax is in AUTO mode (OP006) and READY_FOR_FILTR signal from F1016Ax is received and not Wash Demand (OP001) ON and not Cool Demand (OP004) ON and not FILTER_FAULT and F1057CK/S Hopper number >=1 Lot and not another filter Feed Line Busy then go to step S0300
T0400	Wash Filling		If other F1016A is not in step Filter Washing or Cooling or Core Wash and F1016Ax is in AUTO mode (OP006) and READY_FOR_FILTR signal from F1016Ax is received and Wash Demand (OP001) ON and TT_382C1 <= Max Washing Temp (TP006) and TT_321C1 <= Max Washing Temp (TP006) and TT_321C1 >= Min Washing Temp (TP005) and not FILTER_FAULT and not Cool Demand (OP004) ON and F1082C is in step Prewash Recirculation then go to step S0400
T1600B	Cool Filling		If other F1016A is not in step Filter Washing or Cooling or Core Wash and F1016Ax is in AUTO mode (OP006) and READY_FOR_FILTR signal from F1016Ax is received and Cool Demand (OP004) ON and not FILTER_FAULT and not Wash Demand (OP001) ON and Cooling Type <Water & Oil> or <Water> selected then go to step S1600
T2500B	Open Bombdoor		If F1016Ax is in AUTO mode (OP006) and READY_FOR_FILTR signal from F1016Ax is received and not FILTER_FAULT and Pb Open Filter (OP008) is ON then go to step S2500
S0300	Fill Ramp Up	A0010	Enabling Conditional Logic for DEV1

Step	Description	Action	Action Description
		A0020	Set DEV2 Mode to Remote Cascade
		A0030	Set DEV2 RSP to Fill Ramp Up Pressure
		A0040	Stopping and Disabling Conditional Logic for DEV3
		A0050	Stopping DEV5
		A0060	Opening DEV6
		A0070	Opening DEV7
		A0080	Opening DEV8
		A0090	Closing DEV9
		A0100	Closing and Disabling Conditional Logic for DEV10
		A0110	Closing and Disabling Conditional Logic for DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Enabling Conditional Logic for DEV14
		A0150	Enabling Conditional Logic for DEV15
		A0160	Closing DEV16
		A0170	Closing DEV17
		A0180	Closing DEV18
		A0190	Closing DEV19
		A0200	Set DEV20 Mode to Remote Out
		A0210	Set DEV20 output to Closed
		A0220	Opening DEV21
		A0230	Set Message: Starting Fill Ramp Up Process
		A0240	Command1 active for DEV4
T0500	Fill Constant Pressure		If F1016Ax is in AUTO mode (OP006) and READY_FOR_FILTR signal from F1016Ax is received and not FILTER_FAULT and PT_216AX1 >= Fill Ramp Up Pressure (TP017) or PT_216AX1 >= Fill Ramping End Pressure (TP019) or Pb <Go Squeeze> (OP007) selected then go to step S0500
S0400	Wash Filling	A0010	Stopping DEV1
		A0020	Set DEV2 Mode to Remote Out

Step	Description	Action	Action Description
		A0030	Set DEV2 output to Closed
		A0040	Stopping DEV3
		A0070	Stopping DEV5
		A0080	Closing DEV6
		A0090	Opening DEV7
		A0100	Opening DEV8
		A0110	Closing DEV9
		A0120	Opening DEV10
		A0130	Closing DEV11
		A0140	Closing DEV12
		A0150	Closing DEV13
		A0160	Closing DEV14
		A0170	Closing DEV15
		A0180	Closing DEV16
		A0190	Opening DEV17
		A0200	Closing DEV18
		A0210	Closing DEV19
		A0220	Set DEV20 Mode to Remote Out
		A0230	Set DEV20 output to Closed
		A0240	Opening DEV21
		A0250	Set Message: Starting Wash Filling Process
		A0260	Checking Conditions
		A0270	Set Wash Filling Time
		A0280	Start Wash Filling Time
		A0290	Wash Filling Time Message
T0600	Wash Recirculate		If F1016Ax is in AUTO mode (OP006) and READY_FOR_FILTER signal from F1016Ax is received and not FILTER_FAULT and Wash Filling time (TP002) is over and F1082C is in step <PF1082C to Filter> then go to step S0600
S0500	Fill Constant Pressure	A0010	Enabling Conditional Logic for DEV1
		A0020	Set DEV2 Mode to Remote Cascade
		A0030	Set DEV2 RSP to Fill Ramp Up Pressure
		A0040	Stopping and Disabling Conditional Logic for DEV3
		A0070	Stopping DEV5

Step	Description	Action	Action Description
		A0080	Opening DEV6
		A0090	Opening and Disabling Conditional Logic for DEV7
		A0100	Opening and Disabling Conditional Logic for DEV8
		A0110	Closing DEV9
		A0120	Closing and Disabling Conditional Logic for DEV10
		A0130	Closing and Disabling Conditional Logic for DEV11
		A0140	Closing and Disabling Conditional Logic for DEV12
		A0150	Closing and Disabling Conditional Logic for DEV13
		A0160	Enabling Conditional Logic for DEV14
		A0170	Enabling Conditional Logic for DEV15
		A0180	Closing DEV16
		A0190	Closing DEV17
		A0200	Closing DEV18
		A0210	Closing DEV19
		A0220	Set DEV20 Mode to Remote Out
		A0230	Set DEV20 output to Closed
		A0240	Opening DEV21
		A0250	Set Message: Starting Fill Constant Process Process
		A0270	Set Fill Constant Pressure Time
		A0280	Start Fill Constant Pressure Time
		A0290	Wash Recirculate Time Message
T0300B	Fill Ramp Up		If F1016Ax is in MAN mode (OP006) and READY_FOR_FILTR signal from F1016Ax is not received and FILTER_FAULT then go to step S0300
T0700A	Fill Ramping		If F1016Ax is in AUTO mode (OP006) and READY_FOR_FILTR signal from F1016Ax is received and not FILTER_FAULT and (PT_216AX1 >= Fill Ramp Up Pressure (TP017) and Fill Constant Pressure Timer elapsed) or (PT_216AX1 >= Fill Ramping End Pressure (TP019) or Pb <Go Squeeze> (OP007) selected) then go to step S0700

Step	Description	Action	Action Description
S0600	Wash Recirculate	A0010	Stopping DEV1
		A0020	Set DEV2 Mode to Remote Out
		A0030	Set DEV2 output to Closed
		A0040	Stopping DEV3
		A0070	Stopping DEV5
		A0080	Closing DEV6
		A0090	Opening DEV7
		A0100	Opening DEV8
		A0110	Closing DEV9
		A0120	Opening DEV10
		A0130	Closing DEV11
		A0140	Closing DEV12
		A0150	Closing DEV13
		A0160	Closing DEV14
		A0170	Closing DEV15
		A0180	Opening DEV16
		A0190	Opening DEV17
		A0200	Closing DEV18
		A0210	Closing DEV19
		A0220	Set DEV20 Mode to Remote Out
		A0230	Set DEV20 output to Closed
		A0240	Opening DEV21
		A0250	Set Message: Starting Wash Recirculate Process
		A0260	Checking Conditions
		A0270	Closing DEV17
		A0280	Set Wash Recirculate Time
		A0290	Start Wash Recirculate Time
		A0300	Wash Recirculate Time Message
T0800	Wash Drain		If F1016Ax is in AUTO mode (OP006) and READY_FOR_FILTR signal from F1016Ax is received and not FILTER_FAULT and Wash Recirculate time over then go to step S0800
S0700	Fill Ramping	A0010	Enabling Conditional Logic for DEV1
		A0020	Set DEV2 Mode to Remote Cascade

Step	Description	Action	Action Description
		A0030	Set DEV2 RSP to Fill Ramp Up Pressure
		A0040	Stopping and Disabling Conditional Logic for DEV3
		A0070	Stopping DEV5
		A0080	Opening DEV6
		A0090	Opening and Disabling Conditional Logic for DEV7
		A0100	Opening and Disabling Conditional Logic for DEV8
		A0110	Closing DEV9
		A0120	Closing and Disabling Conditional Logic for DEV10
		A0130	Closing and Disabling Conditional Logic for DEV11
		A0140	Closing and Disabling Conditional Logic for DEV12
		A0150	Closing and Disabling Conditional Logic for DEV13
		A0160	Enabling Conditional Logic for DEV14
		A0170	Enabling Conditional Logic for DEV15
		A0180	Closing DEV16
		A0190	Closing DEV17
		A0200	Closing DEV18
		A0210	Closing DEV19
		A0220	Set DEV20 Mode to Remote Out
		A0230	Set DEV20 output to Closed
		A0240	Opening DEV21
		A0250	Set Message: Starting Fill Ramping Process
T0300C	Fill Ramp Up		If F1016Ax is in MAN mode (OP006) and READY_FOR_FILTR signal from F1016Ax is not received and FILTER_FAULT then go to step S0300
T0900A	Squeeze Ramp 1		If F1016Ax is in AUTO mode (OP006) and READY_FOR_FILTR signal from F1016Ax is received and not FILTER_FAULT and (PT_216AX1 >= Fill Ramping End Pressure (TP019) with 5 secs delay or Pb <Go Squeeze> (OP007) selected) then go to step S0900 and S3200
S0800	Wash Drain	A0010	Stopping DEV1

Step	Description	Action	Action Description
		A0020	Set DEV2 Mode to Remote Out
		A0030	Set DEV2 output to Closed
		A0040	Stopping DEV3
		A0070	Stopping DEV5
		A0080	Closing DEV6
		A0090	Opening DEV7
		A0100	Opening DEV8
		A0110	Closing DEV9
		A0120	Opening DEV10
		A0130	Closing DEV11
		A0140	Closing DEV12
		A0150	Closing DEV13
		A0160	Closing DEV14
		A0170	Closing DEV15
		A0180	Opening DEV16
		A0190	Opening DEV17
		A0200	Closing DEV18
		A0210	Closing DEV19
		A0220	Set DEV20 Mode to Remote Out
		A0230	Set DEV20 output to Closed
		A0240	Opening DEV21
		A0250	Set Message: Starting Wash Drain Process
		A0260	Set Wash Drain Time
		A0270	Start Wash Drain Time
		A0280	Wash Drain Time Message
T1000	Wash Olein Blow		If F1016Ax is in AUTO mode (OP006) and READY_FOR_FILTR signal from F1016Ax is received and not FILTER_FAULT and Wash Drain time over then go to step S1000
S0900	Squeeze Ramp 1	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Set DEV2 Mode to Remote Out
		A0030	Set DEV2 output to Closed
		A0040	Enabling Conditional Logic for DEV3
		A0050	Setpoint for DEV3
		A0060	Stopping DEV5

Step	Description	Action	Action Description
		A0070	Closing DEV6
		A0080	Closing and Disabling Conditional Logic for DEV7
		A0090	Closing and Disabling Conditional Logic for DEV8
		A0100	Closing DEV9
		A0110	Closing and Disabling Conditional Logic for DEV10
		A0120	Closing and Disabling Conditional Logic for DEV11
		A0130	Closing and Disabling Conditional Logic for DEV12
		A0140	Closing and Disabling Conditional Logic for DEV13
		A0150	Enabling Conditional Logic for DEV14
		A0160	Enabling Conditional Logic for DEV15
		A0170	Closing DEV16
		A0180	Closing DEV17
		A0190	Closing DEV18
		A0200	Closing DEV19
		A0210	Set DEV20 Mode to Remote Out
		A0220	Set DEV20 output to Full Open
		A0230	Waiting for 120 sec
		A0240	Set DEV20 Mode to Remote Cascade
		A0250	Set DEV20 setpoint to Squeeze Ramp 1 Rate
		A0260	Closing DEV21
		A0270	Set Message: Starting Squeeze Ramp 1 Process
T1100	Squeeze Ramp 2		If F1016Ax is in AUTO mode (OP006) and READY_FOR_FILTR signal from F1016Ax is received and not FILTER_FAULT and DEV20 PV >= Squeeze Ramp 1 end pressure (TP028) and DEV20 RSP >= Squeeze Ramp 1 end pressure (TP028) then go to step S1100
S1000	Wash Olein Blow	A0010	Stopping DEV1
		A0020	Set DEV2 Mode to Remote Out
		A0030	Set DEV2 output to Closed

Step	Description	Action	Action Description
		A0040	Stopping DEV3
		A0050	Stopping DEV5
		A0060	Closing DEV6
		A0070	Closing DEV7
		A0080	Closing DEV8
		A0090	Closing DEV9
		A0100	Closing DEV10
		A0110	Closing DEV11
		A0120	Closing DEV12
		A0130	Enabling Conditional Logic for DEV13
		A0140	Closing DEV14
		A0150	Closing DEV15
		A0160	Closing DEV18
		A0170	Closing DEV19
		A0180	Set DEV20 Mode to Remote Out
		A0190	Set DEV20 output to Closed
		A0200	Opening DEV21
		A0210	Set Message: Starting Wash Olein Blow Process
		A0220	Closing DEV16
		A0230	Opening DEV17
		A0240	Closing DEV17
		A0250	Opening DEV16
		A0260	Setting Pulse ON/OFF timer for DEV13
		A0270	Pausing Wash Timer
T1200	Wash Core Blow		If F1016Ax is in AUTO mode (OP006) and READY_FOR_FILTER signal from F1016Ax is received and not FILTER_FAULT and Wash Olein Blow time over then go to step S1200
S1100	Squeeze Ramp 2	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Set DEV2 Mode to Remote Out
		A0030	Set DEV2 output to Closed
		A0040	Enabling Conditional Logic for DEV3
		A0050	Setpoint for DEV3
		A0060	Stopping DEV5

Step	Description	Action	Action Description
		A0070	Closing DEV6
		A0080	Closing and Disabling Conditional Logic for DEV7
		A0090	Closing and Disabling Conditional Logic for DEV8
		A0100	Closing DEV9
		A0110	Closing DEV10
		A0120	Closing and Disabling Conditional Logic for DEV11
		A0130	Closing DEV12
		A0140	Closing and Disabling Conditional Logic for DEV13
		A0150	Enabling Conditional Logic for DEV14
		A0160	Enabling Conditional Logic for DEV15
		A0170	Closing DEV16
		A0180	Closing DEV17
		A0190	Closing DEV18
		A0200	Closing DEV19
		A0210	Set DEV20 Mode to Remote Cascade
		A0220	Set DEV20 setpoint to Squeeze Ramp 2 Rate
		A0230	Closing DEV21
		A0240	Set Message: Starting Squeeze Ramp 2 Process
T1300A	Squeeze Ramp 3		If F1016Ax is in AUTO mode (OP006) and READY_FOR_FILTR signal from F1016Ax is received and not FILTER_FAULT and DEV20 PV >= Squeeze Ramp 2 end pressure (TP029) and DEV20 RSP >= Squeeze Ramp 2 end pressure (TP029) then go to step S1300
S1200	Wash Core Blow	A0010	Stopping DEV1
		A0020	Set DEV2 Mode to Remote Out
		A0030	Set DEV2 output to Closed
		A0040	Stopping DEV3
		A0050	Stopping DEV5
		A0060	Closing DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Enabling Conditional Logic for DEV8

Step	Description	Action	Action Description
		A0090	Closing DEV9
		A0100	Opening DEV10
		A0110	Closing DEV11
		A0120	Enabling Conditional Logic for DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Closing DEV14
		A0150	Closing DEV15
		A0160	Closing DEV16
		A0170	Closing DEV17
		A0180	Closing DEV18
		A0190	Closing DEV19
		A0200	Set DEV20 Mode to Remote Out
		A0210	Set DEV20 output to Closed
		A0220	Opening DEV21
		A0230	Setting Pulse ON/OFF for DEV12
		A0240	Set Message: Starting Wash Olein Blow Process
		A0250	Checking Conditions
		A0260	Pausing Core Timer
T1400	Wash Done		If F1016Ax is in AUTO mode (OP006) and READY_FOR_FILTR signal from F1016Ax is received and not FILTER_FAULT and Wash Core Blow time over then go to step S1400
S1300	Squeeze Ramp 3	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Set DEV2 Mode to Remote Out
		A0030	Set DEV2 output to Closed
		A0040	Enabling Conditional Logic for DEV3
		A0050	Setpoint for DEV3
		A0060	Stopping DEV5
		A0070	Closing DEV6
		A0080	Closing DEV7
		A0090	Closing DEV8
		A0100	Closing DEV9
		A0110	Closing DEV10
		A0120	Closing DEV11

Step	Description	Action	Action Description
		A0130	Closing DEV12
		A0140	Closing DEV13
		A0150	Enabling Conditional Logic for DEV14
		A0160	Enabling Conditional Logic for DEV15
		A0170	Closing DEV16
		A0180	Closing DEV17
		A0190	Closing DEV18
		A0200	Closing DEV19
		A0210	Set DEV20 Mode to Remote Cascade
		A0220	Set DEV20 setpoint to Squeeze Ramp 3 Rate
		A0230	Closing DEV21
		A0240	Set Message: Starting Squeeze Ramp 3 Process
T1500A	Squeeze Hold 3		If F1016Ax is in AUTO mode (OP006) and READY_FOR_FILTR signal from F1016Ax is received and not FILTER_FAULT and DEV20 PV >= Squeeze Ramp 3 end pressure (TP030) and DEV20 RSP >= Squeeze Ramp 3 end pressure (TP030) then go to step S1500
S1400	Wash Done	A0010	Stopping DEV1
		A0020	Set DEV2 Mode to Remote Out
		A0030	Set DEV2 output to Closed
		A0040	Stopping DEV3
		A0050	Stopping DEV5
		A0060	Closing DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Closing DEV10
		A0110	Closing DEV11
		A0120	Closing and Disabling Conditional Logic for DEV12
		A0130	Closing DEV13
		A0140	Closing DEV14
		A0150	Opening DEV15

Step	Description	Action	Action Description
		A0160	Closing DEV16
		A0170	Closing DEV17
		A0180	Closing DEV18
		A0190	Closing DEV19
		A0200	Set DEV20 Mode to Remote Out
		A0210	Set DEV20 output to Closed
		A0220	Opening DEV21
		A0230	Checking Conditions
T1600A	Cool Filling		If F1016Ax is in AUTO mode (OP006) and READY_FOR_FILTR signal from F1016Ax is received and not FILTER_FAULT and 5 secs delay over and Pb <Wash and Cool> (OP002) is ON and Cooling Type <Water & Oil> or <Water> (OP005) selected then go to step S1600
T5000	Aux Off		If F1016Ax is in AUTO mode (OP006) and READY_FOR_FILTR signal from F1016Ax is received and not FILTER_FAULT and 5 secs delay over and Pb <Wash and Cool> (OP002) is OFF then go to step S5000
S1500	Squeeze Hold 3	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Set DEV2 Mode to Remote Out
		A0030	Set DEV2 output to Closed
		A0040	Enabling Conditional Logic for DEV3
		A0050	Setpoint for DEV3
		A0060	Stopping DEV5
		A0070	Closing DEV6
		A0080	Closing DEV7
		A0090	Closing DEV8
		A0100	Closing DEV9
		A0110	Closing DEV10
		A0120	Closing DEV11
		A0130	Closing DEV12
		A0140	Closing DEV13
		A0150	Enabling Conditional Logic for DEV14
		A0160	Enabling Conditional Logic for DEV15
		A0170	Closing DEV16
		A0180	Closing DEV17

Step	Description	Action	Action Description
		A0190	Closing DEV18
		A0200	Closing DEV19
		A0210	Set DEV20 Mode to Remote Cascade
		A0220	Set DEV20 setpoint to Squeeze Hold 3 Rate
		A0230	Closing DEV21
		A0240	Set Message: Starting Squeeze Hold 3 Process
		A0250	Set Squeeze Hold 3 Time
		A0260	Start Squeeze Hold 3 Time
		A0270	Squeeze Hold 3 Time Message
T1300B	Squeeze Ramp 3		If F1016Ax is in MAN mode (OP006) and READY_FOR_FILTR signal from F1016Ax is not received and FILTER_FAULT then go to step S1300
T1700A	Down Ramp		If F1016Ax is in AUTO mode (OP006) and READY_FOR_FILTR signal from F1016Ax is received and not FILTER_FAULT and Squeeze Hold 3 time over then go to step S1700
S1600	Cool Filling	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Set DEV2 Mode to Remote Out
		A0030	Set DEV2 output to Closed
		A0040	Enabling Conditional Logic for DEV3
		A0050	Stopping DEV5
		A0060	Closing DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Enabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Enabling Conditional Logic for DEV10
		A0110	Closing DEV11
		A0120	Closing and Disabling Conditional Logic for DEV13
		A0130	Closing DEV13
		A0140	Closing DEV14
		A0150	Opening DEV15
		A0160	Closing DEV16
		A0170	Closing DEV17

Step	Description	Action	Action Description
		A0180	Closing DEV18
		A0190	Closing DEV19
		A0200	Enabling Conditional Logic for DEV20
		A0210	Closing DEV21
		A0220	Set Message: Starting Cool Filling Process
		A0230	Checking Conditions
		A0240	Set Cool Filling Time
		A0250	Start Cool Filling Time
		A0260	Cool Filling Time Message
T1800	Cool Recirculate		If F1016Ax is in AUTO mode (OP006) and READY_FOR_FILTR signal from F1016Ax is received and not FILTER_FAULT and [Pb <Water> (OP005) selected or (F1082C is in step <PF1082C to Filter> and Cool Filling time (TP012) over)] then go to step S1800
S1700	Down Ramp	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Set DEV2 Mode to Remote Out
		A0030	Set DEV2 output to Closed
		A0040	Enabling Conditional Logic for DEV3
		A0050	Setpoint for DEV3
		A0060	Stopping DEV5
		A0070	Closing DEV6
		A0080	Closing DEV7
		A0090	Closing DEV8
		A0100	Closing DEV9
		A0110	Closing DEV10
		A0120	Closing DEV11
		A0130	Closing DEV12
		A0140	Closing DEV13
		A0150	Enabling Conditional Logic for DEV14
		A0160	Enabling Conditional Logic for DEV15
		A0170	Closing DEV16
		A0180	Closing DEV17
		A0190	Closing DEV18
		A0200	Closing DEV19
		A0210	Set DEV20 Mode to Remote Cascade

Step	Description	Action	Action Description
		A0230	Set DEV20 setpoint to Down Ramp Rate
		A0240	Closing DEV21
T1900	Depressurize Membrane		If F1016Ax is in AUTO mode (OP006) and READY_FOR_FILTR signal from F1016Ax is received and not FILTER_FAULT and MISC Blow Over and PT_216AS2 <= Down Ramp End Pressure (TP033) then go to step S1900
S1800	Cool Recirculate	A0010	Stopping DEV1
		A0020	Set DEV2 Mode to Remote Out
		A0030	Set DEV2 output to Closed
		A0040	Enabling Conditional Logic for DEV3
		A0050	Stopping DEV5
		A0060	Closing DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Enabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Enabling Conditional Logic for DEV10
		A0110	Closing DEV11
		A0120	Closing and Disabling Conditional Logic for DEV13
		A0130	Closing DEV13
		A0140	Opening DEV14 If Water & Oil Selected
		A0150	Opening DEV15
		A0160	Closing DEV16
		A0170	Closing DEV17
		A0180	Closing DEV18
		A0190	Closing DEV19
		A0200	Enabling Conditional Logic for DEV20
		A0210	Closing DEV21
		A0220	Set Cool Recirculate Time
		A0230	Start Cool Recirculate Time
		A0240	Cool Recirculate Time Message
		A0250	Set Message: Starting Cool Recirculate Process
		A0260	Checking Conditions

Step	Description	Action	Action Description
T2000	Cool Drain		If F1016Ax is in AUTO mode (OP006) and READY_FOR_FILTR signal from F1016Ax is received and not FILTER_FAULT and Cool Recirculate time over then go to step S2000
S1900	Depressurize Membrane	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Set DEV2 Mode to Remote Out
		A0030	Set DEV2 output to Closed
		A0040	Stopping and Disabling Conditional Logic for DEV3
		A0050	Setpoint for DEV3
		A0060	Stopping DEV5
		A0070	Closing DEV6
		A0080	Closing DEV7
		A0090	Closing DEV8
		A0100	Closing DEV9
		A0110	Closing DEV10
		A0120	Closing DEV11
		A0130	Closing DEV12
		A0140	Closing DEV13
		A0150	Opening and Disabling Conditional Logic for DEV14
		A0160	Opening and Disabling Conditional Logic for DEV15
		A0170	Closing DEV16
		A0180	Closing DEV17
		A0190	Closing DEV18
		A0200	Closing DEV19
		A0210	Set Message: Stopping Down Ramp Process
		A0220	Set DEV20 Mode to Remote Out
		A0230	Set DEV20 output to Open
		A0240	Opening DEV21
		A0250	Disabling Action Block Logic
		A0260	Set Message: Depressurizing Membrane

Step	Description	Action	Action Description
T2100	Olein Blow 2 + LMCB		If F1016Ax is in AUTO mode (OP006) and READY_FOR_FILTR signal from F1016Ax is received and not FILTER_FAULT and PT_216AS1 <= PRESET_VALUE3 and PT_216AS2 <= PRESET_VALUE3 then go to step S2100
S2000	Cool Drain	A0010	Stopping DEV1
		A0020	Set DEV2 Mode to Remote Out
		A0030	Set DEV2 output to Closed
		A0040	Stopping and Disabling Conditional Logic for DEV3
		A0050	Stopping DEV5
		A0060	Closing DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Closing DEV10
		A0110	Closing DEV11
		A0120	Closing and Disabling Conditional Logic for DEV13
		A0130	Closing DEV13
		A0140	Opening DEV14 If Water & Oil Selected
		A0150	Opening DEV15
		A0160	Closing DEV16
		A0170	Closing DEV17
		A0180	Closing DEV18
		A0190	Closing DEV19
		A0200	Set DEV20 Mode to Remote Out
		A0210	Set DEV20 output to Closed
		A0220	Opening DEV21
		A0230	Set Cool Drain Time
		A0240	Start Cool Drain Time
		A0250	Cool Drain Time Message
		A0260	Set Message: Starting Cool Drain Process

Step	Description	Action	Action Description
T2200	Cool Olein Blow		If F1016Ax is in AUTO mode (OP006) and READY_FOR_FILTR signal from F1016Ax is received and not FILTER_FAULT and Cool Drain time (TP014) over then go to step S2200
S2100	Olein Blow 2 + LMCB	A0010	Set Message: Starting Olein Blow 2 + LMCB Process
		A0020	Opening DEV15
		A0030	Closing DEV15
		A0040	Opening DEV14
		A0050	Opening DEV7
		A0060	Opening DEV8 & Closing DEV7
		A0070	Opening DEV9
		A0080	Set DEV13 Mode to Cascade
		A0090	Setting TP01 for DEV13
		A0100	Set DEV12 Mode to Cascade
		A0110	Setting TP01 for DEV12
		A0120	Pausing Olien Timer
		A0130	Stopping and Disabling Conditional Logc for DEV1
		A0140	Set DEV2 Mode to Remote Out
		A0150	Set DEV2 output to Closed
		A0160	Stopping and Disabling Conditional Logc for DEV3
		A0170	Setpoint for DEV3
		A0180	Stopping DEV5
		A0190	Closing DEV6
		A0200	Closing DEV10
		A0210	Closing DEV11
		A0220	Closing DEV16
		A0230	Closing DEV17
		A0240	Closing DEV18
		A0250	Closing DEV19
		A0260	Set Message: Stopping Down Ramp Process
		A0270	Set DEV20 Mode to Remote Out
		A0280	Set DEV20 output to Open
		A0290	Opening DEV21

Step	Description	Action	Action Description
		A0300	Opening DEV8 & Closing DEV7
T2300	Check STR Level		If F1016Ax is in AUTO mode (OP006) and READY_FOR_FILTR signal from F1016Ax is received and not FILTER_FAULT and Olein Blow 2 + LMCB time over then go to step S2300
S2200	Cool Olein Blow	A0010	Stopping DEV1
		A0020	Set DEV2 Mode to Remote Out
		A0030	Set DEV2 output to Closed
		A0040	Stopping DEV3
		A0050	Stopping DEV5
		A0060	Closing DEV6
		A0070	Closing DEV7
		A0080	Closing DEV8
		A0090	Closing DEV9
		A0100	Closing DEV10
		A0110	Closing DEV11
		A0120	Closing DEV12
		A0130	Enabling Conditional Logic for DEV13
		A0160	Closing DEV18
		A0170	Closing DEV19
		A0180	Set DEV20 Mode to Remote Out
		A0190	Set DEV20 output to Closed
		A0200	Opening DEV21
		A0210	Set Message: Starting Cool Olein Blow Process
		A0220	Opening DEV15
T2400	Cool Core Blow	A0230	Closing DEV14
		A0240	Closing DEV15
		A0250	Opening DEV14
		A0260	Pausing Cool Timer
			If F1016Ax is in AUTO mode (OP006) and READY_FOR_FILTR signal from F1016Ax is received and not FILTER_FAULT and (Cool Olein Blow time (TP015) over or Pb <Water> (OP005) is selected) then go to step S2400

Step	Description	Action	Action Description
S2300	Check STR Level	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Set DEV2 Mode to Remote Out
		A0030	Set DEV2 output to Closed
		A0040	Stopping and Disabling Conditional Logic for DEV3
		A0050	Setpoint for DEV3
		A0060	Stopping DEV5
		A0070	Closing DEV6
		A0080	Closing DEV7
		A0090	Closing DEV8
		A0100	Closing DEV9
		A0110	Closing DEV10
		A0120	Closing DEV11
		A0130	Closing DEV12
		A0140	Closing DEV13
		A0150	Opening DEV14
		A0160	Opening DEV15
		A0170	Closing DEV16
		A0180	Closing DEV17
		A0190	Closing DEV18
		A0200	Closing DEV19
		A0210	Set Message: Stopping Down Ramp Process
		A0220	Set DEV20 Mode to Remote Out
		A0230	Set DEV20 output to Open
		A0240	Opening DEV21
T2500A	Open Bombdoor		If F1016Ax is in AUTO mode (OP006) and READY_FOR_FILTR signal from F1016Ax is received and not FILTER_FAULT and LT_382ST1 < Enable Filter Discharge Level (F1082ST_TP013) then go to step S2500
S2400	Cool Core Blow	A0010	Stopping DEV1
		A0020	Set DEV2 Mode to Remote Out
		A0030	Set DEV2 output to Closed
		A0040	Stopping DEV3
		A0050	Stopping DEV5

Step	Description	Action	Action Description
		A0060	Closing DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Enabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Opening DEV10
		A0110	Closing DEV11
		A0120	Enabling Conditional Logic for DEV12
		A0130	Closing and Disabling Conditional Logic for DEV13
		A0140	Closing DEV14
		A0150	Closing DEV15
		A0160	Closing DEV16
		A0170	Closing DEV17
		A0180	Closing DEV18
		A0190	Closing DEV19
		A0200	Set DEV20 Mode to Remote Out
		A0210	Set DEV20 output to Closed
		A0220	Opening DEV21
		A0230	Setting TP02 for DEV12
		A0240	Set Message: Starting Cool Core Blow Process
		A0250	Pausing Blow Timer
T2600	Cool Done		If F1016Ax is in AUTO mode (OP006) and READY_FOR_FILTER signal from F1016Ax is received and not FILTER_FAULT and (Cool Core Blow time (TP015) over or Pb <Water> (OP005) is selected) then go to step S2600
S2500	Open Bombdoor	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Set DEV2 Mode to Remote Out
		A0030	Set DEV2 output to Closed
		A0040	Stopping and Disabling Conditional Logic for DEV3
		A0050	Setpoint for DEV3
		A0060	Stopping DEV5
		A0070	Closing DEV6
		A0080	Closing DEV7

Step	Description	Action	Action Description
		A0090	Closing DEV8
		A0100	Closing DEV9
		A0110	Closing DEV10
		A0120	Closing DEV11
		A0130	Closing DEV12
		A0140	Closing DEV13
		A0150	Closing DEV14
		A0160	Opening DEV15
		A0170	Closing DEV16
		A0180	Closing DEV17
		A0190	Closing DEV18
		A0200	Closing DEV19
		A0210	Set DEV20 Mode to Remote Out
		A0220	Set DEV20 output to Open
		A0230	Opening DEV21
		A0240	Set Message: Starting Open Bombdoor Process
T2700	Cake Discharge		If F1016Ax is in AUTO mode (OP006) and Bombdoor opened signal (BOMBDOOR_OPENED) from F1016Ax is received and PT_216AS1 <= PRESET_VALUE4 and PT_216AS2 <= PRESET_VALUE4 then go to step S2700
S2600	Cool Done	A0010	Stopping DEV1
		A0020	Set DEV2 Mode to Remote Out
		A0030	Set DEV2 output to Closed
		A0040	Stopping DEV3
		A0050	Stopping DEV5
		A0060	Closing DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing DEV9
		A0100	Closing DEV10
		A0110	Closing DEV11
		A0120	Closing and Disabling Conditional Logic for DEV13

Step	Description	Action	Action Description
		A0130	Closing DEV13
		A0140	Closing DEV14
		A0150	Opening DEV15
		A0160	Closing DEV16
		A0170	Closing DEV17
		A0180	Closing DEV18
		A0190	Closing DEV19
		A0200	Set DEV20 Mode to Remote Out
		A0210	Set DEV20 output to Closed
		A0220	Opening DEV21
		A0240	Set Message: Starting Cool Done Process
		A0250	Checking Conditions
T2800	Aux Off		If F1016Ax is in AUTO mode (OP006) and READY_FOR_FILTR signal from F1016Ax is received and not FILTER_FAULT and 5 secs delay over then go to step S2800
S2700	Cake Discharge	A0010	Stopping and Disabling Conditional Logc for DEV1
		A0020	Set DEV2 Mode to Remote Out
		A0030	Set DEV2 output to Closed
		A0040	Stopping and Disabling Conditional Logc for DEV3
		A0050	Setpoint for DEV3
		A0060	Stopping DEV5
		A0070	Closing DEV6
		A0080	Closing DEV7
		A0090	Closing DEV8
		A0100	Closing DEV9
		A0110	Closing DEV10
		A0120	Closing DEV11
		A0130	Closing DEV12
		A0140	Closing DEV13
		A0150	Closing DEV14
		A0160	Opening DEV15
		A0170	Closing DEV16
		A0180	Closing DEV17
		A0190	Closing DEV18

Step	Description	Action	Action Description
		A0200	Closing DEV19
		A0210	Set DEV20 Mode to Remote Out
		A0220	Set DEV20 output to Open
		A0230	Opening DEV21
		A0240	Set Message: Starting Cake Discharge Process
T2900	Close Filter		If F1016Ax is in AUTO mode (OP006) and FILTER_OPENED signal from F1016Ax is received then go to step S2900
S2800	Aux Off	A0010	Closing DEV15
		A0020	Closing DEV21
		A0030	Set DEV20 Mode to Remote Out
		A0040	Set DEV20 output to Closed
		A0050	Resetting PB Wash Demand
T4800C	Disable HM and SM		If no pending actions in S2800 then go to step S2900
S2900	Close Filter	A0010	Stopping and Disabling Conditional Logc for DEV1
		A0020	Set DEV2 Mode to Remote Out
		A0030	Set DEV2 output to Closed
		A0040	Stopping and Disabling Conditional Logc for DEV3
		A0050	Setpoint for DEV3
		A0060	Stopping DEV5
		A0070	Closing DEV6
		A0080	Closing DEV7
		A0090	Closing DEV8
		A0100	Closing DEV9
		A0110	Closing DEV10
		A0120	Closing DEV11
		A0130	Closing DEV12
		A0140	Closing DEV13
		A0150	Closing DEV14
		A0160	Opening DEV15
		A0170	Closing DEV16
		A0180	Closing DEV17
		A0190	Closing DEV18

Step	Description	Action	Action Description
		A0200	Closing DEV19
		A0210	Set DEV20 Mode to Remote Out
		A0220	Set DEV20 output to Open
		A0230	Opening DEV21
		A0240	Set Message: Starting Close Filter Process
T3000	Close Bombdoor		If F1016Ax is in AUTO mode (OP006) and FILTER_OPENED signal from F1016Ax is OFF and FILTER_CLOSED signal received then go to step S3000
S3000	Close Bombdoor	A0010	Stopping and Disabling Conditional Logc for DEV1
		A0020	Set DEV2 Mode to Remote Out
		A0030	Set DEV2 output to Closed
		A0040	Stopping and Disabling Conditional Logc for DEV3
		A0050	Setpoint for DEV3
		A0060	Stopping DEV5
		A0070	Closing DEV6
		A0080	Closing DEV7
		A0090	Closing DEV8
		A0100	Closing DEV9
		A0110	Closing DEV10
		A0120	Closing DEV11
		A0130	Closing DEV12
		A0140	Closing DEV13
		A0150	Closing DEV14
		A0160	Opening DEV15
		A0170	Closing DEV16
		A0180	Closing DEV17
		A0190	Closing DEV18
		A0200	Closing DEV19
		A0210	Set DEV20 Mode to Remote Out
		A0220	Set DEV20 output to Open
		A0230	Opening DEV21
		A0240	Set Message: Starting Close Bombdoor Process

Step	Description	Action	Action Description
T3100	Aux Off		If F1016Ax is in AUTO mode (OP006) and BOMBDOOR_CLOSED signal from F1016Ax is received then go to step S3100
S3100	Aux Off	A0010	Closing DEV15
		A0020	Set DEV20 Mode to Remote Out
		A0030	Set DEV20 output to Closed
		A0040	Closing DEV21
T4800A	Complete		If no pending actions in S3100 then go to step S3200
S3200	Standby Blow	A0010	Closing DEV6
		A0020	Closing DEV11
		A0030	Closing DEV19
		A0040	Set Message: Starting Additional Blow Process
		A0050	Waiting for Feed Line Busy Signal
T3300	Fill Line Blow		If F1016Ax is in AUTO mode (OP006) and F1016Ax is in step Squeeze Ramp 1/2/3 and not F1016Ax Fill Line Blow Over then go to step S3300
S3300	Fill Line Blow	A0010	Opening DEV6
		A0020	Enabling Conditional Logic for DEV11
		A0030	Opening DEV19
		A0040	Set Message: Starting Fill Line Blow Process
		A0050	Checking Conditions
		A0060	Setting Feed Line Busy
		A0070	Pausing Fill Timer
T3600	Vent Fill Line		If F1016Ax is in AUTO mode (OP006) and Fill Line Blow time (TP034) over then go to step S3600
S3400	Core Blow 1	A0010	Enabling Conditional Logic for DEV7
		A0020	Enabling Conditional Logic for DEV8
		A0030	Opening DEV9
		A0040	Enabling Conditional Logic for DEV12

Step	Description	Action	Action Description
		A0050	Set Message: Starting Core Blow 1 Process
		A0060	Checking Conditions
		A0070	Pausing Blow1 Timer
T3700	Core Blow 1 Over		If F1016Ax is in AUTO mode (OP006) and Core Blow 1 timer (TP036) elapsed then go to step S3700
S3500	Olein Blow 1	A0010	Set Message: Starting Olein Blow 1 Process
		A0020	Opening DEV15
		A0030	Closing DEV15
		A0040	Opening DEV14
		A0050	Enabling Conditional Logic for DEV13
		A0060	Pausing Olien Blow1 Timer
		A0070	Closing DEV7
		A0080	Closing DEV8
		A0090	Closing DEV9
		A0100	Closing DEV12
		A0110	Closing DEV10
		A0120	Closing DEV18
T3800	Core Blow 2		If F1016Ax is in AUTO mode (OP006) and Olein Blow 1 timer (TP037) elapsed then go to step S3800
S3600	Vent Fill Line	A0010	Opening DEV6
		A0020	Closing and Disabling Conditional Logic for DEV11
		A0030	Opening DEV19
		A0040	Set Vent Fill Line Time
		A0050	Start Vent Fill Line Time
		A0060	Vent Fill Line Time Message
T3900	Fill Line Blow Over		If F1016Ax is in AUTO mode (OP006) and Vent Fill Line timer (TP035) elapsed then go to step S3900
S3700	Core Blow 1 Over	A0010	Closing and Disabling Conditional Logic for DEV7
		A0020	Closing and Disabling Conditional Logic for DEV8

Step	Description	Action	Action Description
		A0030	Closing DEV9
		A0040	Closing and Disabling Conditional Logic for DEV12
		A0050	Set Message: Starting Core Blow 1 Over Process
T4000	Aux Off		If F1016Ax is in AUTO mode (OP006) and BOMBDOOR_CLOSED signal from F1016Ax is received and F1016Ax is in step <Close Bombdoor> then go to step S4000
S3800	Core Blow 2	A0010	Set Message: Starting Core Blow 2 Process
		A0020	Closing and Disabling Conditional Logic for DEV13
		A0030	Closing DEV14
		A0040	Closing DEV15
		A0050	Enabling Conditional Logic for DEV7
		A0060	Enabling Conditional Logic for DEV8
		A0070	Opening DEV9
		A0080	Enabling Conditional Logic for DEV12
		A0090	Checking Conditions
		A0100	Pausing Core Blow1 Timer
		A0110	Closing DEV10
		A0120	Closing DEV18
T4100	Core Washing		If F1016Ax is in AUTO mode (OP006) and Core Blow 2 time (TP038) over and no other filter is in step <Washing or cooling or core wash> and Pb Core Wash (OP003) ON and TT_382C1 <= Max Washing Temp (TP006) and TT_321C1 <= Max Washing Temp (TP006) and TT_321C1 >= Min Washing Temp (TP005) and F1082C is in step <Prewash Recirculation> with 10 secs delay then go to step S4100
T4200	Core Blow 3		If F1016Ax is in AUTO mode (OP006) and {not Pb Core Wash (OP003) ON or (Pb Core Wash ON and [any filter (washing or cooling or core wash step)])} then go to step S4200
S3900		A0010	Closing DEV6

Step	Description	Action	Action Description
	Fill Line Blow Over	A0020	Closing DEV11
		A0030	Closing DEV19
T4300	Aux Off		If F1016Ax is in AUTO mode (OP006) and F1016Ax is in step <Close Bombdoor> then go to step S4300
S4100	Core Washing	A0010	Set Message: Starting Core Washing Process
		A0020	Opening DEV7
		A0030	Closing DEV8
		A0040	Closing DEV9
		A0050	Opening DEV10
		A0060	Closing and Disabling Conditional Logic for DEV12
		A0070	Opening DEV18
		A0080	Checking Conditions
		A0090	Set Core Washing Time
		A0100	Start Core Washing Time
		A0110	Core Washing Time Message
T4400	Core Wash Drain		If F1016Ax is in AUTO mode (OP006) and Core Wash time (TP009) over then go to step S4400
S4200	Core Blow 3	A0010	Closing and Disabling Conditional Logic for DEV13
		A0020	Closing DEV14
		A0030	Closing DEV15
		A0040	Enabling Conditional Logic for DEV7
		A0050	Enabling Conditional Logic for DEV8
		A0060	Opening DEV9
		A0070	Closing DEV10
		A0080	Enabling Conditional Logic for DEV12
		A0090	Closing DEV18
		A0100	Checking Conditions
		A0110	Pausing Core Blow 3 Timer
		A0120	Set Message: Starting Core Blow 3 Process

Step	Description	Action	Action Description
T4600B	MISC Blow Over		If F1016Ax is in AUTO mode (OP006) and Core Blow 3 time (TP039) over then go to step S4600
S4300	Aux Off	A0060	Setting Feed Line Busy
T4800D	Disable HM and SM		If F1016Ax is in AUTO mode (OP006) and BOMBDOOR_CLOSED signal from F1016Ax is received then go to step S4800
S4400	Core Wash Drain	A0010	Set Message: Starting Core Wash Drain Process
		A0020	Opening DEV7
		A0030	Closing DEV8
		A0040	Closing DEV9
		A0050	Opening DEV10
		A0060	Closing and Disabling Conditional Logic for DEV12
		A0070	Opening DEV18
		A0080	Checking Conditions
		A0090	Set Core Wash Drain Time
		A0100	Start Core Wash Drain Time
		A0110	Core Wash Drain Time Message
T4500	Core Wash Blow		If F1016Ax is in AUTO mode (OP006) and Core Wash Drain time (TP010) over then go to step S4500
S4500	Core Wash Blow	A0010	Set Message: Starting Core Wash Blow Process
		A0020	Enabling Conditional Logic for DEV7
		A0030	Enabling Conditional Logic for DEV8
		A0040	Closing DEV9
		A0050	Opening DEV10
		A0060	Enabling Conditional Logic for DEV12
		A0070	Pausing Core Wash Blow Timer
		A0080	Closing DEV18
		A0090	Checking Conditions
T4600A	MISC Blow Over		If F1016Ax is in AUTO mode (OP006) and Core Wash Blow time (TP011) over then go to step S4600

Step	Description	Action	Action Description
S4600	MISC Blow Over	A0010	Set Message: Starting MISC Blow Over Process
		A0020	Closing and Disabling Conditional Logic for DEV7
		A0030	Closing and Disabling Conditional Logic for DEV8
		A0040	Closing DEV9
		A0050	Closing DEV10
		A0060	Closing DEV12
		A0070	Closing DEV18
		A0080	Set Message: Waiting for Bombdoor Closed Signal
T4700	Aux Off		If F1016Ax is in AUTO mode (OP006) and F1016Ax is in step <Close Bombdoor> then go to step S4700
S4800	Disable HM and SM	A0010	Disable HM 01 (Module Ownership Lost)
T4900	Release Devices		If no pending actions in S4800 then go to step S4900
S4900	Release Devices	A0010	Release Devices Message
		A0020	Release Shared Devices
		A0030	Release Private Devices
		A0040	Release Private Devices
		A0050	Release Shared Devices
		A0060	Release Private Devices
		A0070	Release Private Devices
		A0080	Release Private Devices
		A0090	Release Private Devices
		A0100	Release Private Devices
		A0110	Release Private Devices
		A0120	Release Private Devices
		A0130	Release Private Devices
		A0140	Release Private Devices
		A0150	Release Private Devices
		A0160	Release Private Devices
		A0170	Release Private Devices

Step	Description	Action	Action Description
		A0180	Release Private Devices
		A0190	Release Shared Devices
		A0200	Release Private Devices
		A0210	Release Private Devices
		A0220	Release Private Devices
T9900	Complete		If no pending actions in S4900 then go to step S9900
S5000	Aux Off	A0010	Closing DEV15
		A0020	Set DEV20 Mode to Remote Out
		A0030	Set DEV20 output to Closed
		A0040	Closing DEV21
		A0050	Resetting PB Wash Demand
T4800B	Disable HM and SM		If no pending actions in S5000 then go to step S4800
S5100	Standby	A0010	Starting Olein Blow 1 Process
T3500	Olein Blow 1		If F1016Ax is in AUTO mode (OP006) and F1016Ax is in step <Down Ramp> then go to step S3500
S5200	Standby	A0010	Starting Core Blow 1 Process
T5300	Standby Filtration		If F1016Ax is in AUTO mode (OP006) and F1016Ax is in step Squeeze Ramp 1/2/3 then go to step S5300
T3400	Core Blow 1		If F1016Ax is in AUTO mode (OP006) and F1016Ax is in step <Fill Line Blow Over> and Pb Core Blow 1 ON and F1016Ax is in step Squeeze Ramp 1/2/3 then go to step S3400
S5300	Standby	A0010	Starting Core Blow 1 Process
T1900	Depressurize Membrane		If F1016Ax is in AUTO mode (OP006) and READY_FOR_FILTR signal from F1016Ax is received and not FILTER_FAULT and MISC Blow Over and PT_216AS2 <= Down Ramp End Pressure (TP033) then go to step S1900
S9900	Complete	A0010	Set Message: Filter Command Complete

Step	Description	Action	Action Description
END	Command Complete		True

Table 40: Filter Sequence

6.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value/Reference
Operating Parameters				
OP001	Wash Demand	Named Set	NA	OFF
OP002	Wash & Cool Demand	Named Set	NA	OFF
OP003	Core Wash	Named Set	NA	OFF
OP004	Cool Demand	Named Set	NA	OFF
OP005	Cooling Type	Named Set	NA	Water
OP006	Auto	Named Set	NA	OFF
OP007	Stop Filtration (Go Squeeze)	Boolean	NA	False
OP008	Open Filter	Boolean	NA	False
OP009	Filter E Stop	Named Set	NA	OFF
OP010	Manual Enable	Named Set	NA	OFF
OP011	Redline Alarm Reset	Named Set	NA	Reset
OP012	Alarm Reset	Named Set	NA	Reset
OP013	Core Blow Ramp Up	Named Set	NA	NO
Tuning Parameters				
TP001	Delay Timer	Floating Point	sec	5
TP002	Filling Time	Floating Point	min	600
TP003	Recirculation Time	Floating Point	min	900
TP004	Drain Time	Floating Point	min	300
TP005	Min. Washing Temp	External Reference	°C	CF_ESM_1/MIN_WASH_TEMP
TP006	Max. Washing Temp	External Reference	°C	CF_ESM_1/MAX_WASH_TEMP
TP007	Wash Olein Blow	Floating Point	sec	6
TP008	Wash Core Blow	Floating Point	sec	3
TP009	Core Washing	Floating Point	sec	100
TP010	Core Wash Drain	Floating Point	sec	100

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Parameter	Description	Type	Units	Default Value/Reference
TP011	Core Wash Blow	Floating Point	sec	100
TP012	Cool Filling	Floating Point	min	600
TP013	Cool Recirculation	Floating Point	min	900
TP014	Cool Drain	Floating Point	min	300
TP015	Cool Olein Blow	Floating Point	sec	60
TP016	Cool Core Blow	Floating Point	sec	60
TP017	Fill Ramp Up Pressure	Floating Point	bar	1.5
TP018	Remain Time	Floating Point	min	0
TP019	Fill Ramping End Pressure	Floating Point	bar/m	2.1
TP020	Fill Ramping Rate	Floating Point	bar	2
TP021	Pressure SP to Open XV_F16A_DOL	Floating Point	bar	0.7
TP022	Initial Speed	Floating Point	%	60
TP023	Final Pressure	Floating Point	bar	10
TP024	PF1078/16 Run Speed	Floating Point	%	0
TP025	Squeez Ramp 1 Rate	Floating Point	bar/m	0.5
TP026	Squeez Ramp 2 Rate	Floating Point	bar/m	1
TP027	Squeez Ramp 3 Rate	Floating Point	bar/m	1.5
TP028	Sqz. Ramp 1 End Pressure	Floating Point	bar	3
TP029	Sqz. Ramp 2 End Pressure	Floating Point	bar	6
TP030	Sqz. Ramp 3 End Pressure	Floating Point	bar	9
TP031	Sqz. Hold 3 Time	Floating Point	min	0
TP032	Down Ramp Rate	Floating Point	bar/m	0
TP033	Down Ramp End Pressure	Floating Point	min	0
TP034	Fill Line Blow	Floating Point	sec	100
TP035	Fill Line Vent Time	Floating Point	sec	310
TP036	Core Blow 1 Time	Floating Point	sec	100
TP037	Olein Blow 1 Time	Floating Point	sec	100
TP038	Core Blow 2 Time	Floating Point	sec	120
TP039	Core Blow 3 Time	Floating Point	sec	100
TP040	Olein Blow 2 + LMCB	Floating Point	sec	100
TP041	Core Blow Pulse On	Floating Point	sec	4
TP042	Core Blow Pulse Off	Floating Point	sec	4



Parameter	Description	Type	Units	Default Value/Reference
TP043	Olein Blow Pulse On	Floating Point	sec	4
TP044	Olein Blow Pulse Off	Floating Point	sec	4
TP045	Cool Temperature SP	External Reference	°C	CF_ESM_1/COOL_TEMP_SP
TP046	Elapsed Filtration	Floating Point	min	0
TP047	Filtration Cycle Count	Floating Point	NA	5
TP048	Washing Cycle Count	Floating Point	NA	10
TP049	Cooling Cycle Count	Floating Point	NA	5
TP050	Total Cycle Time Elapsed	Floating Point	min	0
TP051	Fill Const. Pressure Time	Floating Point	min	600
TP052	Drain Speed	Floating Point	%	50
TP053	Maximum Speed	Floating Point	%	95
TP054	Sqz. Tank Temp	Floating Point	°C	0
TP055	Sqz. Tank Min Level	Floating Point	%	10

Table 41: Operating and Tuning Parameters

6.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value/Reference	
MY_TAG		CF1EQ_1016A1	CF1EQ_1016A2
OPR_CMD_MASK		7	7
F1082ST_TP013	External Reference	CF1EQ_1082ST/TP013_VALUE	
PT_216AX1	External Reference	CF1_PT1216A1 1/AI1/PV	CF1_PT1216A2 1/AI1/PV
LT_382B1	External Reference	CF1_LT1382B1/PV	
LSH_382B1	External Reference	CF1_LSH1382B1/PV_D	
PT_216AS1	External Reference	CF1_PT1216AS 1/AI1/PV	CF1_PT1216AS 3/AI1/PV



Parameter Name	Parameter Type	Value/Reference	
PT_216AS2	External Reference	CF1_PT1216AS 2/AI1/PV	CF1_PT1216AS 4/AI1/PV
LSH_36OV1	External Reference	CF1_LSH1360V1/PV_D	
LT_382ST1	External Reference	CF1_LT1382ST1/PV	
LSH_382ST1	External Reference	CF1_LSH1382ST/PV_D	
TT_382C1	External Reference	CF1_TT1321C1/AI1/PV	
TT_321C1	External Reference	CF1_TT1382C1/AI1/PV	
CF_FILT_X	String	CF1EQ_1016A2	CF1EQ_1016A1
CF_F1057CKS	String	CF1EQ_1057CK	
CF_F1082B	String	CF1EQ_1082B	
CF_F1082C	String	CF1EQ_1082C	
DELAY_TIME1	Floating Point	120	
DELAY_TIME2	Floating Point	30	
DELAY_TIME3	Floating Point	60	
PRESET_VALUE1	Floating Point	6.5	
PRESET_VALUE2	Floating Point	7	
PRESET_VALUE3	Floating Point	0.25	
PRESET_VALUE4	Floating Point	0.3	
PRESET_VALUE5	Floating Point	30	
STEP_STATUS	Boolean	False	
HOLD	Boolean	False	
COMPLETED_TIME	Floating Point		
RESET_COMP_TM	Boolean	False	
STEP_TIME	Floating Point		
MACHINE_SEL	Boolean	False	
FEED_LINE_BUSY	Boolean	False	
READY_FOR_FILTR	Boolean	False	
STEP_FLAG1	Boolean	False	
STEP_FLAG2	Boolean	False	
STEP_FLAG3	Boolean	False	
STEP_FLAG4	Boolean	False	
FILTER_OPENED	Boolean	True	
FILTER_CLOSED	Boolean	False	

Parameter Name	Parameter Type	Value/Reference
BOMBDOOR_OPENED	Boolean	False
BOMBDOOR_CLOSED	Boolean	False
EN_WASHING_STEP	Boolean	False
EN_COOLING_STEP	Boolean	False
EN_COREWASH_STEP	Boolean	False
FILTER_FAULT	Boolean	False
DY_REF_1	Dynamic Reference	
DY_REF_2	Dynamic Reference	
DY_REF_3	Dynamic Reference	
DY_REF_4	Dynamic Reference	
DY_REF_5	Dynamic Reference	
DY_REF_6	Dynamic Reference	
DY_REF_7	Dynamic Reference	
DY_REF_8	Dynamic Reference	

Table 42: Configurable Parameters

6.6 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_CF_FILTRATION	CPKO Fractionation Area: Filtration

Table 43: Named Set Details

Details of states for NS_CF_FILTRATION named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Filter

Table 44: Name Set State Details

7 EQUIPMENT MODULE CLASS DESIGN – EM_CF_PRODUCT1 (F1082ST)

7.1 Functional Description

EM_CF_PRODUCT1 is a class based equipment module that is used for Fractionation Products as a part of Discharge Line Blow. EM_CF_PRODUCT1 is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. All devices used in EM shall be acquired in Recirculate Command and shall be released on completion of Recirculate Command. Hold Request shall be enabled for all acquired devices at the end of HOLD command.

7.2 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name
EM	EM_CF_PRODUCT1	CF1EQ_1082ST
DEV1	_MRSS11_130_SE	CF1_PU1310ST
DEV2	_V12_IP_130_SE	CF1_V1382STDO
DEV3	_V12_IP_130_SE	CF1_V1382STRO
DEV4	_V12_IP_130_SE	CF1_V1382ST
DEV5	_V12_IP_130_SE	CF1_V1321DO1
DEV6	_V12_IP_130_SE	CF1_V1321DO2
DEV7	_V12_IP_130_SE	CF1_V1382STBA
DEV8	_AI_130_SE	CF1_TT13821ST1

Table 45: Instances

7.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

7.3.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Start	A0010	Store Fail Message
		A0020	Reset Hold Request
		A0030	Save Previous Step Time

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Step	Description	Action	Action Description
T1000	Actions Complete or Time Out & Hold=True		Go to step H1000
T2000	Actions Complete or Time Out & Hold=False		Go to step H2000
H1000	Hold	A0010	Stopping DEV1
T9900A	Actions Complete or Time Out		Go to step H9900A
H2000	General Reset	A0010	Stopping DEV1
		A0020	Closing DEV2
		A0030	Closing DEV3
		A0040	Closing DEV4
		A0050	Closing DEV5
		A0060	Closing DEV6
		A0070	Closing DEV7
		A0080	Set DEV8 Mode to Remote Out
		A0090	Set DEV8 output to Closed
T9900B	Actions Complete or Time Out		Go to step H9900B
H9900A	Command Hold End	A0010	Completed
END	Command Complete		True
H9900B	Release Devices	A0010	Release Devices Message
		A0020	Release Private Devices
		A0030	Release Private Devices
		A0040	Release Private Devices
		A0050	Release Private Devices
		A0060	Release Private Devices
		A0070	Release Private Devices
		A0080	Release Private Devices



Step	Description	Action	Action Description
		A0090	Release Private Devices
		A0100	Release Private Devices
		A0110	Reset Parameters
END1	Command Complete		True

Table 46: Hold Sequence

7.3.2

Command 01 – Recirculate

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Recirculate Command
T0010	First Pass = False		If First Pass = False then go to step S0010
T0020A	First Pass = True		If First Pass = True then go to step S0020
S0010	Initializations	A0010	First Pass Logic
		A0020	First Pass = True
		A0030	Acquire EM
T0020B	Acquire Devices		If no pending actions in S0010 then go to step S0020
S0020	Acquire Devices	A0010	Acquire Devices Message
		A0020	Acquire Private Devices
		A0030	Acquire Private Devices
		A0040	Acquire Private Devices
		A0050	Acquire Private Devices
		A0060	Acquire Private Devices
		A0070	Acquire Private Devices
		A0080	Acquire Private Devices
		A0090	Acquire Private Devices
		A0100	Reset HOLD_REQ
		A0110	Reset Actual Blow Count
T0100	Standby		Go to step S0100
T0200B	Heating		If Task Pointer = 2 then go to step S0200
T0300B	Recirculation		If Task Pointer = 3 then go to step S0300
T0400B	Storage		If Task Pointer = 4 then go to step S0400
T0500C	Blow On		If Task Pointer = 5 then go to step S0500



Step	Description	Action	Action Description
T0600B	Blow Off		If Task Pointer = 6 then go to step S0600
S0100	Standby	A0010	Stopping DEV1
		A0020	Closing DEV2
		A0030	Opening DEV3
		A0040	Closing DEV4
		A0050	Closing DEV5
		A0060	Closing DEV6
		A0070	Closing DEV7
		A0080	Set DEV8 Mode to Remote Out
		A0090	Set DEV8 output to Closed
		A0100	Re-Setting Actual Blow Count
		A0110	Checking Conditions
T0200A	Heating		If no pending actions in S0100 then go to step S0200
S0200	Heating	A0010	Stopping DEV1
		A0020	Closing DEV2
		A0030	Opening DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Closing DEV5
		A0060	Closing DEV6
		A0070	Closing DEV7
		A0080	Set DEV8 Mode to Remote Out
		A0090	Set DEV8 output to Closed
		A0100	Set Message: Starting Heating Process
		A0110	Checking Conditions
T0300A	Recirculation		If TT_382ST1 >= Enable Recirculation Temp (TP002) and LT_382ST1 >= Enable Recirculation Level (TP003) then go to step S0300
S0300	Recirculation	A0010	Enabling Conditional Logic for DEV1
		A0020	Opening DEV2
		A0030	Opening DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Closing DEV5
		A0060	Closing DEV6

Step	Description	Action	Action Description
		A0070	Closing DEV7
		A0080	Set DEV8 Mode to Remote CAS
		A0090	Setting DEV8 setpoint to (TT_F21ST-1 Setpoint) and Enabling Conditional Logic
		A0100	Set Message: Starting Recirculation Process
		A0110	Checking Conditions
T0400A	Storage		If TT_382ST1 >= Start Drain to Storage Temp (TP005) and LT_382ST1 >= Start Drain to Storage Level (TP006) then go to step S0400
S0400	Storage	A0010	Enabling Conditional Logic for DEV1
		A0020	Opening DEV2
		A0030	Closing DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Enabling Conditional Logic for DEV6
		A0070	Closing DEV7
		A0080	Set DEV8 Mode to Remote CAS
		A0090	Setting DEV8 setpoint to (TT_F21ST-1 Setpoint) and Enabling Conditional Logic
		A0100	Set Message: Starting Storage Process
		A0110	Checking Conditions
T0500A	Blow On		If LT_382ST1 <= Stop Drain to Storage Level (TP007) or LSL_382ST then go to step S0500
S0500	Blow On	A0010	Set Message: Starting Blow On Process
		A0020	Starting Blow Timer
		A0030	Stopping and Disabling Conditional Logic for DEV1
		A0040	Closing DEV2
		A0050	Closing DEV3
		A0060	Enabling Conditional Logic for DEV4
		A0070	Enabling Conditional Logic for DEV5
		A0080	Enabling Conditional Logic for DEV6
		A0090	Enabling Conditional Logic for DEV7
		A0100	Set DEV8 Mode to Remote Out

Step	Description	Action	Action Description
		A0110	Set DEV8 output to Closed
		A0120	Pausing Blow On Timer
T0600A	Blow Off		If Blow On time (TP008) is over then go to step S0600
S0600	Blow Off	A0010	Set Message: Starting Blow Off Process
		A0020	Increment Actual Blow count
		A0030	Set Message
		A0040	Stopping and Disabling Conditional Logic for DEV1
		A0050	Closing DEV2
		A0060	Closing DEV3
		A0070	Enabling Conditional Logic for DEV4
		A0080	Enabling Conditional Logic for DEV5
		A0090	Enabling Conditional Logic for DEV6
		A0100	Closing and Disabling Conditional Logic for DEV7
		A0110	Set DEV8 Mode to Remote Out
		A0120	Set DEV8 output to Closed
		A0130	Time save
		A0140	Set Blow off Time
		A0150	Start Blow off Time
		A0160	Blow off Time Message
T0500B	Blow On		If Blow Off time (TP009) is over and Actual Blow Count (TP010) < Blow Preset Count (TP011) then go to step S0500
T0700	Aux Off		If Blow Off time (TP009) is over and Actual Blow Count (TP010) >= Blow Preset Count (TP011) then go to step S0700
S0700	Aux Off	A0010	Stopping DEV1
		A0020	Closing DEV2
		A0030	Opening DEV3
		A0040	Closing DEV4
		A0050	Closing DEV5
		A0060	Closing DEV6
		A0070	Closing DEV7
		A0080	Set DEV8 Mode to Remote Out
		A0090	Set DEV8 output to Closed

Step	Description	Action	Action Description
T0800	Disable HM and SM		If no pending actions in S0700 then go to step S0800
S0800	Disable HM and SM	A0010	Disable HM 01 (Module Ownership Lost)
T0900	Release Devices		If no pending actions in S0800 then go to step S0900
S0900	Release Devices	A0010	Release Devices Message
		A0020	Release Private Devices
		A0030	Release Private Devices
		A0040	Release Private Devices
		A0050	Release Private Devices
		A0060	Release Private Devices
		A0070	Release Private Devices
		A0080	Release Private Devices
		A0090	Release Private Devices
		A0100	Release Private Devices
T9900	Complete		If no pending actions in S0900 then go to step S9900
S9900	Complete	A0010	Set Message: Recirculate Command Complete
		A0020	Self Release the EM & Reset First Pass
END	Command Complete		True

Table 47: Recirculate Sequence

7.4

Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value/Reference
Operating Parameters				
OP001	Start Blow	Named Set	NA	YES
OP002	Drain to Storage	Named Set	NA	Product
Tuning Parameters				
TP001	Acquire Timer	Floating Point	sec	20

Parameter	Description	Type	Units	Default Value/Reference
TP002	Enable Recirculation Temp.	Floating Point	°C	55
TP003	Enable Recirculation Level	Floating Point	%	70
TP004	Not Used	NA	NA	NA
TP005	Start Drain to Storage Temp.	Floating Point	°C	70
TP006	Start Drain to Storage Level	Floating Point	%	65
TP007	Stop Drain to Storage Level	Floating Point	%	20
TP008	Blow On Time	Floating Point	sec	120
TP009	Blow Off Time	Floating Point	sec	100
TP010	Actual Count	Floating Point	NA	0
TP011	Preset Count	Floating Point	NA	5
TP012	Coil Heating Temp.	Floating Point	°C	70
TP013	Enable Filter Discharge Level	Floating Point	%	70

Table 48: Operating and Tuning Parameters

7.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value/Reference
MY_TAG		CF1EQ_1082ST
OPR_CMD_MASK		3
LSL_382ST	External Reference	CF1_LSL1382ST/DI1/DISC_ACT
LT_382ST1_PV	External Reference	CF1_LT1382ST1/PV
LT_382ST1_LL_ALM	External Reference	CF1_LT1382ST1/LO_LO_ALM
TT_382ST1	External Reference	CF1_TT1382ST1/AI1/PV
HOLD	External Reference	CF1EQ_1002A/OP003_VALUE
COMPLETED_TIME	Floating Time	0
RESET_COMP_TM	Boolean	False
STEP_TIME	Floating Point	0

Table 49: Configurable Parameters

7.6 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_CF_PRODUCT1	CPKO Fractionation Area: Fractionation Products

Table 50: Named Set Details

Details of states for NS_CF_ PRODUCT1 named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Recirculate

Table 51: Name Set State Details

8 EQUIPMENT MODULE CLASS DESIGN – EM_CF_PRODUCT2 (F1082C)

8.1 Functional Description

EM_CF_PRODUCT2 is a class based equipment module that is used for Fractionation Products as a part of Discharge Line Blow. EM_CF_PRODUCT2 is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. All devices used in EM shall be acquired in Wash Command and shall be released on completion of Wash Command. Hold Request shall be enabled for all acquired devices at the end of Hold command.

8.2 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name
EM	EM_CF_PRODUCT2	CFXEQ_1082C
DEV1	_MRSS11_130_SE	CF1_PU13082C
DEV2	_V12_IP_130_SE	CF1_V1382CDOC
DEV3	_V12_IP_130_SE	CF1_V13BDOB
DEV4	_V12_IP_130_SE	CF1_V1382CRO
DEV5	_V12_IP_130_SE	CF1_V1321FIW
DEV6	_V12_IP_130_SE	CF1_V1321RTIW
DEV7	_V12_IP_130_SE	CF1_V1321CS
DEV8	_V12_IP_130_SE	CF1_V1321LPO
DEV9	_V12_IP_130_SE	CF1_V1321CBA
DEV10	_CSD_130_SE	CF1_TT1321C1
DEV11	_V12_IP_130_SE	CF1_V1357CKD

Table 52: Instances

8.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

8.3.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Command Hold Start	A0010	Store Fail Message
		A0020	Set Hold Request
T1000	Actions Complete or		Go to step H1000

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Step	Description	Action	Action Description
	Time Out & Hold=True		
T2000	Actions Complete or Time Out & Hold=False		Go to step H2000
H1000	Hold	A0010	Stopping DEV1
T9900A	Actions Complete or Time Out		Go to step H9900A
H2000	General Reset	A0010	Stopping DEV1
		A0020	Closing DEV2
		A0030	Closing DEV3
		A0040	Opening DEV4
		A0050	Closing DEV5
		A0060	Closing DEV6
		A0070	Closing DEV7
		A0080	Closing DEV8
		A0090	Closing DEV9
		A0100	Set DEV10 Mode to Remote Out
		A0110	Set DEV10 output to Closed
		A0120	Enabling Conditional Logic for DEV11
T9900B	Actions Complete or Time Out		Go to step H9900B
H9900A	Command Hold End	A0010	Completed
END	Command Complete		True
H9900B	Release Devices	A0010	Release Devices Message
		A0020	Release Private Devices
		A0030	Release Private Devices
		A0040	Release Private Devices
		A0050	Release Private Devices
		A0060	Release Private Devices
		A0070	Release Private Devices

Step	Description	Action	Action Description
		A0080	Release Private Devices
		A0090	Release Private Devices
		A0100	Release Private Devices
		A0110	Release Private Devices
		A0120	Release Private Devices
		A0130	Reset Parameters
END1	Command Complete		True

Table 53: Hold Sequence

8.3.2 Command 01 – Wash

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Wash Command
T0010	First Pass = False		If First Pass = False then go to step S0010
T0020A	First Pass = True		If First Pass = True then go to step S0020
S0010	Initializations	A0010	First Pass Logic
		A0020	First Pass = True
		A0030	Acquire EM
T0020B	Acquire Devices		If no pending actions in S0010 then go to step S0020
S0020	Acquire Devices	A0010	Acquire Devices Message
		A0020	Acquire Private Devices
		A0030	Acquire Private Devices
		A0040	Acquire Private Devices
		A0050	Acquire Private Devices
		A0060	Acquire Private Devices
		A0070	Acquire Private Devices
		A0080	Acquire Private Devices
		A0090	Acquire Private Devices
		A0100	Acquire Private Devices
		A0110	Acquire Private Devices
		A0120	Acquire Private Devices
		A0130	Reset HOLD_REQ
T0100	Standby		If Task Pointer = 1 then go to step S0100
T0200A	Wash Level Ready		If Task Pointer = 2 then go to step S0200

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Step	Description	Action	Action Description
T0300A	Prewash Recirculation		If Task Pointer = 3 then go to step S0300
T0400C	PF1082C to Filter		If Task Pointer = 5 then go to step S0400
T0500B	Filter Wash / Cool Drain		If Task Pointer = 6 then go to step S0500
T0600B	Drain		If Task Pointer = 4 then go to step S0600
S0100	Standby	A0010	Stopping DEV1
		A0020	Closing DEV2
		A0030	Closing DEV3
		A0040	Opening DEV4
		A0050	Closing DEV5
		A0060	Closing DEV6
		A0070	Closing DEV7
		A0080	Closing DEV8
		A0090	Closing DEV9
		A0100	Set DEV10 Mode to Remote Out
		A0110	Set DEV10 output to Closed
		A0120	Enabling Conditional Logic for DEV11
		A0130	Checking Conditions
T0200B	Wash Level Ready		If no pending actions in S0100 then go to step S0200
S0200	Wash Level Ready	A0010	Stopping DEV1
		A0020	Closing DEV2
		A0030	Closing DEV3
		A0040	Opening DEV4
		A0050	Closing DEV5
		A0060	Closing DEV6
		A0070	Closing DEV7
		A0080	Closing DEV8
		A0090	Closing DEV9
		A0100	Set DEV10 Mode to Remote Out
		A0110	Set DEV10 output to Closed
		A0120	Enabling Conditional Logic for DEV11
T0300B	Prewash Recirculation		If not PF1082C fault and [F1016A Pb Wash Demand ON or (F1016Ax step Squeeze Hold 3 and Pb Core Wash ON)] then go to step S0300
T0400B	PF1082C to Filter		If F1016Ax is in step <Cool Filling> and Pb Water /7 Oil is ON then go to step S0400



Step	Description	Action	Action Description
T0600A	Drain		If F1082B is in step <Drain F1082C Demand> (F1082B_ACT_STEP) then go to step S0600
S0300	Prewash Recirculation	A0010	Enabling Conditional Logic for DEV1
		A0020	Opening DEV2
		A0030	Closing DEV3
		A0040	Opening DEV4
		A0050	Closing DEV5
		A0060	Closing DEV6
		A0070	Opening DEV7
		A0080	Opening DEV8
		A0090	Enabling Conditional Logic for DEV9
		A0100	Set DEV10 Mode to RCAS
		A0110	Set DEV10 setpoint to RSP for TCV_F21C-1 (°C)
		A0120	Enabling Conditional Logic for DEV11
		A0130	Set Message: Starting Prewash Recirculation Process
T0400A	PF1082C to Filter		If not PF1082C fault and 10secs delay and (F1016Ax is in step <Wash Filling/Wash Recirculate/Core Washing> then go to step S0400
S0400	PF1082C to Filter	A0010	Enabling Conditional Logic for DEV1
		A0020	Enabling Conditional Logic for DEV2
		A0030	Enabling Conditional Logic for DEV3
		A0040	Closing DEV4
		A0050	Enabling Conditional Logic for DEV5
		A0060	Enabling Conditional Logic for DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Enabling Conditional Logic for DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Set DEV10 Mode to RCAS
		A0110	Set DEV10 setpoint to RSP for TCV_F21C-1 (°C)
		A0120	Enabling Conditional Logic for DEV11
		A0130	Set Message: Starting PF1082C to Filter Process
T0500A	Filter Wash / Cool Drain		If F1016Ax is not in step <Wash Filling and Wash Recirculate and Core Washing and Cool Filling and Cool Recirculate> then go to step S0500

Step	Description	Action	Action Description
S0500	Filter Wash / Cool Drain	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Opening DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Set DEV10 Mode to Remote Out
		A0110	Set DEV10 output to Closed
		A0120	Enabling Conditional Logic for DEV11
		A0130	Set Message: Starting Filter Wash / Cool Drain Process
T0700B	Aux Off		If F1016Ax is not in step <Wash Drain and Wash Core Blow and Wash Olein Blow and Core Wash Drain and Core Wash Blow and Cool Drain and Cool Core Blow and Cool Olein Blow> then go to step S0700
S0600	Drain	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Opening DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Set DEV10 Mode to Remote Out

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Step	Description	Action	Action Description
		A0110	Set DEV10 output to Closed
		A0120	Enabling Conditional Logic for DEV11
		A0130	Set Message: Starting Drain Process
T0700A	Aux Off		If F1082B is not in step <Drain F1082C Demand> (F1082B_ACT_STEP) then go to step S0700
S0700	Aux Off	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Closing and Disabling Conditional Logic for DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Opening DEV4
		A0050	Closing and Disabling Conditional Logic for DEV5
		A0060	Closing and Disabling Conditional Logic for DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Closing and Disabling Conditional Logic for DEV8
		A0090	Closing and Disabling Conditional Logic for DEV9
		A0100	Set DEV10 Mode to Remote Out
		A0110	Set DEV10 output to Closed
		A0120	Enabling Conditional Logic for DEV11
T0800	Disable HM and SM		If no pending actions in S0700 then go to step S0800
S0800	Disable HM and SM	A0010	Disable HM 01 (Module Ownership Lost)
T0900	Release Devices		If no pending actions in S0800 then go to step S0900
S0900	Release Devices	A0010	Release Devices Message
		A0020	Release Private Devices
		A0030	Release Private Devices
		A0040	Release Private Devices
		A0050	Release Private Devices
		A0060	Release Private Devices
		A0070	Release Private Devices
		A0080	Release Private Devices
		A0090	Release Private Devices
		A0100	Release Private Devices



Step	Description	Action	Action Description
		A0110	Release Private Devices
		A0120	Release Private Devices
T9900	Complete		If no pending actions in S0900 then go to step S9900
S9900	Complete	A0010	Set Message: Wash Command Complete
		A0020	Self Release the EM & Reset First Pass
END	Command Complete		True

Table 54: Wash Sequence

8.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value/Reference
Operating Parameters				
OP001	Drain CF1_PU13082C Demand	Named Set	NA	NO
OP002	Fill CF1_PU13082C Demand	Named Set	NA	NO
OP003	Empty F1082C	Named Set	NA	NO
Tuning Parameters				
TP001	Delay Timer	Floating Point	sec	5
TP002	Required Wash Level	Floating Point	%	30
TP003	Start Drain Out Level	Floating Point	%	40
TP004	Not Used	NA	NA	NA
TP005	Cool Temp SP	Floating Point	°C	0
TP006	Empty Level	Floating Point	%	0

Table 55: Operating and Tuning Parameters

8.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value/Reference
MY_TAG		CF1EQ_1082C
OPR_CMD_MASK		3
HOLD	External Reference	CF1EQ_1002A/OP003_VALUE

Parameter Name	Parameter Type	Value/Reference
LT_382C1_PV	External Reference	CF1_LT1382C1/AI1/PV
LT_382C1_LL_ALM	External Reference	CF1_LT1382C1/LO_LO_ALM
LSL_383B1	External Reference	CF1_LSL1382B1/DI1/DISC_ACT
LSL_382C1	External Reference	CF1_LSL1382C1/DI1/DISC_ACT
TT_321C1	External Reference	CF1_TT1321C1/AI1/PV
EM_CF_FILT_01	String	CF1EQ_1016A1
EM_CF_FILT_02	String	CF1EQ_1016A2
F1082B_ACT_STEP	External Reference	CF1EQ_1082B/COMMAND_0000 1/ACT_STEPS
DY_REF_1	Dynamic Reference	
DY_REF_2	Dynamic Reference	
DY_REF_3	Dynamic Reference	
DY_REF_4	Dynamic Reference	
DY_REF_5	Dynamic Reference	
DY_REF_6	Dynamic Reference	
DY_REF_7	Dynamic Reference	
DY_REF_8	Dynamic Reference	
DY_REF_9	Dynamic Reference	
DY_REF_10	Dynamic Reference	
F1016AX1	External Reference	CF1EQ_1016A1/COMMAND_000 01/S1600/ACTIVE
F1016AX2	External Reference	CF1EQ_1016A2/COMMAND_000 01/S1600/ACTIVE

Table 56: Configurable Parameters

8.6 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_CF_PRODUCT2	CPKO Fractionation Area: Fractionation Products

Table 57: Named Set Details

Details of states for NS_CF_ PRODUCT2 named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Wash

Table 58: Name Set State Details

9 EQUIPMENT MODULE CLASS DESIGN – EM_CF_PRPDUCT3(F1082B)

9.1 Functional Description

EM_CF_PRPDUCT3 is a class based equipment module that is used for fractionation. EM_CF_PRPDUCT3 is derived from the Command Driven equipment module class. Refer FDS for the functional design requirements including faceplate and detail display. All devices used in EM shall be acquired in Fill Command and shall be released on completion of Fill Command. Hold Request shall be enabled for all acquired devices at the end of HOLD command.

9.2 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name
EM	EM_CF_PRPDUCT3	CF1EQ_1082B
DEV1	_MRSS11_130_SE	CF1_PU13082B
DEV2	_V12_IP_130_SE	CF1_V1382BDO
DEV3	_V12_IP_130_SE	CF1_V1382CDO
DEV4	_V12_IP_130_SE	CF1_V13BDO2
DEV5	_V12_IP_130_SE	CF1_V1382BR
DEV6	_V12_IP_130_SE	CF1_V1382CFO
DEV7	_V12_IP_130_SE	CF1_V13BDO1

Table 59: Instances

9.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

9.3.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Command Hold Start	A0010	Store Fail Message
		A0020	Set Hold Request
T1000	Actions Complete or Time Out & Hold=True		Go to step H1000

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Step	Description	Action	Action Description
T2000	Actions Complete or Time Out & Hold=False		Go to step H2000
H1000	Hold	A0010	Stopping and Disabling Conditional Logic for DEV1
T9900A	Actions Complete or Time Out		Go to step H9900A
H2000	General Reset	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Closing DEV2
		A0030	Closing DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Opening DEV5
		A0060	Closing DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
T9900B	Actions Complete or Time Out		Go to step H9900B
H9900A	Command Hold End	A0010	Completed
END	Command Complete		True
H9900B	Release Devices	A0010	Release Devices Message
		A0020	Release Private Devices
		A0030	Release Private Devices
		A0040	Release Private Devices
		A0050	Release Private Devices
		A0060	Release Private Devices
		A0070	Release Private Devices
		A0080	Release Private Devices
		A0090	Release Private Devices
		A0110	Reset Parameters



Step	Description	Action	Action Description
END1	Command Complete		True

Table 60: Hold Sequence

9.3.2 Command 01 – Fill

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Fill Command
T0010	First Pass = False		If First Pass = False then go to step S0010
T0020A	First Pass = True		If First Pass = True then go to step S0020
S0010	Initializations	A0010	First Pass Logic
		A0020	First Pass = True
T0020B	Acquire Devices		If no pending actions in S0010 then go to step S0020
S0020	Acquire Devices	A0010	Acquire Devices Message
		A0020	Acquire Private Devices
		A0030	Acquire Private Devices
		A0040	Acquire Private Devices
		A0050	Acquire Private Devices
		A0060	Acquire Private Devices
		A0070	Acquire Private Devices
		A0080	Acquire Private Devices
		A0090	Reset HOLD_REQ
T0100	Standby		If Task Pointer = 1 then go to step S0100
T0200B	Filling		If Task Pointer = 2 then go to step S0200
T0300B	Recirculation		If Task Pointer = 3 then go to step S0300
T0400A	Storage		If Task Pointer = 4 then go to step S0400
T0600A	Drain F1082C Demand		If Task Pointer = 5 then go to step S0600
T0500B	Fill F1082C Demand		If Task Pointer = 6 then go to step S0500
S0100	Standby	A0010	Stopping and Disabling Conditional Logc for DEV1
		A0020	Closing DEV2
		A0030	Closing DEV3



Step	Description	Action	Action Description
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Opening DEV5
		A0060	Closing DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Checking Conditions
T0200A	Filling		If no pending actions in S0100 then go to step S0200
S0200	Filling	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Closing DEV2
		A0030	Closing DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Opening DEV5
		A0060	Closing DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Set Message: Starting Filling Process
T0300A	Recirculation		If LT_382B1 >= Stop Drain Out Level (TP003) and not F1082C Pb Drain F1082C Demand or LT_382C1 < F1082C Start Drain Out Level then go to stop S0300
T0600C	Drain F1082C Demand		If F1082C Pb Drain F1082C Demand or LT_382C1 >= F1082C Start Drain Out Level then go to step S0600
S0300	Recirculation	A0010	Enabling Conditional Logic for DEV1
		A0020	Opening DEV2
		A0030	Closing DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Opening DEV5
		A0060	Closing DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Set Message: Starting Recirculation Process
		A0090	Checking Conditions

Step	Description	Action	Action Description
T0400B	Storage		If LT_382C1 < Required Wash Level and LT_382B1 >= F1082B Start Drain Out Level (TP002) then go to step S0400
T0600B	Drain F1082C Demand		If LT_382B1 < F1082B Start Drain Out Level (TP002) and (F1082C Pb Drain F1082C Demand ON or LT_382C1 >= F1082C Start Drain Out Level) then go to step S0600
T0500A	Fill F1082C Demand		If (LT_382C1 < Required Wash Level or F1082C Pb Fill F1082C Demand ON) and LT_382B1 >= F1082B Start Drain Out Level (TP002) and F1016Ax is not in step <Washing and Cooling and Core Wash> then go to step S0500
S0400	Storage	A0010	Enabling Conditional Logic for DEV1
		A0020	Opening DEV2
		A0030	Closing DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Closing DEV5
		A0060	Closing DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Set Message: Starting Storage Process
		A0090	Checking Conditions
T0700A	Aux Off		If LT_382B1 <= Stop Drain Out Level (TP003) + 2 then go to step S0700
S0500	Fill F1082C Demand	A0010	Enabling Conditional Logic for DEV1
		A0020	Opening DEV2
		A0030	Closing DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Closing DEV5
		A0060	Opening DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
		A0080	Set Message: Starting Fill F1082C Demand Process
		A0090	Checking Conditions
T0700C	Aux Off		If LT_382C1 >= F1082C Required Wash Level + 2 or LSH_382C1 triggered or LSL_382B1 triggered then go to step S0700

Step	Description	Action	Action Description
S0600	Drain F1082C Demand	A0010	Enabling Conditional Logic for DEV1
		A0020	Closing DEV2
		A0030	Opening DEV3
		A0040	Enabling Conditional Logic for DEV4
		A0050	Closing DEV5
		A0060	Closing DEV6
		A0070	Enabling Conditional Logic for DEV7
		A0080	Set Message: Starting Drain F1082C Demand Process
		A0090	Checking Conditions
T0700B	Aux Off		If (Pb Empty F1082C is OFF and LT_382C1 <= F1082C Required Wash Level + 2 or LSL_382C1 triggered) or (Pb Empty F1082C is ON and LT_382C1 < F1082C Empty Level or LSL_382C1 triggered) then go to step S0700
S0700	Aux off	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Closing DEV2
		A0030	Closing DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Opening DEV5
		A0060	Closing DEV6
		A0070	Closing and Disabling Conditional Logic for DEV7
T0800	Disable HM and SM		If no pending actions in S0700 then go to step S0800
S0800	Disable HM and SM	A0010	Disable HM 01 (Module Ownership Lost)
		A0020	Resetting Dynamic Reference
T0900	Release Devices		If no pending actions in S0800 then go to step S0900
S0900	Release Devices	A0010	Release Devices Message
		A0020	Release Private Devices
		A0030	Release Private Devices
		A0040	Release Private Devices

Step	Description	Action	Action Description
		A0050	Release Private Devices
		A0060	Release Private Devices
		A0070	Release Private Devices
		A0080	Release Private Devices
T9900	Complete		If no pending actions in S0900 then go to step S9900
S9900	Complete	A0010	Set Message: Fill Command Complete
		A0020	Self Release EM
		A0030	Reset First Pass
END	Command Complete		True

Table 61: Fill Sequence

9.4

Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value/Reference
Operating Parameters				
OP001	Drain to storage	Named Set	NA	PRODUCT
Tuning Parameters				
TP001	Enable Filtration at Level	Floating Point	%	30
TP002	Start Drain Out Level	Floating Point	%	40
TP003	Stop Drain Out Level	Floating Point	%	20

Table 62: Operating and Tuning Parameters

9.5

Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value/Reference
MY_TAG	String	CF1EQ_1082B
OPR_CMD_MASK	32 bit unsigned integer	3
LT_382B1_PV	External Reference	CF1_LT1382B1/PV
LT_382B1_LL_ALM	External Reference	CF1_LT1382B1/LO_LO_ALM
LT_382C1_PV	External Reference	CF1_LT1382C1/PV

Parameter Name	Parameter Type	Value/Reference
LT_382C1_LL_ALM	External Reference	CF1_LT1382C1/LO_LO_ALM
LSL_382B1	External Reference	CF1_LSL1382B1/DI1/DISC_ACT
LSL_382C1	External Reference	CF1_LSL1382C1/DI1/DISC_ACT
LSH_382C1	External Reference	CF1_LSH1382C1/DI1/DISC_ACT
EM_CF_FILT_01	String	CF1EQ_1016A1
EM_CF_FILT_02	String	CF1EQ_1016A2
EM_CF_PRODUCT2	String	CF1EQ_1082C
HOLD	External Reference	CF1EQ_1002A/OP003_VALUE
DY_REF_1	Dynamic Reference	
DY_REF_2	Dynamic Reference	
DY_REF_3	Dynamic Reference	
DY_REF_4	Dynamic Reference	
DY_REF_5	Dynamic Reference	
DY_REF_6	Dynamic Reference	
DY_REF_7	Dynamic Reference	
DY_REF_8	Dynamic Reference	
DY_REF_9	Dynamic Reference	
DY_REF_10	Dynamic Reference	

Table 63: Configurable Parameters

9.6

Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_CF_PRDUCT3	CPKO Fractionation Area: Fractationation

Table 64: Named Set Details

Details of states for NS_CF_PRODUCT3 named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Fill

Table 65: Name Set State Details

10 EQUIPMENT MODULE CLASS DESIGN – EM_CF_PRODUCT4 (F1046OV)

10.1 Functional Description

EM_CF_PRODUCT4 is a class based equipment module that is used for fractionation. EM_CF_PRODUCT4 is derived from the Command Driven equipment module class EM_CF_PRODUCT4. Refer FDS for the functional design requirements including faceplate and detail display. All devices used in EM shall be acquired in Fill Command and shall be released on completion of the same. Hold Request shall be enabled for all acquired devices at the end of Hold command.

10.2 Instances

The following Table contains the module instances of the equipment module class.

Device Name	Module Class	Instance Name
EM	EM_CF_PRODUCT4	CFXE_1046OV
DEV1	_MRSS11_130_SE	CF1_PU1316OV
DEV2	_V12_IP_130_SE	CF1_V13460VDO
DEV3	_V12_IP_130_SE	CF1_V13460V1
DEV4	_V12_IP_130_SE	CF1_V13460VBA

Table 66: Instances

10.3 Control Logic

The following control logic will be implemented for equipment module class. Unless specified, the control logic will be implemented as part of the SFC for each equipment module command.

10.3.1 Command 00 – Hold

Step	Description	Action	Action Description
H0010	Command Hold Start	A0010	Store Fail Message
		A0020	Reset Hold Request
		A0030	Save Previous Step Time
T1000	Actions Complete or Time Out & Hold=True		Go to step H1000
T2000	Actions Complete or Time Out & Hold=False		Go to step H2000

Step	Description	Action	Action Description
H1000	Hold	A0010	Stopping and Disabling Conditional Logic for DEV1
T9900A	Actions Complete or Time Out		Go to step H9900A
H2000	General Reset	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Closing DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
T9900B	Actions Complete or Time Out		Go to step H9900B
H9900A	Command Hold End	A0010	Completed
END	Command Complete		True
H9900B	Release Devices	A0010	Release Devices Message
		A0020	Release Private Devices
		A0030	Release Private Devices
		A0040	Release Private Devices
		A0050	Release Private Devices
		A0060	Reset Parameters
END1	Command Complete		True

Table 67: Hold Sequence

10.3.2 Command 01 – Fill

Step	Description	Action	Action Description
S0000	Start	A0010	Set Message: Starting Fill Command
T0010	First Pass = False		If First Pass = False then go to step S0010
T0020A	First Pass = True		If First Pass = True then go to step S0020
S0010	Initializations	A0010	First Pass Logic
		A0020	First Pass = True
T0020B	Acquire Devices		If no pending actions in S0010 then go to step S0020

Step	Description	Action	Action Description
S0020	Acquire Devices	A0010	Acquire Devices Message
		A0020	Acquire Private Devices
		A0030	Acquire Private Devices
		A0040	Acquire Private Devices
		A0050	Acquire Private Devices
		A0060	Reset HOLD_REQ
		A0070	Set Message: Reset Actual Blow Count
T0100	Standby		If Task Pointer = 1 then go to step S0100
T0200B	Filling		If Task Pointer = 2 then go to step S0200
T0300B	Drain		If Task Pointer = 3 then go to step S0300
T0400C	Blow On		If Task Pointer = 4 then go to step S0400
S0100	Standby	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Closing DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Set Message: Starting Fill Command
		A0060	Resetting Actual Blow Count and timers
T0200A	Filling		If no pending actions in S0100 then go to step S0200
S0200	Filling	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Closing DEV2
		A0030	Enabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Set Message: Starting Filling Process
		A0060	Checking Conditions
T0300A	Drain		If Pb Drain (OP001) is ON or (LSM_360V1 triggered & TT_360V1 >= Heating Temp SP(TP001)) or LSH_360V1 triggered then go to step S0300
S0300	Drain	A0010	Enabling Conditional Logic for DEV1
		A0020	Open DEV2
		A0030	Enabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Set Message: Starting Drain Process

Step	Description	Action	Action Description
		A0060	Checking Conditions
T0400A	Blow On		If LSL_360V1 is in alarm then go to step S0400
S0400	Blow On	A0010	Stopping and Disabling Conditional Logic for DEV1 (DEV1)
		A0020	Closing DEV2
		A0030	Enabling Conditional Logic for DEV3 (DEV3)
		A0040	Enabling Conditional Logic for DEV4 (DEV4)
		A0050	Pausing Blow Timer
		A0060	Set Message
T0500	Blow OFF		If Blow ON Time (TP002) Over then go to step S0500
S0500	Blow OFF	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Closing DEV2
		A0030	Enabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
		A0050	Set Blow off Time
		A0060	Start Blow off Time
		A0070	Blow off Time Message
		A0080	Set Message: Starting Blow OFF Process
		A0090	Increment Actual Blow Count
		A0100	Set Message
T0400B	Blow On		If Blow OFF Time (TP003) Over & Actual Blow count (TP005) < Preset Blow Count (TP004) then go to step S0400
T0600	Aux Off		If Blow OFF Time (TP003) Over & Actual Blow count (TP005) >= Preset Blow Count (TP004) then go to step S0600
S0600	Aux Off	A0010	Stopping and Disabling Conditional Logic for DEV1
		A0020	Closing DEV2
		A0030	Closing and Disabling Conditional Logic for DEV3
		A0040	Closing and Disabling Conditional Logic for DEV4
T0700	Disable HM and SM		Go to step S0700
S0700	Disable HM and SM	A0010	Disable HM 01 (Module Ownership Lost)

Step	Description	Action	Action Description
T0800	Release Devices		Go to step S0800
S0800	Release Devices	A0010	Release Devices Message
		A0020	Release Private Devices
		A0030	Release Private Devices
		A0040	Release Private Devices
		A0050	Release Private Devices
T9900	Complete		Go to step S9900
S9900	Complete	A0010	Set Message: Fill Command Complete
		A0020	Self Release EM
		A0030	Reset First Pass
END	Command Complete		True

Table 68: Fill Sequence

10.4 Parameters

The following table defines the operating, tuning parameters that provide an interface to this equipment module:

Parameter	Description	Type	Units	Default Value/Reference
Operating Parameters				
OP001	Drain	Named Set		OFF
Tuning Parameters				
TP001	Heating Temp	Floating point	°C	50
TP002	Blow On time	Floating point	sec	20
TP003	Blow Off time	Floating point	sec	5
TP004	Preset Count	Floating point	NA	10
TP005	Actual Count	Floating point	sec	0
TP006	Remaining Time	Floating point	sec	0

Table 69: Operating and Tuning Parameters

10.5 Configurable Parameters

The following table provides the information of configurable parameters for the equipment module instances:

Parameter Name	Parameter Type	Value/Reference
MY_TAG	String	CFXEQ_1046OV

Parameter Name	Parameter Type	Value/Reference
OPR_CMD_MASK	32 bit unsigned integer	3
TT_360V1	External Reference	CF1_TT1360V1/PV
LSL_360V1	External Reference	CF1_LSL1360V1/DI1/DISC_ACT
LSM_360V1	External Reference	CF1_LSM1360V1/DI1/DISC_ACT
LSH_360V1	External Reference	CF1_LSH1360V1/DI1/DISC_ACT
HOLD	External Reference	CF1EQ_1002A/OP003_VALUE
COMPLETED_TIME	Floating Point	0
RESET_COMP_TM	Boolean	False
STEP_TIME	Floating Point	0

Table 70: Configurable Parameters

10.6 Named Set

Following Named set shall be used for configuration of EM.

Name	Description
NS_CF_PRODUCT4	CPKO Fractionation Area: Fractationation

Table 71: Named Set Details

Details of states for NS_CF_PRODUCT4 named set are as follows,

Named Set Value	Named Set Name
00	Hold
01	Fill

Table 72: Name Set State Details

11 MODULE INTERLOCKS

11.1 Functional Description

Module	Name	Description	Expression	Force SP State	Delay On (s)
CF1_PU1478D	Condition 1	Auto Logic	'/MODE.ACTUAL' = CAS AND<CR>'//CF1EQ_1002A/OP004_VALUE.CV' = TRUE AND <CR>'//CF1_LSL1478D1/DI1/DISC_ACT.CV' != TRUE)	START	NO
CF1_PU1478D	Condition 2	Auto Logic	'/MODE.ACTUAL' = CAS AND<CR>'//CF1EQ_1002A/OP004_VALUE.CV' != TRUE OR <CR>'//CF1_LSL1478D1/DI1/DISC_ACT.CV' = TRUE)	STOP	NO
CF1_PU1478C1	Condition 1	Auto Logic	'/MODE.ACTUAL' = CAS AND<CR>'//CF1EQ_1002A/OP004_VALUE.CV' = TRUE AND <CR>'//CF1_LSL1478C/DI1/DISC_ACT.CV' != TRUE) AND ('/CF_ESM_1/PF1078C1_C2.CV' = 'NS_F1078B_C_CMD:PF1078C1')	START	NO
CF1_PU1478C1	Condition 2	Auto Logic	'/MODE.ACTUAL' = CAS AND<CR>('//CF1EQ_1002A/OP004_VALUE.CV' != TRUE OR <CR>'//CF1_LSL1478C/DI1/DISC_ACT.CV' = TRUE OR '//CF_ESM_1/PF1078C1_C2.CV' != 'NS_F1078B_C_CMD:PF1078C1')	STOP	NO
CF1_PU125016	Condition 1	Auto Logic	'/MODE.ACTUAL' = CAS AND<CR>'//CF1EQ_1002A/OP004_VALUE.CV' = TRUE AND<CR>'//CF1_LT1278161/AI1/LO_LO_ACT.CV' != TRUE	START	NO
CF1_PU125016	Condition 2	Auto Logic	'/MODE.ACTUAL' = CAS AND<CR>'//CF1EQ_1002A/OP004_VALUE.CV' != TRUE OR <CR>'//CF1_LT1278161/AI1/LO_LO_ACT.CV' = TRUE	STOP	NO
CF1_PU1116AS	Condition 1	Auto Logic	(/TP01.CV' = TRUE)AND <CR>'//CF1_PT1116AS1/AI1/HI_HI_ACT.CV' != TRUE AND	START	NO

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Module	Name	Description	Expression	Force SP State	Delay On (s)
			<CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' != TRUE AND <CR>'//CF1_LSL1109S1/DI1/DISC_ACT.CV' != TRUE AND <CR>'//CF1_LT1382B1/PV.CV' <= '//CF1EQ_1082B/TP001_VALUE.CV' AND <CR>(('//CF1_V1216A1FO/PV_D' = '_EDC_VLV_PV:OPEN' OR '//CF1_V1216A2FO/PV_D' = '_EDC_VLV_PV:OPEN') AND <CR>(('//CF1_V1216A101/PV_D' = '_EDC_VLV_PV:OPEN' OR '//CF1_V1216A2F1/PV_D' = '_EDC_VLV_PV:OPEN') AND<CR>(('//CF1_V1216A102/PV_D' = '_EDC_VLV_PV:OPEN' OR '//CF1_V1261A2F2/PV_D' = '_EDC_VLV_PV:OPEN') AND<CR>(('//CF1_V1216A1DH/PV_D' = '_EDC_VLV_PV:OPEN' OR '//CF1_V1216A2DH/PV_D' = '_EDC_VLV_PV:OPEN') OR<CR>(('//CF1_V1216A1OL/PV_D' = '_EDC_VLV_PV:OPEN' OR '//CF1_V1216A2DL/PV_D' = '_EDC_VLV_PV:OPEN'))))		
CF1_PU1116AS	Condition 2	Auto Logic	('TP01.CV' = TRUE)AND <CR>(('//CF1_PT1116AS1/AI1/HI_HI_ACT.CV' = TRUE OR <CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' = TRUE OR<CR>'//CF1_LSL1109S1/DI1/DISC_ACT.CV' = TRUE OR <CR>'//CF1_LT1382B1/PV.CV' > '//CF1EQ_1082B/TP001_VALUE.CV' OR <CR>(('//CF1_V1216A1FO/PV_D' = '_EDC_VLV_PV:CLOSED' AND '//CF1_V1216A2FO/PV_D' = '_EDC_VLV_PV:CLOSED') OR <CR>(('//CF1_V1216A101/PV_D' = '_EDC_VLV_PV:CLOSED' AND '//CF1_V1216A2F1/PV_D' = '_EDC_VLV_PV:CLOSED') OR<CR>(('//CF1_V1216A102/PV_D' = '_EDC_VLV_PV:CLOSED' AND '//CF1_V1261A2F2/PV_D' = '_EDC_VLV_PV:CLOSED') OR<CR>(('//CF1_V1216A1DH/PV_D' = '_EDC_VLV_PV:CLOSED' AND '//CF1_V1216A2DH/PV_D' = '_EDC_VLV_PV:CLOSED' AND '//CF1_V1216A1OL/PV_D' = '_EDC_VLV_PV:CLOSED' AND '//CF1_V1216A2DL/PV_D' = '_EDC_VLV_PV:CLOSED'))))	STOP	NO

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CF1_PF1078C2	Condition 1	Auto Logic	'/MODE.ACTUAL' = CAS AND<CR>'//CF_ESM_1/RUN_AUX_CKT.CV' = TRUE AND <CR>'//CF1_LSL1478C/DI1/DISC_ACT.CV' != TRUE) AND ('//CF_ESM_1/PF1078C1_C2.CV' = 'NS_F1078B_C_CMD:PF1078C2')	START	NO
CF1_PF1078C2	Condition 2	Auto Logic	'/MODE.ACTUAL' = CAS AND<CR>('//CF_ESM_1/RUN_AUX_CKT.CV' != TRUE OR <CR>'//CF1_LSL1478C/DI1/DISC_ACT.CV' = TRUE OR '//CF_ESM_1/PF1078C1_C2.CV' = 'NS_F1078B_C_CMD:PF1078C2')	STOP	NO
CF1_FN127816 2	Condition 1	Auto Logic	('/TP01.CV' = TRUE) AND<CR>'//CF1_LT1382B1/PV' <='//CF1EQ_1082B/TP001_VALUE.CV' AND <CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' != TRUE AND <CR>'//CF1EQ_1016A2/OP006_VALUE.CV' = '_DO_OFF_ON:ON' AND <CR>'//CF1_V1216A2DL/PV_D' = '_EDC_VLV_PV:OPEN' OR <CR>'//CF1_V1216A2DH/PV_D' = '_EDC_VLV_PV:OPEN';	START	NO
CF1_FN127816 2	Condition 2	Auto Logic	('/TP01.CV' = TRUE) AND<CR>'//CF1_LT1382B1/PV' >'//CF1EQ_1082B/TP001_VALUE.CV' OR <CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' = TRUE OR <CR>'//CF1EQ_1016A2/OP006_VALUE.CV' = '_DO_OFF_ON:OFF' OR <CR>'//CF1_V1216A2DL/PV_D' = '_EDC_VLV_PV:CLOSED' AND <CR>'//CF1_V1216A2DH/PV_D' = '_EDC_VLV_PV:CLOSED';	STOP	NO
CF1_FN127816 2	Condition 3	Auto Logic	('/TP02.CV' = TRUE) AND<CR>'//CF1EQ_1016A2/OP006_VALUE.CV' = '_DO_OFF_ON:ON' AND <CR>'//CF1_LT1382B1/PV' <='//CF1EQ_1082B/TP001_VALUE.CV' AND <CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' != TRUE AND <CR>'//CF1_V1216A2DL/PV_D' = '_EDC_VLV_PV:OPEN' OR <CR>'//CF1_V1216A2DH/PV_D' = '_EDC_VLV_PV:OPEN';	START	NO
CF1_FN127816 2	Condition 4	Auto Logic	('/TP02.CV' = TRUE) AND<CR>'//CF1EQ_1016A2/OP006_VALUE.CV' = '_DO_OFF_ON:OFF' OR <CR>'//CF1_LT1382B1/PV' >'//CF1EQ_1082B/TP001_VALUE.CV'	STOP	NO

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Module	Name	Description	Expression	Force SP State	Delay On (s)
			OR<CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' = TRUE OR<CR>'//CF1_V1216A1OL/PV_D' = '_EDC_VLV_PV:CLOSED' AND<CR>'//CF1_V1216A1DH/PV_D' = '_EDC_VLV_PV:CLOSED';		
CF1_FN127816 1	Condition 1	Auto Logic	('/TP01.CV' = TRUE) AND<CR>'//CF1_LT1382B1/PV' <='//CF1EQ_1082B/TP001_VALUE.CV' AND <CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' != TRUE AND <CR>'//CF1EQ_1016A1/OP006_VALUE.CV' = '_DO_OFF_ON:ON' AND <CR>'//CF1_V1216A1OL/PV_D' = '_EDC_VLV_PV:OPEN' OR <CR>'//CF1_V1216A1DH/PV_D' = '_EDC_VLV_PV:OPEN';	START	NO
CF1_FN127816 1	Condition 2	Auto Logic	('/TP01.CV' = TRUE) AND<CR>'//CF1_LT1382B1/PV' >'//CF1EQ_1082B/TP001_VALUE.CV' OR <CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' = TRUE OR <CR>'//CF1EQ_1016A1/OP006_VALUE.CV' = '_DO_OFF_ON:OFF' OR <CR>'//CF1_V1216A1OL/PV_D' = '_EDC_VLV_PV:CLOSED' AND<CR>'//CF1_V1216A1DH/PV_D' = '_EDC_VLV_PV:CLOSED';	STOP	NO
CF1_FN127816 1	Condition 3	Auto Logic	('/TP02.CV' = TRUE) AND<CR>'//CF1EQ_1016A1/OP006_VALUE.CV' = '_DO_OFF_ON:ON' AND <CR>'//CF1_LT1382B1/PV' <='//CF1EQ_1082B/TP001_VALUE.CV' AND <CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' != TRUE AND <CR>'//CF1_V1216A1OL/PV_D' = '_EDC_VLV_PV:OPEN' OR <CR>'//CF1_V1216A1DH/PV_D' = '_EDC_VLV_PV:OPEN';	START	NO
CF1_FN127816 1	Condition 4	Auto Logic	('/TP02.CV' = TRUE) AND<CR>'//CF1EQ_1016A1/OP006_VALUE.CV' = '_DO_OFF_ON:OFF' OR <CR>'//CF1_LT1382B1/PV' >'//CF1EQ_1082B/TP001_VALUE.CV' OR <CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' = TRUE OR<CR>'//CF1_V1216A1OL/PV_D' = '_EDC_VLV_PV:CLOSED' AND<CR>'//CF1_V1216A1DH/PV_D' = '_EDC_VLV_PV:CLOSED';	STOP	NO
CF1_PU1310ST	Condition 1	Auto Logic	('/TP01.CV' = TRUE) AND <CR>'//CF1_V1382STDO/PV_D' = '_EDC_VLV_PV:OPEN' AND	START	NO

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			<CR>'//CF1_LSL1382ST/DI1/DISC_ACT.CV' != TRUE AND <CR>'//CF1_LT1382ST1/AI1/LO_LO_ACT.CV' != TRUE)		
CF1_PU1310ST	Condition 2	Auto Logic	('/TP01.CV' = TRUE) AND <CR>'//CF1_V1382STDO/PV_D' = '_EDC_VLV_PV:CLOSED' OR<CR>'//CF1_LSL1382ST/DI1/DISC_ACT.CV' = TRUE OR <CR>'//CF1_LT1382ST1/AI1/LO_LO_ACT.CV' = TRUE)	STOP	NO
CF1_PU13082C	Condition 1	Auto Logic	('/TP01.CV' = TRUE)AND<CR>'//CF1_V1382CDOC/PV_D' = '_EDC_VLV_PV:OPEN' AND '/CF1_V1382CRO/PV_D' = '_EDC_VLV_PV:OPEN' AND<CR>'//CF1_LSL1382C1/DI1/DISC_ACT.CV' = FALSE AND '//CF1_LT1382C1/AI1/LO_LO_ACT.CV' = FALSE)	START	NO
CF1_PU13082C	Condition 2	Auto Logic	('/TP01.CV' = TRUE) AND<CR>'//CF1_V1382CDOC/PV_D' = '_EDC_VLV_PV:CLOSED' OR '/CF1_V1382CRO/PV_D' = '_EDC_VLV_PV:CLOSED' OR<CR>'//CF1_LSL1382C1/DI1/DISC_ACT.CV' = TRUE OR '//CF1_LT1382C1/AI1/LO_LO_ACT.CV' = TRUE)	STOP	NO
CF1_PU13082C	Condition 3	Auto Logic	('/TP02.CV' = TRUE)AND <CR>(((('/CF1EQ_1016A1/EN_WASHING_STEP.CV' = TRUE OR '//CF1EQ_1016A1/EN_COREWASH_STEP.CV' = TRUE OR '//CF1EQ_1016A2/EN_WASHING_STEP.CV' = TRUE OR '//CF1EQ_1016A2/EN_COREWASH_STEP.CV' = TRUE) AND <CR>'//CF1_LSL1382C1/DI1/DISC_ACT.CV' != TRUE AND '//CF1_V1382CDOC/PV_D' = '_EDC_VLV_PV:OPEN'))<CR> OR <CR>(((('/CF1EQ_1016A1/EN_COOLING_STEP.CV' = TRUE OR '//CF1EQ_1016A2/EN_COOLING_STEP.CV' = TRUE) AND '//CF1_LSL1382B1/DI1/DISC_ACT.CV' != TRUE AND '//CF1_V13BDOB/PV_D' = '_EDC_VLV_PV:OPEN'))	START	NO
CF1_PU13082C	Condition 4	Auto Logic	('//CF1_V1216A1CH/PV_D' = '_EDC_VLV_PV:CLOSED' AND '//CF1_V1216A1CL/PV_D' = '_EDC_VLV_PV:CLOSED' AND '//CF1EQ_1016A1/EN_COOLING_STEP.CV' = TRUE)	STOP	NO

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Module	Name	Description	Expression	Force SP State	Delay On (s)
			OR<CR>('//CF1_V1216A1DH/PV_D' = ' _EDC_VLV_PV:CLOSED' AND '//CF1_V1216A1OL/PV_D' = ' _EDC_VLV_PV:CLOSED' AND '//CF1EQ_1016A1/EN_WASHING_STEP.CV' = TRUE) OR<CR>('//CF1_V1216A1FC/PV_D' = ' _EDC_VLV_PV:CLOSED' AND '//CF1_V1216A1FO/PV_D' = ' _EDC_VLV_PV:CLOSED' AND '//CF1EQ_1016A1/EN_COREWASH_STEP' = TRUE) OR<CR>('//CF1_V1216A2CO/PV_D' = ' _EDC_VLV_PV:CLOSED' AND '//CF1_V1216A2CL/PV_D' = ' _EDC_VLV_PV:CLOSED' AND '//CF1EQ_1016A2/EN_COOLING_STEP.CV' = TRUE) OR<CR>('//CF1_V1216A2DH/PV_D' = ' _EDC_VLV_PV:CLOSED' AND '//CF1_V1216A2DL/PV_D' = ' _EDC_VLV_PV:CLOSED' AND '//CF1EQ_1016A2/EN_WASHING_STEP.CV' = TRUE) OR<CR>('//CF1_V1216A2FC/PV_D' = ' _EDC_VLV_PV:CLOSED' AND '//CF1_V1216A2FO/PV_D' = ' _EDC_VLV_PV:CLOSED' AND '//CF1EQ_1016A2/EN_COREWASH_STEP' = TRUE)		
CF1_PU13082B	Condition 1	Auto Logic	('/TP01.CV' = TRUE) AND <CR>('//CF1_LSL1382B1/DI1/DISC_ACT' != TRUE AND <CR>'//CF1_LT1382B1/AI1/LO_LO_ACT.CV' != TRUE) AND<CR>'//CF1_V1382BDO/PV_D.CV' = ' _EDC_VLV_PV:OPEN' AND <CR>'//CF1_V1382BR/PV_D.CV' = ' _EDC_VLV_PV:OPEN')	START	NO
CF1_PU13082B	Condition 2	Auto Logic	('/TP01.CV' = TRUE) AND <CR>('//CF1_LSL1382B1/DI1/DISC_ACT' = TRUE OR <CR>'//CF1_LT1382B1/AI1/LO_LO_ACT.CV' = TRUE) OR<CR>'//CF1_V1382BDO/PV_D.CV' = ' _EDC_VLV_PV:CLOSED' OR <CR>'//CF1_V1382BR/PV_D.CV' = ' _EDC_VLV_PV:CLOSED')	STOP	NO
CF1_PU13082B	Condition 3	Auto Logic	('/TP02.CV' = TRUE) AND <CR>('//CF1_LSL1382B1/DI1/DISC_ACT' != TRUE AND <CR>'//CF1_LT1382B1/AI1/LO_LO_ACT.CV' != TRUE) AND<CR>'//CF1_V1382BDO/PV_D.CV' = ' _EDC_VLV_PV:OPEN');	START	NO
CF1_PU13082B	Condition 4	Auto Logic	('/TP02.CV' = TRUE) AND <CR>('//CF1_LSL1382B1/DI1/DISC_ACT' = TRUE OR<CR>'//CF1_LT1382B1/AI1/LO_LO_ACT.CV' = TRUE) OR<CR>'//CF1_V1382BDO/PV_D.CV' = ' _EDC_VLV_PV:CLOSED');	STOP	NO

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Module	Name	Description	Expression	Force SP State	Delay On (s)
CF1_PU1119S1	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>('CF1_LSH1109S1/DI1/DISC_ACT.CV' != TRUE)	START LEFT	NO
CF1_PU1119S1	Condition 2	Auto Logic	('TP01.CV' = TRUE OR 'TP02.CV' = TRUE) AND <CR>('CF1_LSH1109S1/DI1/DISC_ACT.CV' = TRUE)	STOP	NO
CF1_PU1119S1	Condition 3	Auto Logic	('TP02.CV' = TRUE) AND <CR>('CF1_LSH1109S1/DI1/DISC_ACT.CV' != TRUE)	START RIGHT	NO
CF1_PU1109S2	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>('CF1_LSH1109S1/DI1/DISC_ACT.CV' != TRUE)	START LEFT	NO
CF1_PU1109S2	Condition 2	Auto Logic	('TP01.CV' = TRUE OR 'TP02.CV' = TRUE) AND <CR>('CF1_LSH1109S1/DI1/DISC_ACT.CV' = TRUE)	STOP	NO
CF1_PU1109S2	Condition 3	Auto Logic	('TP02.CV' = TRUE) AND <CR>('CF1_LSH1109S1/DI1/DISC_ACT.CV' != TRUE)	START RIGHT	NO
CF1_FN1119S1	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>('CF1_LSH1109S1/DI1/DISC_ACT.CV' != TRUE)	START	NO
CF1_FN1119S1	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>('CF1_LSH1109S1/DI1/DISC_ACT.CV' = TRUE)	STOP	NO
CF1_FN1109S2	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>('CF1_LSH1109S1/DI1/DISC_ACT.CV' != TRUE)	START	NO
CF1_FN1109S2	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>('CF1_LSH1109S1/DI1/DISC_ACT.CV' = TRUE)	STOP	NO
CF1_PU1316OV	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>('CF1_V13460VDO/PV_D' = 'EDC_VLV_PV:OPEN' AND <CR>('CF1_V13460VBA/PV_D' = 'EDC_VLV_PV:CLOSED'))	START	NO
CF1_PU1316OV	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>('CF1_V13460VDO/PV_D' = 'EDC_VLV_PV:CLOSED' OR <CR>('CF1_V13460VBA/PV_D' = 'EDC_VLV_PV:OPEN'))	STOP	NO
CF1_PU127816 2	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND<CR>('CF1_LT1382B1/PV' <='CF1EQ_1082B/TP001_VALUE.CV' AND <CR>('CF1_LSH1382B1/DI1/DISC_ACT.CV' != TRUE AND	START	NO

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Module	Name	Description	Expression	Force SP State	Delay On (s)
			<CR>'//CF1EQ_1016A2/OP006_VALUE.CV' = '_DO_OFF_ON:ON') AND <CR>'//CF1_V1216A2DL/PV_D' = '_EDC_VLV_PV:OPEN' OR <CR>'//CF1_V1216A2DH/PV_D' = '_EDC_VLV_PV:OPEN');		
CF1_PU127816 2	Condition 2	Auto Logic	('//TP01.CV' = TRUE) AND<CR>'//CF1_LT1382B1/PV' >'//CF1EQ_1082B/TP001_VALUE.CV' OR <CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' = TRUE OR <CR>'//CF1EQ_1016A2/OP006_VALUE.CV' = '_DO_OFF_ON:OFF')OR<CR>'//CF1_V1216A2DL/PV_D' = '_EDC_VLV_PV:CLOSED' AND <CR>'//CF1_V1216A2DH/PV_D' = '_EDC_VLV_PV:CLOSED');	STOP	NO
CF1_PU127816 2	Condition 3	Auto Logic	('//TP02.CV' = TRUE) AND<CR>'//CF1EQ_1016A2/OP006_VALUE.CV' = '_DO_OFF_ON:ON' AND <CR>'//CF1_LT1382B1/PV' <='//CF1EQ_1082B/TP001_VALUE.CV' AND <CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' != TRUE)AND <CR>'//CF1_V1216A2DL/PV_D' = '_EDC_VLV_PV:OPEN' OR <CR>'//CF1_V1216A2DH/PV_D' = '_EDC_VLV_PV:OPEN');	START	NO
CF1_PU127816 2	Condition 4	Auto Logic	('//TP02.CV' = TRUE) AND<CR>'//CF1EQ_1016A2/OP006_VALUE.CV' = '_DO_OFF_ON:OFF' OR <CR>'//CF1_LT1382B1/PV' >'//CF1EQ_1082B/TP001_VALUE.CV' OR<CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' = TRUE) OR <CR>'//CF1_V1216A2DL/PV_D' = '_EDC_VLV_PV:CLOSED' AND <CR>'//CF1_V1216A2DH/PV_D' = '_EDC_VLV_PV:CLOSED');	STOP	NO
CF1_PU127816 2	Condition 5	Auto Logic	('//TP03.CV' = TRUE) AND<CR>'//CF1EQ_1016A2/OP006_VALUE.CV' = '_DO_OFF_ON:ON' AND ('//CF1EQ_1016A2/OP005_VALUE.CV' = 'NS_COOLING_TYPE:Water and Oil' OR <CR>'//CF1EQ_1016A2/OP005_VALUE.CV' = 'NS_COOLING_TYPE:Water'))	START	NO

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Module	Name	Description	Expression	Force SP State	Delay On (s)
CF1_PU127816 2	Condition 6	Auto Logic	('TP03.CV' = TRUE) AND<CR>('//CF1EQ_1016A2/OP006_VALUE.CV' = '_DO_OFF_ON:OFF' OR ('//CF1EQ_1016A2/OP005_VALUE.CV' = 'NS_COOLING_TYPE:Water and Oil' OR <CR>'//CF1EQ_1016A2/OP005_VALUE.CV' = 'NS_COOLING_TYPE:Water'))	STOP	NO
CF1_PU127816 1	Condition 1	Auto Logic	('TP01.CV' = TRUE)AND<CR>('//CF1_LT1382B1/PV' <='//CF1EQ_1082B/TP001_VALUE.CV' AND <CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' != TRUE AND <CR>'//CF1EQ_1016A1/OP006_VALUE.CV' = '_DO_OFF_ON:ON') AND <CR>('//CF1_V1216A1OL/PV_D' = '_EDC_VLV_PV:OPEN' OR <CR>'//CF1_V1216A1DH/PV_D' = '_EDC_VLV_PV:OPEN');	START	NO
CF1_PU127816 1	Condition 2	Auto Logic	('TP01.CV' = TRUE)AND<CR>('//CF1_LT1382B1/PV' >'//CF1EQ_1082B/TP001_VALUE.CV' OR <CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' = TRUE OR <CR>'//CF1EQ_1016A1/OP006_VALUE.CV' = '_DO_OFF_ON:OFF') OR<CR>('//CF1_V1216A1OL/PV_D' = '_EDC_VLV_PV:CLOSED' AND '//CF1_V1216A1DH/PV_D' = '_EDC_VLV_PV:CLOSED');	STOP	NO
CF1_PU127816 1	Condition 3	Auto Logic	('TP02.CV' = TRUE) AND<CR>('//CF1EQ_1016A1/OP006_VALUE.CV' = '_DO_OFF_ON:ON' AND <CR>'//CF1_LT1382B1/PV' <='//CF1EQ_1082B/TP001_VALUE.CV' AND <CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' != TRUE)AND <CR>('//CF1_V1216A1OL/PV_D' = '_EDC_VLV_PV:OPEN' OR <CR>'//CF1_V1216A1DH/PV_D' = '_EDC_VLV_PV:OPEN');	START	NO
CF1_PU127816 1	Condition 4	Auto Logic	('TP02.CV' = TRUE)AND<CR>('//CF1EQ_1016A1/OP006_VALUE.CV' = '_DO_OFF_ON:OFF' OR <CR>'//CF1_LT1382B1/PV' >'//CF1EQ_1082B/TP001_VALUE.CV' OR <CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' = TRUE	STOP	NO

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Module	Name	Description	Expression	Force SP State	Delay On (s)
			)OR<CR>('//CF1_V1216A1OL/PV_D' = ' _EDC_VLV_PV:CLOSED' AND<CR>'//CF1_V1216A1DH/PV_D' = ' _EDC_VLV_PV:CLOSED');		
CF1_PU127816 1	Condition 5	Auto Logic	('TP03.CV' = TRUE)AND<CR>('//CF1EQ_1016A1/OP006_VALUE.CV' = ' _DO_OFF_ON:ON' AND ('//CF1EQ_1016A1/OP005_VALUE.CV' = 'NS_COOLING_TYPE:Water and Oil' OR <CR>'//CF1EQ_1016A1/OP005_VALUE.CV' = 'NS_COOLING_TYPE:Water'))	START	NO
CF1_PU127816 1	Condition 6	Auto Logic	('TP03.CV' = TRUE) AND<CR>('//CF1EQ_1016A1/OP006_VALUE.CV' = ' _DO_OFF_ON:OFF' OR ('//CF1EQ_1016A1/OP005_VALUE.CV' = 'NS_COOLING_TYPE:Water and Oil' OR <CR>'//CF1EQ_1016A1/OP005_VALUE.CV' = 'NS_COOLING_TYPE:Water'))	STOP	NO
CF1_PU1002A	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//CF1_LT1002A1/PV.CV' >= '//CF1EQ_1002A/TP004_VALUE.CV';	START	NO
CF1_PU1002A	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//CF1_LT1002A1/PV.CV' < '//CF1EQ_1002A/TP004_VALUE.CV';	STOP	NO
CF1_PF1001A_ B	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>('//CF1_LT1002A1/PV' < '//CF1EQ_1002A/TP003_VALUE.CV' AND<CR>'//CF1_V1002AFO/PV_D' = ' _EDC_VLV_PV:OPEN' AND <CR>'//CF1_V1001AD/PV_D' = ' _EDC_VLV_PV:OPEN' AND <CR>'//CF1_LSH1002A1/DI1/DISC_ACT.CV' != TRUE)	START	NO
CF1_PF1001A_ B	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>('//CF1_LT1002A1/PV' >= '//CF1EQ_1002A/TP003_VALUE.CV' OR <CR>'//CF1_V1002AFO/PV_D' != ' _EDC_VLV_PV:OPEN' OR '//CF1_V1001AD/PV_D' != ' _EDC_VLV_PV:OPEN'<CR>OR '//CF1_LSH1002A1/DI1/DISC_ACT.CV' = TRUE)	STOP	NO
CF1_V148THW	Condition 1	Auto Logic	'//CF1_PU125016/PV_D.CV' = ' _EDC_MTR2_PV:RUNNING' AND<CR>'//CF1_TT121161/PV.CV' < ('//CF_ESM_1/SQZ_TANK_TEMP.CV' + 2)	OPEN	NO

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Module	Name	Description	Expression	Force SP State	Delay On (s)
CF1_V148THW	Condition 2	Auto Logic	'//CF1_PU125016/PV_D.CV' = ' _EDC_MTR2_PV:RUNNING' <CR>'//CF1_TT121161/PV.CV' >= '//CF_ESM_1/SQZ_TANK_TEMP.CV'	CLOSE	NO
CF1_V1482BF	Condition 1	Auto Logic	'//CF1EQ_1002A/OP004_VALUE.CV' = FALSE AND <CR>'//CF1_LSL1478B1/DI1/DISC_ACT.CV' = TRUE	OPEN	NO
CF1_V1482BF	Condition 2	Auto Logic	'//CF1EQ_1002A/OP004_VALUE.CV' = TRUE OR <CR>'//CF1_LSH1478B1/DI1/DISC_ACT.CV' = TRUE	CLOSE	NO
CF1_V1478DRT	Condition 1	Auto Logic	'//CF1_PU1478D/PV_D.CV' = ' _EDC_MTR2_PV:RUNNING' AND <CR>'//CF1_V1002AHW/PV_D.CV' = ' _VLVNC_PV:CLOSED' AND<CR>'//CF1_V1102S1FW/PV_D.CV' = ' _VLVNC_PV:CLOSED' AND<CR>'//CF1_V1121S2FW/PV_D.CV' = ' _VLVNC_PV:CLOSED' AND<CR>'//CF1_V11TRFIW/PV_D.CV' = ' _VLVNC_PV:CLOSED' AND<CR>'//CF1_V1216FW/PV_D.CV' = ' _VLVNC_PV:CLOSED'	OPEN	NO
CF1_V1478DRT	Condition 2	Auto Logic	'//CF1_PU1478D/PV_D.CV' != ' _EDC_MTR2_PV:RUNNING' OR <CR>'//CF1_V1002AHW/PV_D.CV' != ' _VLVNC_PV:CLOSED' OR <CR>'//CF1_V1102S1FW/PV_D.CV' != ' _VLVNC_PV:CLOSED' OR <CR>'//CF1_V1121S2FW/PV_D.CV' != ' _VLVNC_PV:CLOSED' OR <CR>'//CF1_V11TRFIW/PV_D.CV' != ' _VLVNC_PV:CLOSED' OR <CR>'//CF1_V1216FW/PV_D.CV' != ' _VLVNC_PV:CLOSED'	CLOSE	NO
CF1_V1478DF	Condition 1	Auto Logic	'//CF1EQ_1002A/OP004_VALUE.CV' = FALSE AND <CR>'//CF1_LSL1478D1/DI1/DISC_ACT.CV' = TRUE	OPEN	NO
CF1_V1478DF	Condition 2	Auto Logic	'//CF1EQ_1002A/OP004_VALUE.CV' = TRUE OR <CR>'//CF1_LSH1478D1/DI1/DISC_ACT.CV' = TRUE	CLOSE	NO
CF1_V1478D	Condition 1	Auto Logic	('//CF1_PU1478C1/PV_D.CV' = ' _EDC_MTR2_PV:RUNNING' OR '//CF1_PF1078C2/PV_D.CV' = ' _EDC_MTR2_PV:RUNNING') AND <CR>('//CF1_LSH1478B1/DI1/DISC_ACT.CV' = TRUE)	OPEN	NO
CF1_V1478D	Condition 2	Auto Logic	('//CF1_PU1478C1/PV_D.CV' != ' _EDC_MTR2_PV:RUNNING' AND '//CF1_PF1078C2/PV_D.CV' != ' _EDC_MTR2_PV:RUNNING') OR <CR>('//CF1_LSH1478B1/DI1/DISC_ACT.CV' != TRUE)	CLOSE	NO

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Module	Name	Description	Expression	Force SP State	Delay On (s)
CF1_V1478CRT	Condition 1	Auto Logic	('//CF1_PU1478C1/PV_D.CV' = '_EDC_MTR2_PV:RUNNING' OR '//CF1_PF1078C2/PV_D.CV' = '_EDC_MTR2_PV:RUNNING') AND<CR>'//CF1_V1002AIW/PV_D.CV' = '_VLVNC_PV:CLOSED' AND<CR>'//CF1_V11TRFIW/PV_D.CV' = '_VLVNC_PV:CLOSED' AND<CR>'//CF1_V1216FIW/PV_D.CV' = '_VLVNC_PV:CLOSED' AND<CR>'//CF1_V1321FIW/PV_D.CV' = '_VLVNC_PV:CLOSED' AND<CR>'//CF1_V1478D/PV_D.CV' = '_VLVNC_PV:CLOSED' AND<CR>'//CF1_TT1150S11/OUT.CV' < 0.001 AND<CR>'//CF1_TT1150S21/OUT.CV' < 0.001	OPEN	NO
CF1_V1478CRT	Condition 2	Auto Logic	('//CF1_PU1478C1/PV_D.CV' != '_EDC_MTR2_PV:RUNNING' AND '//CF1_PF1078C2/PV_D.CV' != '_EDC_MTR2_PV:RUNNING') OR <CR>'//CF1_V1002AIW/PV_D.CV' != '_VLVNC_PV:CLOSED' OR <CR>'//CF1_V11TRFIW/PV_D.CV' != '_VLVNC_PV:CLOSED' OR <CR>'//CF1_V1216FIW/PV_D.CV' != '_VLVNC_PV:CLOSED' OR <CR>'//CF1_V1321FIW/PV_D.CV' != '_VLVNC_PV:CLOSED' OR <CR>'//CF1_V1478D/PV_D.CV' != '_VLVNC_PV:CLOSED' OR <CR>'//CF1_TT1150S11/OUT.CV' >= 0.001 OR <CR>'//CF1_TT1150S21/OUT.CV' >= 0.001	CLOSE	NO
CF1_V1316A2D1	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//CF1EQ_1082B/OP001_VALUE.CV' = 'NS_PRODUCT_FEED:PRODUCT';	OPEN	NO
CF1_V1316A2D1	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//CF1EQ_1082B/OP001_VALUE.CV' = 'NS_PRODUCT_FEED:FEED';	CLOSE	NO
CF1_V1216RTIW	Condition 1	Auto Logic	'//CF1_PU125016/PV_D.CV' = '_EDC_MTR2_PV:RUNNING' AND<CR>'//CF1_TT121161/PV.CV' > ('//CF_ESM_1/SQZ_TANK_TEMP.CV' + 2)	OPEN	NO
CF1_V1216RTIW	Condition 2	Auto Logic	'//CF1_PU125016/PV_D.CV' = '_EDC_MTR2_PV:RUNNING' AND<CR>'//CF1_TT121161/PV.CV' <= '//CF_ESM_1/SQZ_TANK_TEMP.CV'	CLOSE	NO

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CF1_V1216FIW	Condition 1	Auto Logic	'//CF1_PU125016/PV_D.CV' = ' _EDC_MTR2_PV:RUNNING' AND<CR>'//CF1_TT121161/PV.CV' > ('//CF_ESM_1/SQZ_TANK_TEMP.CV' + 2)	OPEN	NO
CF1_V1216FIW	Condition 2	Auto Logic	'//CF1_PU125016/PV_D.CV' = ' _EDC_MTR2_PV:RUNNING' AND<CR>'//CF1_TT121161/PV.CV' <= '//CF_ESM_1/SQZ_TANK_TEMP.CV'	CLOSE	NO
CF1_V1216FW	Condition 1	Auto Logic	'//CF1_PU125016/PV_D.CV' = ' _EDC_MTR2_PV:RUNNING' AND<CR>'//CF1_TT121161/PV.CV' < ('//CF_ESM_1/SQZ_TANK_TEMP.CV' + 2)	OPEN	NO
CF1_V1216FW	Condition 2	Auto Logic	'//CF1_PU125016/PV_D.CV' = ' _EDC_MTR2_PV:RUNNING' AND<CR>'//CF1_TT121161/PV.CV' >= '//CF_ESM_1/SQZ_TANK_TEMP.CV'	CLOSE	NO
CF1_V1102SBA	Condition 1	Auto Logic	(' /TP01.CV' = TRUE) AND <CR>'//CF1_PT1102S/PV' < 2 AND<CR>'//CF1_LSH1360V1/DI1/DISC_ACT.CV' != TRUE;	OPEN	NO
CF1_V1102SBA	Condition 2	Auto Logic	(' /TP01.CV' = TRUE) AND <CR>'//CF1_PT1102S/PV' >= 2 OR <CR>'//CF1_LSH1360V1/DI1/DISC_ACT.CV' = TRUE;	CLOSE	NO
CF1_V1382STB A	Condition 1	Auto Logic	(' /TP01.CV' = TRUE)AND <CR>'//CF1_PU1310ST/PV_D' = ' _EDC_MTR2_PV:STOPPED';	OPEN	NO
CF1_V1382STB A	Condition 2	Auto Logic	(' /TP01.CV' = TRUE) AND <CR>'//CF1_PU1310ST/PV_D' = ' _EDC_MTR2_PV:RUNNING';	CLOSE	NO
CF1_V1382ST	Condition 1	Auto Logic	(' /TP01.CV' = TRUE) AND ('//CF1_TT1382ST1/PV' <= '//CF1EQ_1082ST/TP012_VALUE.CV')	OPEN	NO
CF1_V1382ST	Condition 2	Auto Logic	(' /TP01.CV' = TRUE) AND ('//CF1_TT1382ST1/PV' > '//CF1EQ_1082ST/TP012_VALUE.CV' + 2)	CLOSE	NO
CF1_V1321DO2	Condition 1	Auto Logic	(' /TP01.CV' = TRUE) AND <CR>'//CF1EQ_1082ST/OP002_VALUE.CV' ='NS_PRODUCT_FEED:FEED';	OPEN	NO
CF1_V1321DO2	Condition 2	Auto Logic	(' /TP01.CV' = TRUE) AND <CR>'//CF1EQ_1082ST/OP002_VALUE.CV' !='NS_PRODUCT_FEED:FEED';	CLOSE	NO

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CF1_V1321DO1	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//CF1EQ_1082ST/OP002_VALUE.CV' ='NS_PRODUCT_FEED:PRODUCT';	OPEN	NO
CF1_V1321DO1	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//CF1EQ_1082ST/OP002_VALUE.CV' !='NS_PRODUCT_FEED:PRODUCT';	CLOSE	NO
CF1_V13BDOB	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>('//CF1EQ_1016A1/EN_COOLING_STEP.CV' = TRUE OR '//CF1EQ_1016A2/EN_COOLING_STEP.CV' = TRUE)	OPEN	NO
CF1_V13BDOB	Condition 2	Auto Logic	('TP01.CV' = TRUE)AND <CR>('//CF1EQ_1016A1/EN_COOLING_STEP.CV' != TRUE AND '//CF1EQ_1016A2/EN_COOLING_STEP.CV' != TRUE)	CLOSE	NO
CF1_V1382CDOC	Condition 1	Auto Logic	('TP01.CV' = TRUE)AND ('//CF1EQ_1016A1/EN_WASHING_STEP.CV' = TRUE OR '//CF1EQ_1016A1/EN_COREWASH_STEP.CV' = TRUE OR '//CF1EQ_1016A2/EN_WASHING_STEP.CV' = TRUE OR '//CF1EQ_1016A2/EN_COREWASH_STEP.CV' = TRUE)	OPEN	NO
CF1_V1382CDOC	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND ('//CF1EQ_1016A1/EN_WASHING_STEP.CV' = FALSE AND '//CF1EQ_1016A1/EN_COREWASH_STEP.CV' = FALSE AND '//CF1EQ_1016A2/EN_WASHING_STEP.CV' = FALSE AND '//CF1EQ_1016A2/EN_COREWASH_STEP.CV' = FALSE)	CLOSE	NO
CF1_V1357CKD	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>('//CF1EQ_1016A1/STEP_FLAG1.CV' = FALSE OR <CR>'//CF1EQ_1016A2/STEP_FLAG1.CV' = FALSE)	OPEN	NO
CF1_V1357CKD	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>('//CF1EQ_1016A1/STEP_FLAG1.CV' != FALSE AND <CR>'//CF1EQ_1016A2/STEP_FLAG1.CV' != FALSE)	CLOSE	NO

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Module	Name	Description	Expression	Force SP State	Delay On (s)
CF1_V1321RTIW	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>('//CF1EQ_1016A1/EN_COOLING_STEP.CV' = TRUE OR '//CF1EQ_1016A2/EN_COOLING_STEP.CV' = TRUE)	OPEN	NO
CF1_V1321RTIW	Condition 2	Auto Logic	('TP01.CV' = TRUE)AND <CR>('//CF1EQ_1016A1/EN_COOLING_STEP.CV' != TRUE AND '//CF1EQ_1016A2/EN_COOLING_STEP.CV' != TRUE)	CLOSE	NO
CF1_V1321LPO	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>('//CF1EQ_1016A1/EN_WASHING_STEP.CV' = TRUE OR '//CF1EQ_1016A1/EN_COREWASH_STEP.CV' = TRUE<CR> OR <CR> '//CF1EQ_1016A2/EN_WASHING_STEP.CV' = TRUE OR '//CF1EQ_1016A2/EN_COREWASH_STEP.CV' = TRUE)	OPEN	NO
CF1_V1321LPO	Condition 2	Auto Logic	('TP01.CV' = TRUE)AND <CR>('//CF1EQ_1016A1/EN_WASHING_STEP.CV' != TRUE AND '//CF1EQ_1016A1/EN_COREWASH_STEP.CV' != TRUE	CLOSE	NO
CF1_V1321FIW	Condition 1	Auto Logic	('TP01.CV' = TRUE)AND <CR>('//CF1_TT1321C1/PV' > ('//CF1EQ_1082C/TP005_VALUE.CV' + 2)) AND<CR>('//CF1EQ_1016A1/EN_COOLING_STEP.CV' = TRUE OR '//CF1EQ_1016A2/EN_COOLING_STEP.CV' = TRUE)	OPEN	NO
CF1_V1321FIW	Condition 2	Auto Logic	('TP01.CV' = TRUE)AND <CR>('//CF1_TT1321C1/PV' <= ('//CF1EQ_1082C/TP005_VALUE.CV' - 1)) OR<CR>('//CF1EQ_1016A1/EN_COOLING_STEP.CV' != TRUE AND '//CF1EQ_1016A2/EN_COOLING_STEP.CV' != TRUE)	CLOSE	NO
CF1_V1321CS	Condition 1	Auto Logic	('TP01.CV' = TRUE)AND <CR>('//CF1EQ_1016A1/EN_WASHING_STEP.CV' = TRUE OR '//CF1EQ_1016A1/EN_COREWASH_STEP.CV' = TRUE<CR> OR <CR> '//CF1EQ_1016A2/EN_WASHING_STEP.CV' = TRUE OR '//CF1EQ_1016A2/EN_COREWASH_STEP.CV' = TRUE)	OPEN	NO
CF1_V1321CS	Condition 2	Auto Logic	('TP01.CV' = TRUE)AND <CR>('//CF1EQ_1016A1/EN_WASHING_STEP.CV' != TRUE AND '//CF1EQ_1016A1/EN_COREWASH_STEP.CV' != TRUE AND <CR>	CLOSE	NO

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			'//CF1EQ_1016A2/EN_WASHING_STEP.CV' != TRUE AND '//CF1EQ_1016A2/EN_COREWASH_STEP.CV' != TRUE)		
CF1_V1321CBA	Condition 1	Auto Logic	('TP01.CV' = TRUE)AND ('^PV_D' = '_EDC_VLV_PV:CLOSED')	OPEN	NO
CF1_V1321CBA	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND ('^PV_D' = '_EDC_VLV_PV:OPEN')	CLOSE	NO
CF1_V13BDO2	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//CF1EQ_1082B/OP001_VALUE.CV' = 'NS_PRODUCT_FEED:FEED';	OPEN	NO
CF1_V13BDO2	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//CF1EQ_1082B/OP001_VALUE.CV' = 'NS_PRODUCT_FEED:PRODUCT';	CLOSE	NO
CF1_V13BDO1	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//CF1EQ_1082B/OP001_VALUE.CV' = 'NS_PRODUCT_FEED:PRODUCT';	OPEN	NO
CF1_V13BDO1	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//CF1EQ_1082B/OP001_VALUE.CV' = 'NS_PRODUCT_FEED:FEED';	CLOSE	NO
CF1_V11TRTIW	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//CF1_TT1178TR1/PV' >= ('//CF1EQ_1057CK/TP004_VALUE.CV'-0.001);	OPEN	NO
CF1_V11TRTIW	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//CF1_TT1178TR1/PV' < ('//CF1EQ_1057CK/TP004_VALUE.CV'-1);	CLOSE	NO
CF1_V11TRFIW	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//CF1_TT1178TR1/PV' >= ('//CF1EQ_1057CK/TP004_VALUE.CV');	OPEN	NO
CF1_V11TRFIW	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//CF1_TT1178TR1/PV' < ('//CF1EQ_1057CK/TP004_VALUE.CV'-1);	CLOSE	NO
CF1_V11TRFW	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//CF1_TT1178TR1/PV' < ('//CF1EQ_1057CK/TP004_VALUE.CV'-2);	OPEN	NO
CF1_V11TRFW	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//CF1_TT1178TR1/PV' >= ('//CF1EQ_1057CK/TP004_VALUE.CV'-1);	CLOSE	NO
CF1_V1116ASRO	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//CF1_PT1116AS1/AI1/HI_HI_ACT.CV' = TRUE;	OPEN	NO
CF1_V1116ASRO	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//CF1_PT1116AS1/AI1/HI_HI_ACT.CV' != TRUE;	CLOSE	NO

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Module	Name	Description	Expression	Force SP State	Delay On (s)
CF1_V13460VB A	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//CF1_V13460VDO/PV_D' = '_EDC_VLV_PV:CLOSED' AND <CR>'//CF1_PU1316OV/PV_D' = '_EDC_MTR2_PV:STOPPED')	OPEN	NO
CF1_V13460VB A	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//CF1_V13460VDO/PV_D' = '_EDC_VLV_PV:OPEN' OR <CR>'//CF1_PU1316OV/PV_D' = '_EDC_MTR2_PV:RUNNING')	CLOSE	NO
CF1_V13460V1	Condition 1	Auto Logic	'TP01.CV' = TRUE AND<CR>'//CF1_TT1360V1/PV.CV' < '//CF1EQ_1046OV/TP001_VALUE.CV' AND<CR>'//CF1_LSL1360V1/DI1/DISC_ACT.CV' != TRUE)	OPEN	NO
CF1_V13460V1	Condition 2	Auto Logic	'TP01.CV' = TRUE AND<CR>'//CF1_TT1360V1/PV.CV' >= ('//CF1EQ_1046OV/TP001_VALUE.CV' + 2) OR<CR>'//CF1_LSL1360V1/DI1/DISC_ACT.CV' = TRUE)	CLOSE	NO
CF1_V1261A2F 2	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//CF1_V1216A2F1/PV_D' = '_EDC_VLV_PV:CLOSED';	OPEN	NO
CF1_V1261A2F 2	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//CF1_V1261A2F2/PV_D' = '_EDC_VLV_PV:OPEN';	CLOSE	NO
CF1_V1261A2F 2	Condition 1	Auto Logic	('TP02.CV' = TRUE) AND '//CF1EQ_1016A2/OP005_VALUE.CV' = 'NS_COOLING_TYPE:Water and Oil'	OPEN	NO
CF1_V1261A2F 2	Condition 2	Auto Logic	('TP02.CV' = TRUE) AND '//CF1EQ_1016A2/OP005_VALUE.CV' != 'NS_COOLING_TYPE:Water and Oil'	CLOSE	NO
CF1_V1216A2F C	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND '//CF1EQ_1016A2/OP005_VALUE.CV' = 'NS_COOLING_TYPE:Water and Oil'	OPEN	NO
CF1_V1216A2F C	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND '//CF1EQ_1016A2/OP005_VALUE.CV' != 'NS_COOLING_TYPE:Water and Oil'	CLOSE	NO
CF1_V1216A2F 1	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//CF1_V1216A2F2/PV_D' = '_EDC_VLV_PV:CLOSED';	OPEN	NO
CF1_V1216A2F 1	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//CF1_V1216A2F1/PV_D' = '_EDC_VLV_PV:OPEN';	CLOSE	NO

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CF1_V1216A2F1	Condition 1	Auto Logic	('TP02.CV' = TRUE) AND '//CF1EQ_1016A2/OP005_VALUE.CV' = 'NS_COOLING_TYPE:Water and Oil'	OPEN	NO
CF1_V1216A2F1	Condition 2	Auto Logic	('TP02.CV' = TRUE) AND '//CF1EQ_1016A2/OP005_VALUE.CV' != 'NS_COOLING_TYPE:Water and Oil'	CLOSE	NO
CF1_V1216A2DL	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND<CR>'//CF1_PT1216A21/PV' >= '//CF1EQ_1016A2/TP021_VALUE.CV' AND<CR>'//CF1_LT1382B1/PV' <= '//CF1EQ_1082B/TP001_VALUE.CV' AND <CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' != TRUE;	OPEN	NO
CF1_V1216A2DL	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND<CR>'//CF1_PT1216A21/PV' < '//CF1EQ_1016A2/TP021_VALUE.CV' OR <CR>'//CF1_LT1382B1/PV' > '//CF1EQ_1082B/TP001_VALUE.CV' OR <CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' = TRUE;	CLOSE	5
CF1_V1216A2DL	Condition 3	Auto Logic	('TP02.CV' = TRUE) AND<CR>'//CF1_LT1382B1/PV' <= '//CF1EQ_1082B/TP001_VALUE.CV' AND <CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' != TRUE AND <CR>'//CF1EQ_1016A2/OP006_VALUE.CV' = '_DO_OFF_ON:ON'AND<CR>'//CF1_V1216A2DL/PV_D' = '_EDC_VLV_PV:OPEN' OR <CR>'//CF1_V1216A2DH/PV_D' = '_EDC_VLV_PV:OPEN';	OPEN	NO
CF1_V1216A2DL	Condition 4	Auto Logic	('TP02.CV' = TRUE) AND<CR>'//CF1_LT1382B1/PV' >'//CF1EQ_1082B/TP001_VALUE.CV' OR <CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' = TRUE OR <CR>'//CF1EQ_1016A2/OP006_VALUE.CV' = '_DO_OFF_ON:OFF'OR<CR>'//CF1_V1216A2DL/PV_D' = '_EDC_VLV_PV:CLOSED' AND <CR>'//CF1_V1216A2DH/PV_D' = '_EDC_VLV_PV:CLOSED';	CLOSE	5
CF1_V1216A2DL	Condition 5	Auto Logic	('TP03.CV' = TRUE) AND<CR>'//CF1_LT1382B1/PV' <= '//CF1EQ_1082B/TP001_VALUE.CV' AND <CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' != TRUE;	OPEN	NO

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CF1_V1216A2DL	Condition 6	Auto Logic	('TP03.CV' = TRUE) AND<CR>'//CF1_LT1382B1/PV' >'//CF1EQ_1082B/TP001_VALUE.CV' OR<CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' = TRUE;	CLOSE	NO
CF1_V1216A2DL	Condition 7	Auto Logic	('TP04.CV' = TRUE) AND <CR>'//CF1_V1216A2DH/PV_D' = ' EDC_VLV_PV:CLOSED';	OPEN	NO
CF1_V1216A2DL	Condition 8	Auto Logic	('TP04.CV' = TRUE) AND <CR>'//CF1_V1216A2DH/PV_D' = ' EDC_VLV_PV:OPEN';	CLOSE	NO
CF1_V1216A2DH	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//CF1_LT1382B1/PV' <='//CF1EQ_1082B/TP001_VALUE.CV' AND <CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' != TRUE;	OPEN	NO
CF1_V1216A2DH	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//CF1_LT1382B1/PV' >'//CF1EQ_1082B/TP001_VALUE.CV' OR<CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' = TRUE);	CLOSE	5
CF1_V1216A2DH	Condition 3	Auto Logic	('TP02.CV' = TRUE) AND<CR>'//CF1_LT1382B1/PV' <='//CF1EQ_1082B/TP001_VALUE.CV' AND <CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' != TRUE AND <CR>'//CF1EQ_1016A2/OP006_VALUE.CV' = ' DO_OFF_ON:ON'AND<CR>'//CF1_V1216A2DL/PV_D' = ' EDC_VLV_PV:OPEN' OR <CR>'//CF1_V1216A2DH/PV_D' = ' EDC_VLV_PV:OPEN');	OPEN	NO
CF1_V1216A2DH	Condition 4	Auto Logic	('TP02.CV' = TRUE) AND<CR>'//CF1_LT1382B1/PV' >'//CF1EQ_1082B/TP001_VALUE.CV' OR <CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' = TRUE OR <CR>'//CF1EQ_1016A2/OP006_VALUE.CV' = ' DO_OFF_ON:OFF'OR<CR>'//CF1_V1216A2DL/PV_D' = ' EDC_VLV_PV:CLOSED') AND<CR>'//CF1_V1216A2DH/PV_D' = ' EDC_VLV_PV:CLOSED');	CLOSE	5
CF1_V1216A2DH	Condition 5	Auto Logic	('TP03.CV' = TRUE) AND<CR>'//CF1_LT1382B1/PV' <='//CF1EQ_1082B/TP001_VALUE.CV' AND <CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' != TRUE);	OPEN	NO

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CF1_V1216A2DH	Condition 6	Auto Logic	('TP03.CV' = TRUE) AND<CR>('CF1_LT1382B1/PV' >'CF1EQ_1082B/TP001_VALUE.CV' OR<CR>'CF1_LSH1382B1/DI1/DISC_ACT.CV' = TRUE);	CLOSE	NO
CF1_V1216A2CL	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>'CF1_V1216A2CO/PV_D' = ' _EDC_VLV_PV:CLOSED';	OPEN	NO
CF1_V1216A2CL	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>'CF1_V1216A2CO/PV_D' = ' _EDC_VLV_PV:OPEN';	CLOSE	NO
CF1_V1216A2BO	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>('CF1_LSH1382B1/DI1/DISC_ACT.CV' != TRUE AND<CR>('CF1_V1216A2DL/PV_D' != ' _EDC_VLV_PV:CLOSED' AND <CR>'CF1_V1216A2DH/PV_D' != ' _EDC_VLV_PV:CLOSED') AND 'CF1EQ_1016A1/DEV13_ENABLE_BIT.CV' = 1);	OPEN	NO
CF1_V1216A2BO	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>('CF1_LSH1382B1/DI1/DISC_ACT.CV' = TRUE OR<CR>('CF1_V1216A2DL/PV_D' = ' _EDC_VLV_PV:CLOSED' OR<CR>'CF1_V1216A2DH/PV_D' = ' _EDC_VLV_PV:CLOSED')OR 'CF1EQ_1016A2/DEV13_ENABLE_BIT.CV' = 0);	CLOSE	NO
CF1_V1216A2BO	Condition 3	Auto Logic	('TP02.CV' = TRUE) AND <CR>('CF1_LSH1382C1/DI1/DISC_ACT.CV' != TRUE AND<CR>('CF1_V1216A2CL/PV_D' != ' _EDC_VLV_PV:CLOSED' OR <CR>'CF1_V1216A2CO/PV_D' != ' _EDC_VLV_PV:CLOSED')AND 'CF1EQ_1016A1/DEV13_ENABLE_BIT.CV' = 1);	OPEN	NO
CF1_V1216A2BO	Condition 4	Auto Logic	('TP02.CV' = TRUE) AND <CR>('CF1_LSH1382C1/DI1/DISC_ACT.CV' = TRUE OR<CR>('CF1_V1216A2CL/PV_D' = ' _EDC_VLV_PV:CLOSED' AND<CR>'CF1_V1216A2CO/PV_D' = ' _EDC_VLV_PV:CLOSED') OR 'CF1EQ_1016A2/DEV13_ENABLE_BIT.CV' = 0);	CLOSE	NO
CF1_V1216A2BO	Condition 5	Auto Logic	('TP01.CV' = TRUE) AND <CR>('CF1_LSH1382B1/DI1/DISC_ACT.CV' != TRUE	OPEN	NO

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			AND<CR>'//CF1_LT1382B1/AI1/PV' <= ' //CF1EQ_1082B/TP001_VALUE.CV' AND<CR>('//CF1_V1216A2DL/PV_D' != ' _EDC_VLV_PV:CLOSED' AND <CR>'//CF1_V1216A2DH/PV_D' != ' _EDC_VLV_PV:CLOSED') AND <CR>'//CF1EQ_1016A1/DEV13_ENABLE_BIT.CV' = 1);		
CF1_V1216A2B O	Condition 6	Auto Logic	('TP01.CV' = TRUE) AND <CR>('//CF1_LSH1382B1/DI1/DISC_ACT.CV' = TRUE OR<CR>'//CF1_LT1382B1/AI1/PV' > '//CF1EQ_1082B/TP001_VALUE.CV' OR<CR>('//CF1_V1216A2DL/PV_D' = ' _EDC_VLV_PV:CLOSED' OR<CR>'//CF1_V1216A2DH/PV_D' = ' _EDC_VLV_PV:CLOSED')OR <CR>'//CF1EQ_1016A2/DEV13_ENABLE_BIT.CV' = 0);	CLOSE	NO
CF1_V1216A2B C	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>('//CF1_LSH1360V1/DI1/DISC_ACT.CV' != TRUE AND<CR>('//CF1_V1216A2F1/PV_D' != ' _EDC_VLV_PV:CLOSED' AND <CR>'//CF1_V1261A2F2/PV_D' != ' _EDC_VLV_PV:CLOSED')AND '//CF1EQ_1016A2/DEV12_ENABLE_BIT.CV' = 1);	OPEN	NO
CF1_V1216A2B C	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>('//CF1_LSH1360V1/DI1/DISC_ACT.CV' = TRUE OR<CR>('//CF1_V1216A2F1/PV_D' = ' _EDC_VLV_PV:CLOSED' OR<CR>'//CF1_V1261A2F2/PV_D' = ' _EDC_VLV_PV:CLOSED')OR<CR> '//CF1EQ_1016A2/DEV12_ENABLE_BIT.CV' = 0);	CLOSE	NO
CF1_V1216A2B C	Condition 3	Auto Logic	('TP02.CV' = TRUE) AND <CR>('//CF1_LSH1382C1/DI1/DISC_ACT.CV' != TRUE AND <CR>'//CF1_V1382CRO/PV_D' = ' _EDC_VLV_PV:OPEN' AND <CR>('//CF1_V1216A2F1/PV_D' = ' _EDC_VLV_PV:OPEN' OR <CR>'//CF1_V1261A2F2/PV_D' = ' _EDC_VLV_PV:OPEN')AND '//CF1EQ_1016A2/DEV12_ENABLE_BIT.CV' = 1);	OPEN	NO

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CF1_V1216A2B C	Condition 4	Auto Logic	(/TP02.CV' = TRUE) AND <CR>('//CF1_LSH1382C1/DI1/DISC_ACT.CV' = TRUE OR <CR>'//CF1_V1382CRO/PV_D' = ' _EDC_VLV_PV:OPEN' OR <CR>('//CF1_V1216A2F1/PV_D' = ' _EDC_VLV_PV:CLOSED' OR <CR>'//CF1_V1261A2F2/PV_D' = ' _EDC_VLV_PV:CLOSED') OR '//CF1EQ_1016A2/DEV12_ENABLE_BIT.CV' = 0);	CLOSE	NO
CF1_V1216A1B A	Condition 1	Auto Logic	(/TP01.CV' = TRUE) AND <CR>('//CF1_LSH1360V1/DI1/DISC_ACT.CV' != TRUE AND '^/PV_D' = ' _EDC_VLV_PV:CLOSED')	OPEN	NO
CF1_V1216A1B A	Condition 2	Auto Logic	(/TP01.CV' = TRUE) AND <CR>('//CF1_LSH1360V1/DI1/DISC_ACT.CV' = TRUE OR '^/PV_D' = ' _EDC_VLV_PV:OPEN')	CLOSE	NO
CF1_V1216ABC	Condition 1	Auto Logic	(/TP01.CV' = TRUE) AND <CR>(((//CF1_LSH1360V1/DI1/DISC_ACT.CV' != TRUE) AND<CR>('//CF1_V1216A101/PV_D' != ' _EDC_VLV_PV:CLOSED' AND <CR>'//CF1_V1216A102/PV_D' != ' _EDC_VLV_PV:CLOSED')) AND '//CF1EQ_1016A1/DEV12_ENABLE_BIT.CV' = 1);	OPEN	NO
CF1_V1216ABC	Condition 2	Auto Logic	(/TP01.CV' = TRUE) AND <CR>(((//CF1_LSH1360V1/DI1/DISC_ACT.CV' = TRUE)OR<CR>('//CF1_V1216A101/PV_D' = ' _EDC_VLV_PV:CLOSED' OR<CR>'//CF1_V1216A102/PV_D' = ' _EDC_VLV_PV:CLOSED')) OR <CR>'//CF1EQ_1016A1/DEV12_ENABLE_BIT.CV' = 0);	CLOSE	NO
CF1_V1216ABC	Condition 3	Auto Logic	(/TP02.CV' = TRUE)AND <CR>('//CF1_LSH1382C1/DI1/DISC_ACT.CV' != TRUE AND <CR>'//CF1_V1382CRO/PV_D' = ' _EDC_VLV_PV:OPEN' AND <CR>('//CF1_V1216A101/PV_D' = ' _EDC_VLV_PV:OPEN' OR <CR>'//CF1_V1216A102/PV_D' = ' _EDC_VLV_PV:OPEN') AND '//CF1EQ_1016A1/DEV12_ENABLE_BIT.CV' = 1)	OPEN	NO
CF1_V1216ABC	Condition 4	Auto Logic	(/TP02.CV' = TRUE) AND <CR>('//CF1_LSH1382C1/DI1/DISC_ACT.CV' = TRUE OR	CLOSE	NO

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			<CR>'//CF1_V1382CRO/PV_D' != ' _EDC_VLV_PV:OPEN' OR <CR>'//CF1_V1216A101/PV_D' = ' _EDC_VLV_PV:CLOSED' AND<CR>'//CF1_V1216A102/PV_D' = ' _EDC_VLV_PV:CLOSED')OR '//CF1EQ_1016A1/DEV12_ENABLE_BIT.CV' = 0)		
CF1_V1216A1OL	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND<CR>'//CF1_PT1216A11/PV' >= '//CF1EQ_1016A1/TP021_VALUE.CV' AND<CR>'//CF1_LT1382B1/PV' <=/'//CF1EQ_1082B/TP001_VALUE.CV' AND <CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' != TRUE);	OPEN	5
CF1_V1216A1OL	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND<CR>'//CF1_LT1382B1/PV' > '//CF1EQ_1082B/TP001_VALUE.CV' OR <CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' = TRUE);	CLOSE	NO
CF1_V1216A1OL	Condition 3	Auto Logic	('TP02.CV' = TRUE)AND<CR>'//CF1_LT1382B1/PV' <=/'//CF1EQ_1082B/TP001_VALUE.CV' AND <CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' != TRUE)	OPEN	5
CF1_V1216A1OL	Condition 4	Auto Logic	('TP02.CV' = TRUE) AND<CR>'//CF1_LT1382B1/PV' >'//CF1EQ_1082B/TP001_VALUE.CV' OR <CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' = TRUE)	CLOSE	5
CF1_V1216A1OL	Condition 5	Auto Logic	('TP03.CV' = TRUE)AND<CR>'//CF1_LT1382B1/PV' <=/'//CF1EQ_1082B/TP001_VALUE.CV' AND <CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' != TRUE);	OPEN	NO
CF1_V1216A1OL	Condition 6	Auto Logic	('TP03.CV' = TRUE)AND<CR>'//CF1_LT1382B1/PV' >'//CF1EQ_1082B/TP001_VALUE.CV' OR <CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' = TRUE);	CLOSE	NO
CF1_V1216A1OL	Condition 7	Auto Logic	('TP04.CV' = TRUE) AND <CR>'//CF1_V1216A1DH/PV_D' = ' _EDC_VLV_PV:CLOSED';	OPEN	NO
CF1_V1216A1OL	Condition 8	Auto Logic	('TP04.CV' = TRUE)AND <CR>'//CF1_V1216A1DH/PV_D' = ' _EDC_VLV_PV:OPEN';	CLOSE	NO
CF1_V1216A1FC	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND '//CF1EQ_1016A1/OP005_VALUE.CV' = 'NS_COOLING_TYPE:Water and Oil'	OPEN	NO

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CF1_V1216A1FC	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND '/CF1EQ_1016A1/OP005_VALUE.CV' != 'NS_COOLING_TYPE:Water and Oil'	CLOSE	NO
CF1_V1216A1DH	Condition 1	Auto Logic	('TP01.CV' = TRUE)AND <CR>('/CF1_LT1382B1/PV' <='//CF1EQ_1082B/TP001_VALUE.CV' AND <CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' != TRUE);	OPEN	5
CF1_V1216A1DH	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>('/CF1_LT1382B1/PV' > '/CF1EQ_1082B/TP001_VALUE.CV' OR<CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' = TRUE);	CLOSE	5
CF1_V1216A1DH	Condition 3	Auto Logic	('TP02.CV' = TRUE)AND<CR>('/CF1_LT1382B1/PV' <='//CF1EQ_1082B/TP001_VALUE.CV' AND <CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' != TRUE)	OPEN	NO
CF1_V1216A1DH	Condition 4	Auto Logic	('TP02.CV' = TRUE) AND<CR>('/CF1_LT1382B1/PV' >'//CF1EQ_1082B/TP001_VALUE.CV' OR <CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' = TRUE)	CLOSE	5
CF1_V1216A1DH	Condition 5	Auto Logic	('TP03.CV' = TRUE)AND<CR>'//CF1_LT1382B1/PV' <='//CF1EQ_1082B/TP001_VALUE.CV' AND <CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' != TRUE;	OPEN	NO
CF1_V1216A1DH	Condition 6	Auto Logic	('TP03.CV' = TRUE) AND<CR>('/CF1_LT1382B1/PV' >'//CF1EQ_1082B/TP001_VALUE.CV' OR<CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' = TRUE);	CLOSE	NO
CF1_V1216A1BO	Condition 1	Auto Logic	('TP01.CV' = TRUE)AND <CR>('/CF1_LSH1382B1/DI1/DISC_ACT.CV' != TRUE AND<CR>('/CF1_V1216A1OL/PV_D' != '_EDC_VLV_PV:CLOSED' AND <CR>'//CF1_V1216A1DH/PV_D' != '_EDC_VLV_PV:CLOSED')AND '/CF1EQ_1016A1/DEV13_ENABLE_BIT.CV' = 1);	OPEN	2
CF1_V1216A1BO	Condition 2	Auto Logic	('TP01.CV' = TRUE)AND <CR>('/CF1_LSH1382B1/DI1/DISC_ACT.CV' = TRUE OR<CR>('/CF1_V1216A1OL/PV_D' = '_EDC_VLV_PV:CLOSED'	CLOSE	NO

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			OR<CR>'//CF1_V1216A1DH/PV_D' = '_EDC_VLV_PV:CLOSED') AND '//CF1EQ_1016A1/DEV13_ENABLE_BIT.CV' = 0);		
CF1_V1216A1BO	Condition 3	Auto Logic	('TP02.CV' = TRUE) AND <CR>'//CF1_LSH1382C1/DI1/DISC_ACT.CV' != TRUE AND<CR>'//CF1_V1216A1CL/PV_D' != '_EDC_VLV_PV:CLOSED' AND <CR>'//CF1_V1216A1CH/PV_D' != '_EDC_VLV_PV:CLOSED') AND '//CF1EQ_1016A1/DEV13_ENABLE_BIT.CV' = 1);	OPEN	2
CF1_V1216A1BO	Condition 4	Auto Logic	('TP02.CV' = TRUE) AND <CR>'//CF1_LSH1382C1/DI1/DISC_ACT.CV' = TRUE OR<CR>'//CF1_V1216A1CL/PV_D' = '_EDC_VLV_PV:CLOSED' OR<CR>'//CF1_V1216A1CH/PV_D' = '_EDC_VLV_PV:CLOSED') OR '//CF1EQ_1016A1/DEV13_ENABLE_BIT.CV' = 0);	CLOSE	NO
CF1_V1216A1BO	Condition 5	Auto Logic	('TP01.CV' = TRUE)AND <CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' != TRUE AND<CR>'//CF1_LT1382B1/AI1/PV' <= '//CF1EQ_1082B/TP001_VALUE.CV' AND<CR>'//CF1_V1216A1OL/PV_D' != '_EDC_VLV_PV:CLOSED' AND <CR>'//CF1_V1216A1DH/PV_D' != '_EDC_VLV_PV:CLOSED')AND '//CF1EQ_1016A1/DEV13_ENABLE_BIT.CV' = 1);	OPEN	NO
CF1_V1216A1BO	Condition 6	Auto Logic	('TP01.CV' = TRUE)AND <CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' = TRUE OR<CR>'//CF1_LT1382B1/AI1/PV' > '//CF1EQ_1082B/TP001_VALUE.CV' OR<CR>'//CF1_V1216A1OL/PV_D' = '_EDC_VLV_PV:CLOSED' OR<CR>'//CF1_V1216A1DH/PV_D' = '_EDC_VLV_PV:CLOSED') AND <CR>'//CF1EQ_1016A1/DEV13_ENABLE_BIT.CV' = 0);	CLOSE	NO
CF1_V1216A102	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//CF1_V1216A101/PV_D' = '_EDC_VLV_PV:CLOSED'	OPEN	30

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CF1_V1216A102	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//CF1_V1216A102/PV_D' = ' _EDC_VLV_PV:OPEN'	CLOSE	NO
CF1_V1216A102	Condition 3	Auto Logic	('TP02.CV' = TRUE) AND '//CF1EQ_1016A1/OP005_VALUE.CV' = 'NS_COOLING_TYPE:Water and Oil'	OPEN	NO
CF1_V1216A102	Condition 4	Auto Logic	('TP02.CV' = TRUE) AND '//CF1EQ_1016A1/OP005_VALUE.CV' != 'NS_COOLING_TYPE:Water and Oil'	CLOSE	NO
CF1_V1216A101	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//CF1_V1216A102/PV_D' = ' _EDC_VLV_PV:CLOSED'	OPEN	3
CF1_V1216A101	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//CF1_V1216A101/PV_D' = ' _EDC_VLV_PV:OPEN';	CLOSE	NO
CF1_V1216A101	Condition 3	Auto Logic	('TP02.CV' = TRUE)AND '//CF1EQ_1016A1/OP005_VALUE.CV' = 'NS_COOLING_TYPE:Water and Oil'	OPEN	NO
CF1_V1216A101	Condition 4	Auto Logic	('TP02.CV' = TRUE) AND '//CF1EQ_1016A1/OP005_VALUE.CV' != 'NS_COOLING_TYPE:Water and Oil'	CLOSE	NO
CF1_V12161BA	Condition 1	Auto Logic	('TP01.CV' = TRUE)AND <CR>('//CF1_LSH1360V1/DI1/DISC_ACT.CV' != TRUE AND '^/PV_D' = ' _EDC_VLV_PV:CLOSED')	OPEN	5
CF1_V12161BA	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>('^/PV_D' = ' _EDC_VLV_PV:OPEN')	CLOSE	10
CF1_V12161BA	Condition 3	Auto Logic	('TP01.CV' = TRUE) AND <CR>('//CF1_LSH1360V1/DI1/DISC_ACT.CV' = TRUE)	CLOSE	NO
CF1_V1102S2B A	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//CF1_PT1102S21/AI1/PV.CV' < 2 AND<CR>'//CF1_LSH1360V1/DI1/DISC_ACT.CV' != TRUE;	OPEN	NO
CF1_V1102S2B A	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//CF1_PT1102S21/AI1/PV.CV' >= 2 OR <CR>'//CF1_LSH1360V1/DI1/DISC_ACT.CV' = TRUE;	CLOSE	NO
CF1_V1102DW	Condition 1	Auto Logic	'TP01.CV' = TRUE AND<CR>'//CF1EQ_1002S2/TMR1/TM_HOLD.CV' = ' _TIMER_HOLD:Run'	OPEN	NO
CF1_V1102DW	Condition 2	Auto Logic	'TP01.CV' = TRUE AND<CR>'//CF1_TT1150S11/OUT.CV' >= 0.01	CLOSE	NO

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CF1_V1102S1B A	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//CF1_PT1102S11/A11/PV.CV' < 2 AND<CR>'//CF1_LSH1360V1/DI1/DISC_ACT.CV' != TRUE;	OPEN	NO
CF1_V1102S1B A	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//CF1_PT1102S11/A11/PV.CV' >= 2 OR <CR>'//CF1_LSH1360V1/DI1/DISC_ACT.CV' = TRUE;	CLOSE	NO
CF1_V1102S1R W	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//CF1_LT1478D1/PV' > 'CF1EQ_1002S1/TP022_VALUE.CV')	OPEN	NO
CF1_V1102S1R W	Condition 2	Auto Logic	('TP02.CV' = TRUE) AND <CR>'//CF1_TT1150S11/PV' >= ('CF1_TT1150S11/REQ_SP.CV'))	CLOSE	NO
CF1_V1102S1R W	Condition 3	Auto Logic	'TP02.CV' = TRUE AND <CR>'//CF1_TT1150S11/PV' < ('CF1_TT1150S11/REQ_SP.CV'))	CLOSE	NO
CF1_V1102S1F W	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//CF1_TT1150S11/PV' < ('CF1_TT1150S11/REQ_SP.CV' - 1))	OPEN	60
CF1_V1102S1F W	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//CF1_TT1150S11/PV' < ('CF1_TT1150S11/REQ_SP.CV' - 2))	CLOSE	NO
CF1_V1102S1F W	Condition 3	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//CF1_TT1150S11/PV' >= ('CF1_TT1150S11/REQ_SP.CV'))	CLOSE	NO
CF1_V11021DW	Condition 1	Auto Logic	'TP01.CV' = TRUE AND<CR>'//CF1EQ_1002S1/TMR1/TM_HOLD.CV' = 'TIMER_HOLD:Run'	OPEN	NO
CF1_V11021DW	Condition 2	Auto Logic	'TP01.CV' = TRUE AND<CR>'//CF1_V1102S1FW/PV_D.CV' = '_EDC_VLV_PV:OPEN' OR <CR>'//CF1_TT1150S11/OUT.CV' >= 0.01)	CLOSE	NO
CF1_V1002TIW	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//CF1_V1002THW/PV_D' = '_EDC_VLV_PV:CLOSED' AND <CR>'//CF1_V1002AHW/PV_D' = '_EDC_VLV_PV:CLOSED')	OPEN	NO
CF1_V1002TIW	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>'//CF1_V1002THW/PV_D' != '_EDC_VLV_PV:CLOSED' OR <CR>'//CF1_V1002AHW/PV_D' != '_EDC_VLV_PV:CLOSED')	CLOSE	NO

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CF1_V1002THW	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>('CF1_V1002TIW/PV_D' = '_EDC_VLV_PV:CLOSED' AND <CR>'CF1_V1002AIW/PV_D' = '_EDC_VLV_PV:CLOSED')	OPEN	NO
CF1_V1002THW	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>('CF1_V1002TIW/PV_D' != '_EDC_VLV_PV:CLOSED' OR <CR>'CF1_V1002AIW/PV_D' != '_EDC_VLV_PV:CLOSED')	CLOSE	NO
CF1_V1002DO	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>('CF1_LT1002A1/PV' >= '//CF1EQ_1002A/TP001_VALUE.CV' OR <CR>'CF1_PU1102A/PV_D.CV' != '_EDC_MTR2_PV:STOPPED')	OPEN	NO
CF1_V1002DO	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>('CF1_LT1002A1/PV' < '//CF1EQ_1002A/TP001_VALUE.CV' AND <CR>'CF1_PU1102A/PV_D.CV' = '_EDC_MTR2_PV:STOPPED')	CLOSE	NO
CF1_V1002AIW	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>('CF1_V1002THW/PV_D' = '_EDC_VLV_PV:CLOSED' AND <CR>'CF1_V1002AHW/PV_D' = '_EDC_VLV_PV:CLOSED')	OPEN	NO
CF1_V1002AIW	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>('CF1_V1002THW/PV_D' != '_EDC_VLV_PV:CLOSED' OR <CR>'CF1_V1002AHW/PV_D' != '_EDC_VLV_PV:CLOSED')	CLOSE	NO
CF1_V1002AHW	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>('CF1_V1002TIW/PV_D' = '_EDC_VLV_PV:CLOSED' AND <CR>'CF1_V1002AIW/PV_D' = '_EDC_VLV_PV:CLOSED')	OPEN	NO
CF1_V1002AHW	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>('CF1_V1002TIW/PV_D' != '_EDC_VLV_PV:CLOSED' OR <CR>'CF1_V1002AIW/PV_D' != '_EDC_VLV_PV:CLOSED')	CLOSE	NO
CF1_V1002AFO	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>('CF1_LT1002A1/PV' < '//CF1EQ_1002A/TP003_VALUE.CV' AND <CR>'CF1_LSH1002A1/DI1/DISC_ACT.CV' != TRUE)	OPEN	5
CF1_V1002AFO	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>('CF1_LT1002A1/PV' >= '//CF1EQ_1002A/TP003_VALUE.CV' OR <CR>'CF1_LSH1002A1/DI1/DISC_ACT.CV' = TRUE)	CLOSE	NO

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Module	Name	Description	Expression	Force SP State	Delay On (s)
CF1_V1002ADO	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>('CF1_LT1002A1/PV' >= 'CF1EQ_1002A/TP001_VALUE.CV' OR <CR>'CF1_PU1102A/PV_D.CV' != 'EDC_MTR2_PV:STOPPED')	OPEN	NO
CF1_V1002ADO	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>('CF1_LT1002A1/PV' < 'CF1EQ_1002A/TP001_VALUE.CV' AND <CR>'CF1_PU1102A/PV_D.CV' = 'EDC_MTR2_PV:STOPPED')	CLOSE	NO
CF1_V1001AR	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>('CF1_V1102ABY/PV_D' = 'EDC_VLV_PV:OPEN' AND 'CF1_LSH1360V1/DI1/DISC_ACT' != TRUE)	OPEN	NO
CF1_V1001AR	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>('CF1_V1102ABY/PV_D' = 'EDC_VLV_PV:OPEN' OR 'CF1_LSH1360V1/DI1/DISC_ACT' != TRUE)	CLOSE	NO
CF1_V1001AD	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>('CF1_LT1002A1/PV' < 'CF1EQ_1002A/TP003_VALUE.CV' AND <CR>'CF1_LSH1002A1/DI1/DISC_ACT.CV' != TRUE)	OPEN	5
CF1_V1001AD	Condition 2	Auto Logic	('TP01.CV' = TRUE) AND <CR>('CF1_LT1002A1/PV' >= 'CF1EQ_1002A/TP003_VALUE.CV' OR <CR>'CF1_LSH1002A1/DI1/DISC_ACT.CV' = TRUE)	CLOSE	NO
CF1_V1001AD	Condition 3	Auto Logic	('TP02.CV' = TRUE) AND <CR>'CF1_LT1002A1/PV.CV' >= 'CF1EQ_1002A/TP004_VALUE.CV';	OPEN	NO
CF1_V1001AD	Condition 4	Auto Logic	('TP02.CV' = TRUE) AND <CR>'CF1_LT1002A1/PV.CV' < 'CF1EQ_1002A/TP004_VALUE.CV';	CLOSE	NO
CF1_PT1102S	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>('CF1_PT1102S/PV' >= 1 OR <CR>'CF1_LSH1102S1/DI1/DISC_ACT' != FALSE OR 'CF1_LSH1102S2/DI1/DISC_ACT.CV' != FALSE OR <CR>'CF1_LSH11S11/DI1/DISC_ACT.CV' != FALSE OR 'CF1_LSH1102S3/DI1/DISC_ACT.CV' != FALSE OR ('CF1_LT1002A1/PV.CV' < 'CF1EQ_1002S1/TP002_VALUE.CV')) AND<CR>('MODELOCK_OVR.CV' = FALSE)	0	NO

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CF1_TT1321C1	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>('//CF1EQ_1016A1/EN_WASHING_STEP.CV' != TRUE OR '//CF1EQ_1016A1/EN_COREWASH_STEP.CV' != TRUE<CR> OR <CR> '//CF1EQ_1016A2/EN_WASHING_STEP.CV' != TRUE OR '//CF1EQ_1016A2/EN_COREWASH_STEP.CV' != TRUE) AND <CR>('/MODELOCK_OVR.CV' = FALSE)	0	NO
CF1_PT1216AS 4	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND<CR>'//CF1_LT1382B1/PV' >'//CF1EQ_1082B/TP001_VALUE.CV' AND <CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' = TRUE;	0	NO
CF1_PT1216AS 4	Condition 2	Auto Logic	('TP02.CV' = TRUE) AND<CR>'//CF1EQ_1016A2/OP006_VALUE.CV' = '_DO_OFF_ON:ON' AND ('//CF1EQ_1016A2/OP005_VALUE.CV' = '_NS_COOLING_TYPE:Water and Oil' OR <CR>'//CF1EQ_1016A2/OP005_VALUE.CV' = '_NS_COOLING_TYPE:Water')	0	NO
CF1_PT1216AS 2	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND<CR>'//CF1_LT1382B1/PV' >'//CF1EQ_1082B/TP001_VALUE.CV' AND <CR>'//CF1_LSH1382B1/DI1/DISC_ACT.CV' = TRUE;	0	NO
CF1_PT1216AS 2	Condition 2	Auto Logic	('TP02.CV' = TRUE) AND<CR>'//CF1EQ_1016A1/OP006_VALUE.CV' = '_DO_OFF_ON:ON' AND ('//CF1EQ_1016A1/OP005_VALUE.CV' = '_NS_COOLING_TYPE:Water and Oil' OR <CR>'//CF1EQ_1016A1/OP005_VALUE.CV' = '_NS_COOLING_TYPE:Water')	0	NO
CF1_TT1150S2 1	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>('/TP02.CV' = TRUE) AND<CR>('//CF1_V1102DW/PV_D.CV' = '_EDC_VLV_PV:OPEN') AND<CR>('/MODELOCK_OVR.CV' = FALSE)	0	NO
CF1_TT1150S1 1	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND <CR>('/TP02.CV' = TRUE) OR ('//CF1_V1102S1FW/PV_D.CV' = '_EDC_VLV_PV:OPEN' AND '/TP02.CV' = FALSE)) AND<CR>('/MODELOCK_OVR.CV' = FALSE)	0	NO
CF1_TT1050A1	Condition 1	Auto Logic	('TP01.CV' = TRUE AND '/TP03.CV' = FALSE) AND ('/MODELOCK_OVR.CV' = FALSE)	0	NO

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CF1_TT1050A1	Condition 2	Auto Logic	('TP02.CV' = TRUE AND 'TP03.CV' = TRUE) AND ('MODELOCK_OVR.CV' = FALSE)	0	NO
CF1_TT1021A1	Condition 1	Auto Logic	('TP01.CV' = TRUE) AND ('MODELOCK_OVR.CV' = FALSE) AND <CR>('CF1_LT1002A1/PV' >= 'CF1EQ_1002A/TP003_VALUE.CV' OR <CR>'CF1_LSH1002A1/DI1/DISC_ACT.CV' = TRUE)	0	NO

Table 73: Module Force Setpoints